

# Rare Earth Elements Supply and Financial Markets: Evidence from High-Frequency Analysis



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# Rare Earth Elements Supply and Financial Markets: Evidence from High-Frequency Analysis

– *Master thesis* –

Philip Kroos

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## **Abstract**

Rare earth elements (REEs) are bottleneck inputs in many value chains, yet there is little evidence on how firm-level supply disclosures are reflected in REE spot prices and in the financial market. I compile calendar-dated disclosures from major listed producers, align them with daily spot quotes, and construct a firm–metal news shock series within a high-frequency analysis. Event windows display flatter densities with heavier tails, but no systematic rise in volatility. Local projections indicate small, metal-heterogeneous level shifts that emerge quickly and persist for several weeks. I find no statistically significant effects on the financial market, neither for broad equity indexes nor for REE-intensive firms. The paper contributes a new event-time database news shock series for REEs and provides, to my knowledge, the first high-frequency evidence on how firm disclosures are reflected in REE spot prices and the broader financial market.

# 1 Introduction

Over the past two decades, rare earth elements (REEs) have gained increasing importance in both global economic and energy policy debates. They serve as critical inputs for key technologies, including electric motors, wind turbines, digital infrastructure, defense applications, and everyday consumer technologies. As the world accelerates its transition to renewable energy and electric mobility, demand for REEs has risen significantly. According to projections by the International Energy Agency (IEA), clean energy technologies alone are expected to account for nearly 40% of specific REE demand by 2040, compared to just 15% in 2020 under the Sustainable Development Scenario ([International Energy Agency, 2021](#)). This rapid growth highlights structural shifts in the energy sector and raises important questions about the market structure, transparency, and stability of the REE supply chain. REE extraction and processing is technologically demanding, and environmentally costly ([Lederer et al., 2024](#); [Mudd et al., 2017](#)). Due to these challenges, global REE supply has become highly concentrated. Since the late 1980s, China has become the dominant producer and processor along the REE supply chain ([Seaman, 2019](#)). This dominance coincides with thin spot trading, off-exchange transactions, and limited price transparency. In such opaque markets, publicly disclosed supply information by producers can be unusually informative for spot price discovery ([Duffie et al., 2010](#)). Despite the growing economic and strategic relevance of rare earths, the empirical literature remains relatively thin, and existing papers focus primarily on geopolitical dependencies, trade policy risks, and long-term supply security. In contrast, little is known about how information is processed within rare earth markets at the firm-level, particularly in response to supply-side news. This leads to two central questions: first, how quickly and to what extent are firm-level supply announcements reflected in daily REE spot prices? Second, do these announcements have measurable effects on broader financial markets (for example, stock indices or firms with high REE exposure)? To address these questions, I align precisely timed firm announcements with daily REE spot prices in a high-frequency event-study framework. Specifically, each announcement defines a narrow event window to capture immediate price adjustments, and I use local projections to map post-announcement return dynamics over subsequent trading days. Given the market’s concentration and opacity, public supply announcements by major REE producers may carry significant informational value for spot prices. I compile a new, calendar-dated database of REE supply announcements from major listed firms, and align them with the REE spot trading calendar.

A substantial literature examines how informational shocks affect asset prices, with early work focusing on monetary policy and macro announcements ([Altavilla et al., 2019](#); [Flan-](#)

nery and Protopapadakis, 2002; Gürkaynak et al., 2005; Nakamura and Steinsson, 2018). Beyond monetary policy, recent applications include commodity and environmental markets: Känzig (2021) studies OPEC related oil supply surprises, and Känzig (2023) applies similar logic to the EU Emissions Trading System. Other studies highlight the role of unscheduled, qualitative, or firm-level news in settings with pronounced informational frictions (Boyd et al., 2005; Medovikov, 2016). For estimation, I employ local projections (Jordà, 2005) a transparent way to trace dynamics that, under standard conditions, align with VAR-based impulse responses (Plagborg-Møller and Wolf, 2021; Ramey, 2016). Despite these advances, high-frequency evidence on rare earth spot pricing is limited. Most research emphasizes structural, geopolitical, and policy dimensions rather than price discovery around firm news. Consistent with this gap, the local projections I report summarize conditional event-time dynamics and are not interpreted as causal effects in the absence of an identified exogenous shock or instrument. This paper contributes in three ways. First, it builds a precisely dated dataset of firm-level supply disclosures by major listed producers and uses it to construct a high-frequency news shock series for rare earth markets. Second, it quantifies immediate and medium-run price responses using an announcement window design and local projections, reporting standardized cumulative return responses. Third, it explores heterogeneity across elements and documents effects on broader financial markets. Taken together, these contributions extend high-frequency event-study methods to a strategically important but thin commodity market, shedding new light on informational efficiency in REE pricing. To my knowledge this is the first study to analyze rare earth spot prices in a high-frequency analysis around firm announcements.

In summary, the empirical results show the following: Announcement windows exhibit heavier tails in return distributions than control windows, but without a systematic increase in volatility. Local-projection estimates indicate small but persistent level shifts that vary across elements, with effects that typically persist over several weeks; by contrast, there is no statistically significant impact on broad equity indices or on stocks of firms with high REE intensity.

The remainder of the paper is organized as follows. Section 2 provides background on the rare earth market. Section 3 describes data sources and event construction. Section 4 outlines the empirical framework. Section 5 presents the main results. Section 6 presents extensions to the results. Section 7 reports robustness checks. Section 8 discusses limitations and implications. Section 9 concludes.

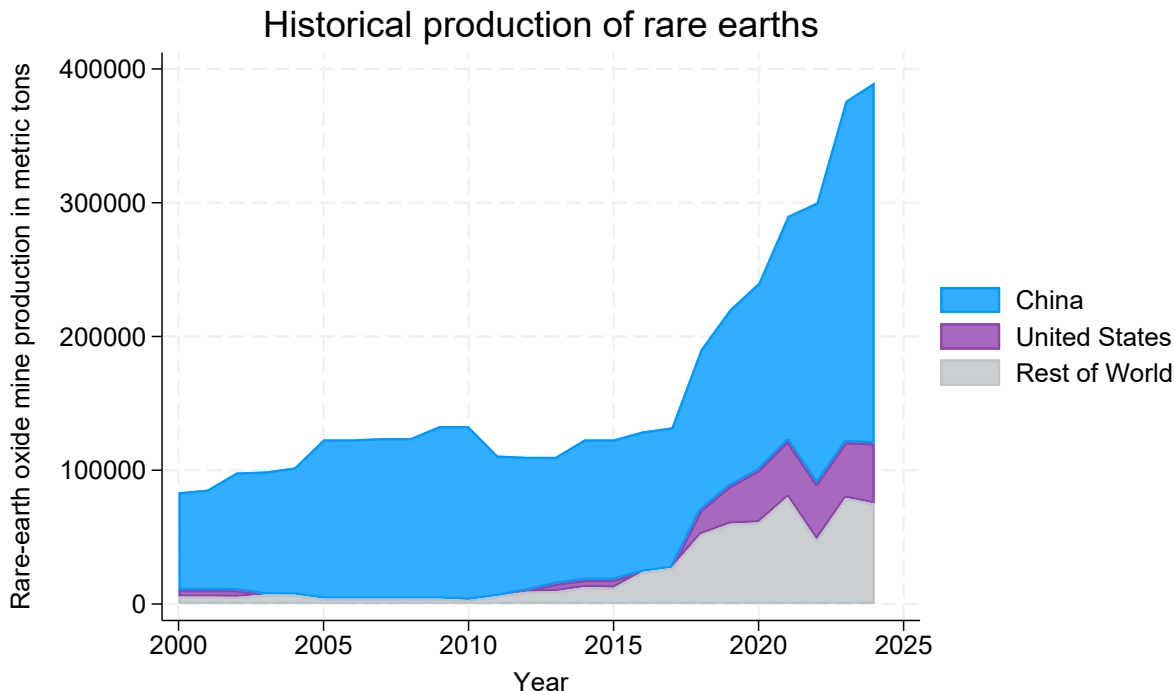
## 2 Background

The strategic importance of rare earth elements (REEs) has become a defining feature of the twenty-first-century global economy. Although largely invisible to the general public, they are the “vitamins” of modern industry, essential in trace amounts yet indispensable to technologies that underpin advanced manufacturing, clean energy, and national defense. Their irreplaceability in high-strength magnets, batteries, catalysts, display phosphors, and critical military applications implies that supply disruptions can have large macroeconomic and geopolitical effects. This combination of high strategic value and thin, opaque spot markets creates a setting where firm-level news could trigger strong price reactions (Humphries, 2010; Kalantzakos, 2017). REEs are a group of seventeen chemically similar metals characterized by unique physical and chemical properties (Fernandez, 2017; Lederer et al., 2024). Although several REEs are as abundant as common base metals such as copper or zinc, they typically co-occur in the same mineral deposits rather than as concentrated single-element ores, making extraction and separation inherently collective and creating interdependencies in supply, pricing, and market dynamics across elements (Mudd et al., 2017; Werner et al., 2017). The extraction and separation processes are technologically demanding and environmentally challenging, including the management of radioactive byproducts, hazardous waste, and intensive water usage (Fernandez, 2017; Lee and Wen, 2017).

The global supply chain for REEs is shaped not only by their unique chemistry but also by an asymmetric geographical and industrial structure. Although substantial REE reserves exist in several countries, including Brazil, Australia, India, and the United States, commercial production has remained highly concentrated in China since the late 1990s. In 2024, China accounted for about 70% of global rare earth mine production and held roughly 44 million metric tons of reserves, far exceeding other producers such as the United States and Myanmar (Kim and Jariwala, 2021; Lederer et al., 2024). Figure 1 plots annual rare earth oxide mine output (metric tons) by country, 2000–2024, to highlight the overwhelming concentration of production in China. The rare earth market is concentrated both geographically and industrially, with China dominating every stage from mining to refining. China holds the world’s largest REE reserves and controls a very large share of midstream separation and refining capacity (often estimated at 80–90%) (Seaman, 2019). Beyond favorable geology and costs, a sustained program of state-directed industrial policy has reinforced this position (Andrews-Speed and Hove, 2023; Morrison and Tang, 2012; Tse, 2011). On the demand side, REEs are essential to a wide range of industrial applications: mature sectors such as catalysts, glassmaking, and metallurgy still account for most global consumption, while demand is expanding rapidly in high-tech industries including renewable energy, elec-



Figure 1: Historical Production of Rare Earth Elements by Country (2000–2024)



*Note:* The figure displays annual mine production of rare earth oxide (in metric tons) for China, the United States, and the rest of the world from 2000 to 2024. The figure highlights China’s increasing dominance in global rare earth production over the past two decades. Own illustration based on official production data from the United States Geological Survey (USGS). Data source: <https://www.usgs.gov/centers/national-minerals-information-center/rare-earths-statistics-and-information>. [accessed 05 July, 2025]

tric vehicles, advanced electronics, and defense systems (Alonso et al., 2012). This growing dependence on REEs is reflected in their formal designation as ”critical raw materials” by the EU and the United States and in concerted efforts by industrialized economies to secure alternative sources and develop more resilient supply chains (Binnemans et al., 2013b; Gulley et al., 2018; Service, 2023; U.S. Department of Energy, 2023). Despite these efforts, most value addition and final processing continue to occur in China, and attempts to establish independent supply chains elsewhere are hindered by high capital costs, strict environmental regulation, and the technical complexity of processing mixed REE concentrates (Ilankoon et al., 2022). Consequently, the continued concentration of mining, separation, and refining in China leaves the global REE market highly exposed to supply shocks, regulatory shifts, and broader uncertainties associated with economic statecraft. In sum, REEs are technologically essential inputs and a central concern in the economic and geopolitical challenges facing advanced economies. The combination of joint production, high processing

costs, and extreme geographical concentration makes rare earths an informative setting to study whether and how firm supply announcements are reflected in spot prices and broader financial markets.

## 2.1 Institutional and Geopolitical Context

Beyond geology and demand, rare earth pricing is shaped by institutions and state policy that influence price discovery and market functioning. As noted above, supply chains are highly concentrated. China’s longstanding policy interventions, including export quotas, licensing regimes, subsidies, and related regulatory controls, have shaped global availability and pricing dynamics (Morrison and Tang, 2012; Tse, 2011; Wübbeke, 2013). China’s use of rare earth policy as a tool of economic statecraft is most clearly illustrated by the 2010 export embargo against Japan. China effectively stopped rare earth shipments to Japan and tightened quotas, demonstrating the vulnerability of supply. This action triggered global supply disruptions and highlighted the strategic vulnerabilities of advanced economies (Vekasi, 2019). In March 2012, the United States, European Union, and Japan filed a complaint at the World Trade Organization (WTO) challenging China’s export quotas, licensing restrictions, and dual pricing practices. These measures were widely viewed as inconsistent with the WTO rules, distorting international competition by favoring domestic firms and artificially restricting global supply (Proelss et al., 2018; Humphries, 2010). To structure the temporal dynamics of this institutional dispute, I distinguish between three phases, each reflecting a different stage in the evolution of market conditions and legal expectations.

- (i) **Phase 1: Pre-Complaint (pre-2012):** This period was characterized by a tightly controlled export regime, including strict quotas, high tariffs, and differential pricing. These policies segmented the global market, undermined price transparency, and introduced substantial inefficiencies in cross-border trade (Müller et al., 2016). In 2010, following a diplomatic incident with Japan, China was widely perceived to have reduced rare earth exports. This prompted Japan to secure alternative suppliers and may have contributed to subsequent price declines. In 2012, the US, EU, and Japan filed a formal WTO dispute against China’s export restrictions, arguing that the regime distorted global prices by favoring domestic consumers and restricting foreign access (Kim and Jariwala, 2021; Wübbeke, 2013).
- (ii) **Phase 2: Complaint Pending (2012–2014):** Following the WTO complaint in March 2012, formal restrictions remained in place, but legal uncertainty increased. Even before the ruling, empirical evidence suggests that expectations of regulatory change

had already begun to influence market behavior. For example, [Proelss et al. \(2018\)](#) report signs of improved informational efficiency and reduced price distortions during this interim period.

- (iii) **Phase 3:** Ruling and Policy Adjustment (post-2014): After the WTO’s 2014 ruling in favor of the complainants (European Commission, 2014), China began dismantling export quotas and easing licensing rules in 2015.<sup>1</sup> This shift marked a partial liberalization of the market and was associated with measurable improvements in price dynamics and transparency ([Mancheri, 2015](#))

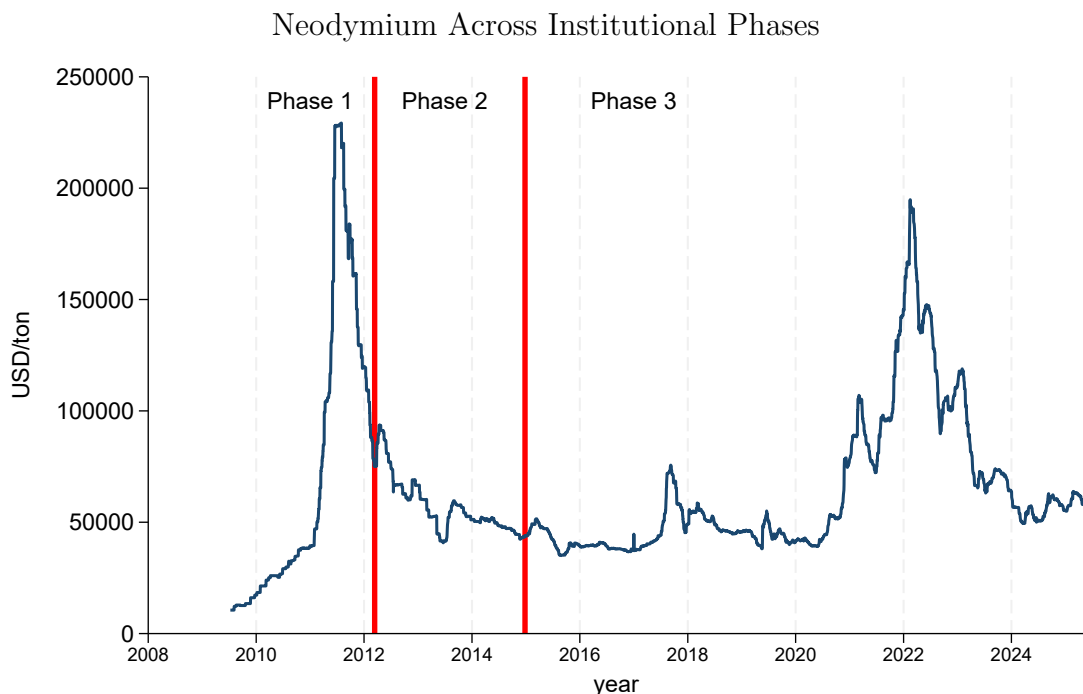
Figure 2 plots spot price trajectories for a representative element, segmented by the three phases defined above. The observed shifts provide a visual summary of how large regulatory and policy changes correspond to market-wide pricing and expectations. While the empirical analysis focuses on the immediate impact of firm-level announcements rather than on policy shifts or phase transitions, this context remains important. It helps interpret the informational environment and the nature of price discovery in rare earth markets. The high-frequency analysis allows to assess how quickly and how strongly firm specific information is incorporated into prices within this evolving market structure.

Notably, prices begin to decline before the WTO complaint filed by the US, EU, and Japan against China in March 2012, suggesting either anticipatory behavior or strategic demand reductions by key importers. This decline may reflect a response to unsustainable price levels and increasing concerns about China’s dominant market position. It is also consistent with a recent study from [Alfaro et al. \(2025\)](#), who find that the 2010 Chinese export restrictions on rare earth elements (REEs) led downstream users outside China to adapt first; REE-related patenting and exports in REE-intensive industries rose outside China, while upstream supply concentration persisted. After this sharp correction late in Phase 1 and early in Phase 2, spot prices remained comparatively low through Phase 2 and much of Phase 3 relative to the 2010–2011 spike ([Fernandez, 2017](#)). From late 2020 onward, prices rose and peaked in early 2022, in line with recent evidence on co-movement with clean energy and technology markets and on the growing role of geopolitical risk ([Hanif et al., 2023](#); [Li et al., 2023](#); [Yang et al., 2024](#)). Although the WTO ruling marked a critical shift in formal trade policy, it did not dismantle China’s structural dominance of the supply chain. While export quotas were removed after the WTO case, China retained effective control over domestic supply through annual mining and smelting/separation production quotas. These measures are different from export controls and regulate internal production ([Mancheri et al., 2019](#); [Shen et al.,](#)

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<sup>1</sup>The WTO issued its reports in Aug 2014; China then abolished rare earth export quotas in early January 2015 and ended export duties by 1 May 2015, so Phase 3 starts from January 2015.

Figure 2: Rare Earth Spot Prices: Across Institutional Phases



*Note:* This figure shows the monthly spot price (USD per ton) of one of the rare earths (neodymium) from end of 2009 to 2024. The three vertical red lines mark the transition between distinct phases in the rare earth market. The patterns observed here are representative of broader dynamics across rare earth elements, which are presented in Appendix A.1 Figures A1-A2. Daily spot prices for rare-earth oxides (USD per metric ton). (Source: Author's own illustration) with Shanghai Metals Market (SMM) prices accessed via LSEG Datastream.

2020). China's continued lead in processing technology, infrastructure, and know-how leaves advanced economies exposed to strategic risks (Eggert et al., 2016; Jowitt et al., 2018). Initiatives outside China, including new mining projects, strategic stockpiles, and recycling have so far achieved only limited diversification of global supply and risk mitigation (Binnemans et al., 2013b; Gulley et al., 2018; Massari and Ruberti, 2013). As a result, the REE market remains vulnerable to policy shocks, supply disruptions, and informational asymmetries. The structural characteristics of the rare earth market, high concentration, discretionary policy interventions, and limited price transparency, create conditions conducive to informational frictions. In such an environment, firm-level disclosures could carry significant informational value and prompt revisions in investor expectations. Observing how quickly and by how much rare earth prices incorporate this information offers a direct lens on market efficiency in this strategically important commodity sector.

### 3 Data Sources

This empirical analysis uses daily spot price data for rare earth oxides (REOs)<sup>2</sup>, available from 2009 onward. The dataset includes prices for five of the most economically important REEs: dysprosium, lanthanum, neodymium, praseodymium, and terbium (Binnemans et al., 2013a; Ganguli and Cook, 2018).<sup>3</sup> Table 1 summarizes the main industrial applications of the rare earth elements included in the analysis, along with representative company examples.

Table 1: Key Uses and Company Examples for Rare Earth Elements

| Element      | Key Uses                                    | Example Company/Product                                                   |
|--------------|---------------------------------------------|---------------------------------------------------------------------------|
| Neodymium    | Magnets for EVs, wind turbines, electronics | Tesla (drive motors); Nvidia/AMD (server/GPU cooling)                     |
| Dysprosium   | Magnets for high-temp, defense, wind power  | Lockheed Martin (F-35 jet magnets); Siemens (wind turbine magnets)        |
| Lanthanum    | Batteries, catalysts, optics                | BYD (EV batteries); Apple (iPhone camera lenses)                          |
| Praseodymium | Magnets, alloys for aircraft, turbines      | Boeing (aircraft alloys); Siemens (wind turbines); Apple (device magnets) |
| Terbium      | Phosphors (LED, displays), magnets, lasers  | Apple (iPhone/MacBook displays); Nvidia (LED chips)                       |

*Note:* Table summarizes the principal industrial uses of the rare earth elements analyzed in this study, with verified company/product examples. Sources: (Alonso et al., 2012; Alves Dias et al., 2020; Apple Inc., 2023; Ganguli and Cook, 2018; U.S. Department of Energy, 2022; Wilburn, 2012; Nvidia report 2024; AMD report 2024)

Spot prices are quoted in USD per metric ton from Shanghai Metals Market (SMM) via LSEG Datastream. The data set starts in July 2009 because reliable, consistently reported daily spot prices are scarce before that date, reflecting historical opacity, limited trading, and low liquidity in rare earths (Klossek et al., 2016). In a market with limited transparency and negotiated pricing, firm-level disclosures can constitute genuine informational shocks. They often introduce material news that spot prices do not incorporate immediately (Duffie et al.,

<sup>2</sup>Oxide price series are critical since they reflect market conditions more accurately, capturing both industrial demand and the processing costs involved in rare earth refinement.

<sup>3</sup>By economically important rare earths I refer to elements with widespread current industrial demand and significant value added along the value chain, particularly in high-value applications (such as catalysts and permanent magnets) (Omodara et al., 2019).

2010). The analysis is strictly restricted to trading days, excluding weekends and public holidays when markets are closed and reliable spot price data are not available. Accurate spot price series are crucial for the empirical strategy, allowing to observe market responses to informational news shocks and credibly assess the rare earth market’s informational efficiency.

### 3.1 Firm Coverage and Event Definition

Events are defined as firm-specific announcements by major rare earth producers that convey new, market-sensitive information. These events primarily include quarterly reports and press releases disclosing strategic operational changes such as planned increases or reductions in mining activity, major capacity expansions, significant operational disruptions, or notable financial performance updates. I do not further classify announcements by content or direction. Instead, each event is treated as a potential informational shock on the basis that such disclosures typically contain news relevant to the rare earth sector. This approach follows established event study practice for settings where announcement timing is known but the precise news content or directional component varies (e.g., MacKinlay (1997)). For illustration, an excerpt from the Lynas Rare Earths Q1 2023 quarterly report shows how an issuer disclosure can constitute a news shock (Lynas Q1 FY2023 quarterly report, pp. 6–7) .

*“During the quarter, production was affected by significant water supply outages which limited production at the Lynas Malaysia plant. [...] In early September 2022, a catastrophic equipment failure [...] led to a water supply outage affecting all users in the Kuantan area. [...] The water supply disruption has resulted in a very frustrating performance for the quarter.”*

This matters for prices because such disclosures may update expectations about firm-level output and near-term supply conditions. All events are dated by the official release date of the first publication on issuer or regulatory channels.<sup>4</sup> I assembled events from multiple sources: firms’ investor relations webpages, regulatory filings, major financial news outlets. Coverage varies by firm because not all publish disclosures suitable for identifying news shocks.<sup>5</sup>

For each firm, I recorded the dates of quarterly reports and qualifying press releases (Appendix Table A1 reports the exact count of news shock days by firm). The covered firms are China Northern Rare Earth Group High-Tech, Lynas Rare Earths, MP Materials, Shenghe

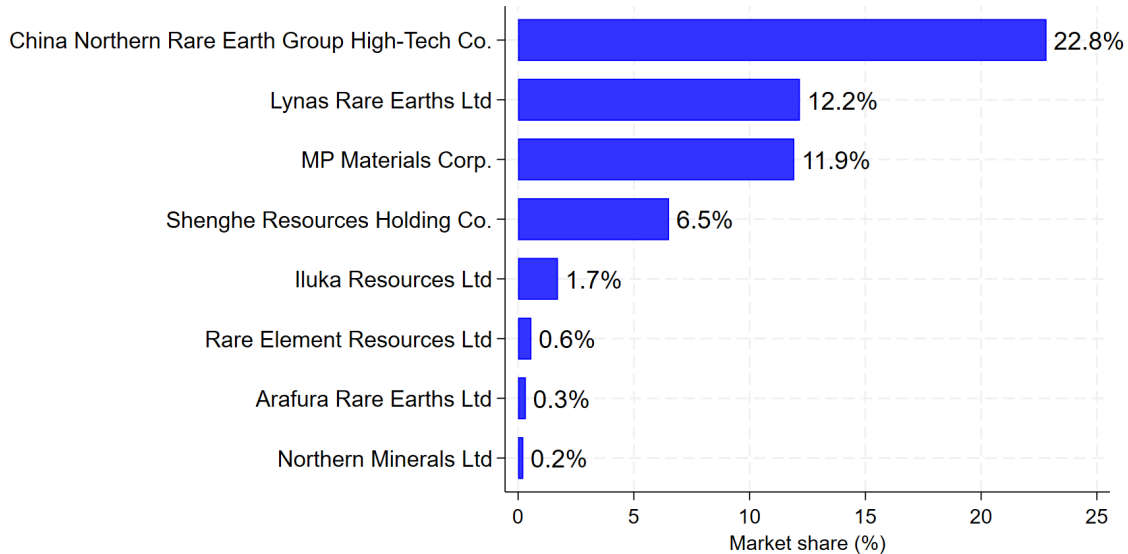
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<sup>4</sup>Only the calendar date is available, the exact time of release is not recorded.

<sup>5</sup>Examples include MarketScreener (<https://marketscreener.com/>), Reuters (<https://www.reuters.com/>), and MarketWatch (<https://www.marketwatch.com/>) and from the firms websites directly.

Resources Holding, Iluka Resources, Rare Element Resources, Arafura Rare Earths, and Northern Minerals. Together, these firms account for over 50% of sector market capitalization as of June 2025 (Figure 3).<sup>6</sup>

Figure 3: Market Capitalization Shares of Selected Rare Earth Companies (31 July 2025)



*Note:* This figure displays the relative market capitalization shares of eight major rare earth companies as of 31 July 2025. Values represent each firm’s market cap as a percentage of the sector total, proxied by the MVIS Global Rare Earth/Strategic Metals Index (USD 82.56 billion). The four largest firms account for over half the sector’s market value. Sources: ([CompaniesMarketCap, 2025](#); [MarketVector Indexes GmbH, 2025](#)).

## 4 Empirical Framework

**Identification Strategy** The identification strategy builds on the structure of the rare earth market. The industry is highly concentrated; China’s production, regulation, and processing capacity largely determine global supply. In this setting, firm announcements (earnings disclosures, capacity expansions, production halts) plausibly shift expectations and may move prices. The question is how supply-side firm news affects REE spot quotes and, potentially, broader financial markets. The design follows the high-frequency approach established in monetary policy work (e.g., [Altavilla et al., 2019](#); [Gürkaynak et al., 2005](#); [Nakamura and](#)

<sup>6</sup>Given the structure of the rare earth market, it may be important to consider firm origin: China Northern Rare Earth Group and Shenghe Resources are Chinese firms; Lynas, Iluka, Arafura, and Northern Minerals are Australian firms; MP Materials and Rare Element Resources are American firms.

Steinsson, 2018) and extended to commodities and emissions markets (Känzig, 2021, 2023). Calendar-dated firm disclosures are treated as news shocks. I then study post-news dynamics using local projections. Coefficients are reported as cumulative log return percentage points per one standard deviation of the surprise. These dynamics are descriptive event-time responses, not causal effects. Finally, I examine effects on the financial market by regressing broad market returns on the news shock measure to assess whether financial markets react around disclosure days.

This section sets out the identification and empirical framework. It defines event timing, constructs a firm by metal news shock series, and then quantifies responses using kernel density comparisons, Brown–Forsythe variance tests, and a local projection. Full specifications follow in the subsequent subsections, and results appear in the next section. Throughout, an event window is the symmetric three-day interval  $[t-1, t, t+1]$  centered on the announcement date  $t$  (the news shock day). The news shock series assigns to  $t$  the three-day log return from  $t-1$  to  $t+1$  and equals zero on all no news shock days.

#### 4.1 Event Window and Construction of the News Shock Series

In high-frequency event studies of financial markets, event windows are often as short as 30–60 minutes (Nakamura and Steinsson, 2018). Such narrow windows are infeasible, because price discovery in thin spot markets is slower and quotes are sparse.<sup>7</sup> Following Känzig (2021), who uses a one-day window for liquid oil futures,<sup>8</sup> I extend to a symmetric three-day window  $(-1, \text{news shock day}, +1)$  around firm announcements. This choice reflects the low liquidity, infrequent trading, and greater opacity of rare earth spot markets, which are characterized by slow price discovery. The first day ( $t-1$ ) captures potential information leakage and time zone mismatches. The last day ( $t+1$ ) captures disclosures released after market close on the news shock day ( $t$ ) and their delayed incorporation on the next trading day. Therefore, the symmetric three-day trading window is supposed to capture both anticipatory and delayed market responses (Kothari and Warner, 2007).

**Event timing, sample construction** For each firm  $F$ , I date-stamp public disclosures from investor relations channels, regulatory filings, and reputable news wires. I then align each announcement to the trading calendar and apply the symmetric three-day event window  $(t-1, t, t+1)$  around the announcement date  $t$ , allowing windows to overlap when

<sup>7</sup>Also the SMM spot price series on LSEG Datastream is only updated once per trading day (no intraday updates).

<sup>8</sup>The surprise is constructed as the difference between the oil futures price at the close of the announcement day and the close of the previous trading day



announcements occur within three trading days. Announcements posted on non-trading days are assigned to the next trading day in the spot price series. Control days are restricted to strictly three-day no news windows.<sup>9</sup> Because many disclosures (e.g., quarterly reports) occur on regular schedules, I test timing predictability by estimating firm-level logits of the daily news shock indicator on two lags of each rare earth return (dysprosium, lanthanum, neodymium, praseodymium, terbium), including weekday and month dummies and clustering standard errors by calendar month (Granger, 1969; Cameron and Miller, 2015). Then I conduct a joint Wald test that all lagged return coefficients are zero (Granger-style timing-predictability test). As Table A2 in Appendix A2 shows, I find only very limited evidence that lagged returns predict the timing of news shocks (occasional, marginal 10% rejections that do not persist across specifications), which supports the assumption that event timing is not systematically predictable from past rare earth returns.<sup>10</sup>

**Shock construction** Let  $P_t^M$  denote the daily (trading-day) spot price of metal  $M$ . Define the log return over the three-day event window as <sup>11</sup>

$$R_{t,[-1,+1]}^M \equiv 100 [\log P_{t+1}^M - \log P_{t-1}^M] = r_t^M + r_{t+1}^M,$$

where  $r_t^M \equiv 100 [\log P_t^M - \log P_{t-1}^M]$ .

Let  $\text{NewsShock}_t^F \in \{0, 1\}$  denote an announcement by firm  $F$  on date  $t$ . Define the firm–metal news shock series as

$$S_t^{M,F} \equiv R_{t,[-1,+1]}^M \text{NewsShock}_t^F.$$

So  $S_t^{M,F} = 0$  on all non event days and, on event days, equals the three-day log return in percentage points (pp),  $R_{t,[-1,+1]}^M$ . Positive (negative) spikes indicate price increases (decreases) associated with the disclosure. This follows surprise-based measurement used in policy and commodity event studies.

**Standardization.** To make shocks comparable across metals with different volatilities, I

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<sup>9</sup>No news windows are days on which none of the firms in the sample has an identified news shock. Given sparse coverage, I cannot rule out shocks at non-covered firms or via unobserved channels.

<sup>10</sup>See Appendix Section A.2, Table A3– A10, report Granger-style timing-predictability tests for each firm–metal pair. Only a few firm–metal–lag coefficients are individually significant; most are not, and isolated rejections are consistent with chance under multiple testing.

<sup>11</sup>I use log returns because they approximate percentage changes for small moves and are additive over time, which simplifies multi-day aggregation.

standardize the three-day return for each rare earth.

$$z_t^M \equiv \frac{R_{t,[-1,+1]}^M - \bar{R}_{\text{event}}^M}{\hat{\sigma}_{\text{event}}^M},$$

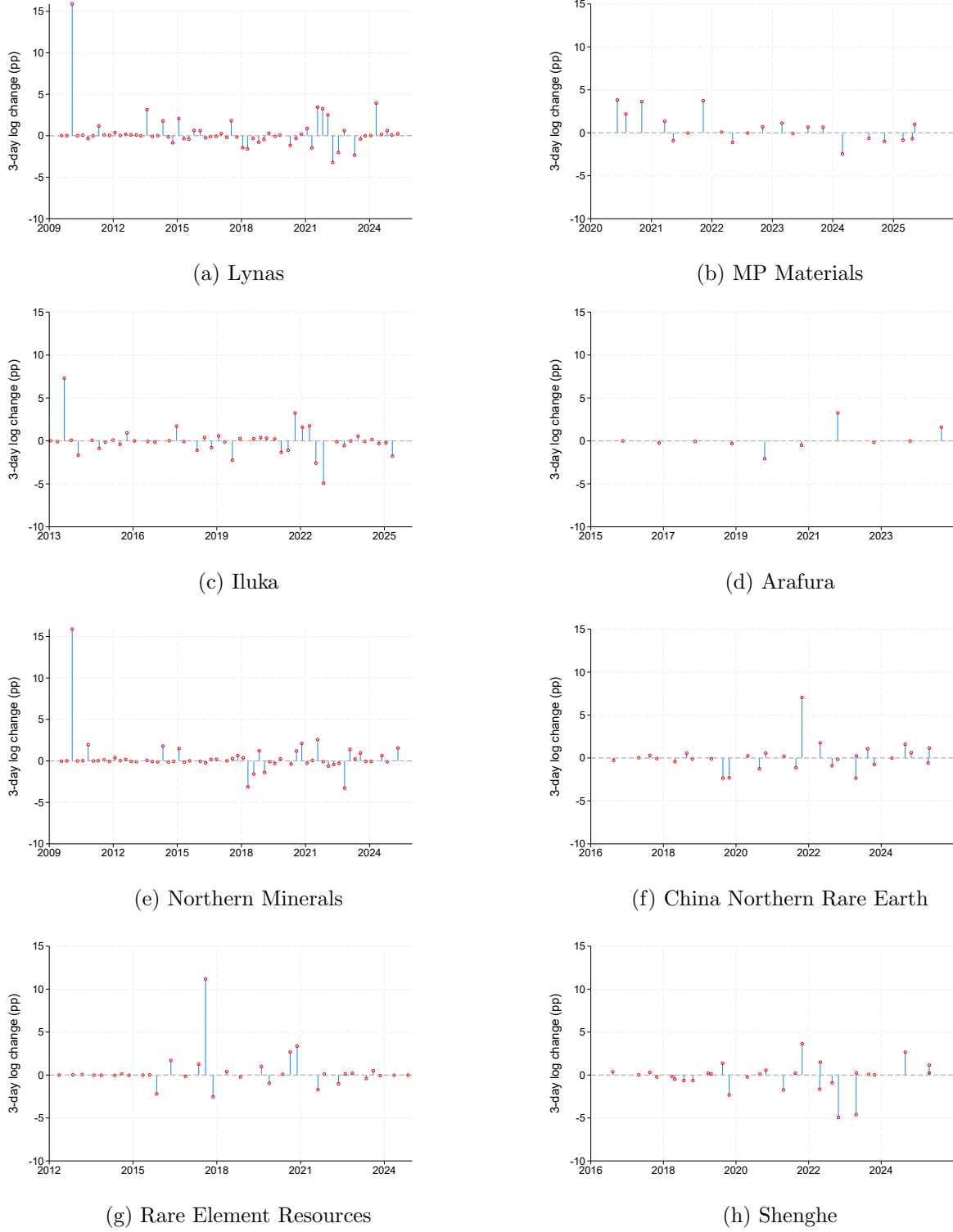
where  $\bar{R}_{\text{event}}^M$  and  $\hat{\sigma}_{\text{event}}^M$  are computed across news shock days for metal  $M$  pooled over firms. Then set

$$s_t^{M,F} = z_t^M \cdot \text{NewsShock}_t^F.$$

So  $s_t^{M,F} = 0$  off events and, on the news shock day for firm  $F$ , equals the within-metal standardized news shock window return of  $R_{t,[-1,+1]}^M$ . This event-conditional scaling places all metals on a common unit (one standard deviation within event news shock days).

**Visualization and diagnostics** For transparency, I plot spike graphs for each firm and rare earth pair  $(M, F)$ , marking  $s_t^{M,F}$  on event dates and zero elsewhere. The figures make identification tangible by showing (i) timing, (ii) sign and magnitude, and (iii) any calendar clustering. Figure 4 presents the news shock series defined above in the sample of firm announcements for one of the rare earths (neodymium) the rest is provided in Appendix Section A.3 Figure A3 - A6.

Figure 4: Neodymium news shock series across firms.



*Note:* Each panel plots the firm-metal shock series  $S_t^{M,F} = R_{t,[-1,+1]}^M \text{NewsShock}_t^F$ , where  $R_{t,[-1,+1]}^M = 100 [\log P_{t+1}^M - \log P_{t-1}^M]$  is the three-day event-window log return (the vertical axis reports values in percentage points). Values are nonzero only on news shock days. The figure shows neodymium shock series, the other rare earths appear in the Appendix Section A.3 Figure A3 - A6.

**Interpretation** Figure 4 shows that most announcement windows are close to zero, while a few display large positive or negative realizations. A plausible explanation is heterogeneity in surprise content. I do not categorize news shocks by type in this paper. Consequently, some disclosures are routine or anticipated, while others contain new or state-contingent information that markets had not priced in. Standard event study logic implies that prices move mainly on the unexpected component of news, consistent with many near-zero observations and a small number of large spikes (Jarociński and Karadi, 2020).

## 4.2 Empirical Methods

To assess how firm-level announcements are incorporated into prices and informational efficiency in the rare earth market. I use a three-step econometric approach to detect immediate, volatility-related, and dynamic price responses. The methodology consists of **(i) kernel density analysis**, **(ii) variance testing**, and **(iii) local projection estimation**.

### (i) Distributional Analysis: Kernel Density Plots

I compare the distribution of three-day event window log returns  $R_{t,[-1,+1]}^M$  between news shock windows and strict no news shock controls windows, using kernel density estimation (Silverman, 1986; MacKinlay, 1997). Densities are estimated with an Epanechnikov kernel.<sup>12</sup> To ensure comparability within each metal, I use a single common bandwidth per panel, computed on the pooled news and control observations by the normal reference formula of Silverman (1986),  $h = 1.06 \hat{\sigma} n^{-1/5}$  where  $\hat{\sigma}$  denotes the sample standard deviation of  $R_{t,[-1,+1]}^M$  computed over the pooled news and control observations for the given metal (and, in the per-firm figures, for the corresponding firm-metal panel), and  $n$  is the pooled sample size. The distributions for news shock windows are compared with strictly no news control day windows. This approach reveals whether news shocks are associated with a higher likelihood of extreme price changes or broader dispersion in returns, thereby revealing the market’s sensitivity to new information. When density plots for news shock days appear flatter or show heavier tails compared to controls, this signals higher variance and a greater probability of large price movements. These distributional diagnostics are especially informative for thin and opaque markets, because they require minimal parametric assumptions and remain sensitive to higher-order distributional changes (Cont, 2001).

### (ii) Variance Analysis

To test for volatility differences, I use a Levene-type test for equality of variances,

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<sup>12</sup>Results are qualitatively unchanged when using a Gaussian kernel

specifically the Brown-Forsythe test<sup>13</sup> (Brown and Forsythe, 1974a,b). This approach uses absolute deviations from the median rather than the mean, improving robustness to non-normality in the return distributions. This test assesses whether return volatility is higher in news shock windows than in control windows. This would be expected in markets with significant informational frictions and strategic concentration. I report summary statistics, including news shock numbers, standard deviations, and variance test statistics for each firm-metal combination. This serves as an additional, nonparametric check on kernel density findings.

### (iii) Local Projections impulse responses

To describe dynamic price changes around standardized news shocks, I estimate impulse responses<sup>14</sup> developed by Jordà (2005). This method allows me to trace the cumulative response of metal prices over time while flexibly controlling for confounding dynamics. The interpretation mirrors Jordà et al. (2013) local projections describe conditional paths in the absence of an identified structural shock. Specifically, I estimate cumulative price responses, for each metal  $M$ , firm  $F$ , and horizon  $h \in \{0, 1, \dots, 20\}$ , where the trading days are plotted as calendar days 2 to 22. I start the cumulation two days after the news shock date ( $t+2$ ) to avoid mechanical overlap with the construction of the shock variable  $s_t^{M,F}$ . If the outcome window began at  $t$  or  $t+1$ , it would mechanically include returns used to construct the shock, thereby inducing autocorrelation and overstating significance. The ( $t+2$ ) construction avoids this issue while still capturing early price adjustment dynamics. I follow Jordà and Taylor (2024) in using a difference from baseline specification, estimating changes instead of levels. Specifically, I estimate the following regression:

$$Y_{t,h}^M = \alpha_h + \beta_h s_t^{M,F} + \sum_{\ell=1}^{12} \gamma_{h,\ell} r_{t-\ell}^M + \varepsilon_{t,h}^M \quad (1)$$

where the dependent variable  $Y_{t,h}^M$  denotes the overlapping cumulative log return from

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<sup>13</sup>The Brown-Forsythe test is a median-based modification of Levene’s test and is considered more robust to non-normality and outliers in the data. See also Rosa (2016), who applies this approach in a similar context.

<sup>14</sup>I report impulse response functions (IRFs) via local projections (Jordà, 2005). The local projection framework is well suited for this setting. It is robust to model misspecification and structural breaks (Plagborg-Møller and Wolf, 2021), accommodates sparse and event-driven data, and avoids the strong assumptions required by structural VAR models (Ramey, 2016). This flexibility is especially important given the rare earth market’s exposure to regime changes and policy interventions. Without an identified exogenous shock or instrument, IRFs are descriptive (non-structural), not causal (see Plagborg-Møller and Wolf, 2021); see also Jordà et al. (2013) for an application without an identified exogenous shock.

trading day  $(t+2)$  through  $(t+h+2)$ . It is defined as

$$Y_{t,h}^M \equiv \sum_{j=2}^{h+2} r_{t+j}^M = 100 [\log P_{t+h+2}^M - \log P_{t+1}^M],$$

and  $r_t^M \equiv 100 [\log P_t^M - \log P_{t-1}^M]$  is the daily log return (percentage points),  $\alpha_h$  is the intercept: it equals the expected cumulative log return over horizon  $h$  when  $s_t^{M,F} = 0$  and  $r_{t-\ell}^M = 0$  for all  $\ell$ , i.e., the baseline no news change. The regressor  $s_t^{M,F}$  is the standardized news shock for firm  $F$ : it equals 0 on non event days and, on the event news shock day, equals the standardized three-day log return  $100[\log P_{t+1}^M - \log P_{t-1}^M]$ , within the respective rare earth metal. The main results use a pooled specification at the metal day level by aggregating firm shocks as  $s_t^{M,\text{pool}} \equiv \sum_F s_t^{M,F}$ . In this case I replace  $s_t^{M,F}$  by  $s_t^{M,\text{pool}}$  to obtain a single  $\beta_h^{\text{pool}}$  per metal and horizon.  $\sum_{\ell=1}^{12} \gamma_{h,\ell} r_{t-\ell}^M$  collects twelve lags of the same metal's daily log return (where  $\ell$  indexes the lag); these horizon-specific controls absorb short-run momentum/reversal and other predictable dynamics, so  $\beta_h$  identifies the response to the news shock net of such autocorrelation. The standardized three-day event return is constructed from stock prices and is therefore not an exogenous shock in the structural-VAR sense. Accordingly,  $\beta_h$  should be read as a non structural response function tracing typical cumulative rare earth returns after event-window surprises, not as a causal impulse response (Jordà et al., 2013; Plagborg-Møller and Wolf, 2021). I estimate (1) for all firms pooled together and for each metal–firm pair  $(M, F)$  separately. Because  $s_t^{M,F}$  is standardized within metal, pooling across firms places shocks on comparable units. Hence,  $\beta_h^{\text{pool}}$  is the cumulative log return (pp) at horizon  $h$  (under the  $t+2$  construction) in response to a one standard deviation news shock in the pooled (construction-sample) units. Off event days serve as no news observations by construction. Horizons advance over the sequential trading-day index (where only observed trading days enter). I do not include additional controls for other firms. Finally,  $\varepsilon_{t,h}^M$  is the LP residual. I use a 20-trading-day (about one-month) projection horizon to capture short-run price discovery after firm news while limiting contamination. Related evidence points to adjustments over a several-week window, and recent work implements month scale LP windows (e.g., van Binsbergen and Grotteria, 2024; Gorea et al., 2025). For financial market effects, I apply the same LP design to daily equity returns (S&P 500 and selected rare earth-intensive firms), using the pooled cross-metal shock  $s_t^{\text{pool}} \equiv \sum_M \sum_F s_t^{M,F}$ . Because  $Y_{t,h}^M$  overlaps across  $t$ , residuals exhibit autocorrelation in time, I therefore compute Newey–West heteroskedasticity- and autocorrelation-robust (HAC) standard

errors with a horizon-specific bandwidth  $b_h = \min\{h, 12\}$  (with  $b_0 = 1$ , capped at 12), and report 68% and 90% confidence intervals.

This three-step empirical approach uses distributional analysis, variance testing, and local projections to examine how firm-level news is incorporated into rare earth prices. Kernel density estimation and variance tests are well suited to detect asymmetries and heavy tails in return distributions (Silverman, 1986). Local projections offer flexibility and robustness in estimating dynamic price effects. After presenting the main results, I perform several robustness checks to confirm the stability and validity of the news shock series and the findings.

## 5 Empirical Results

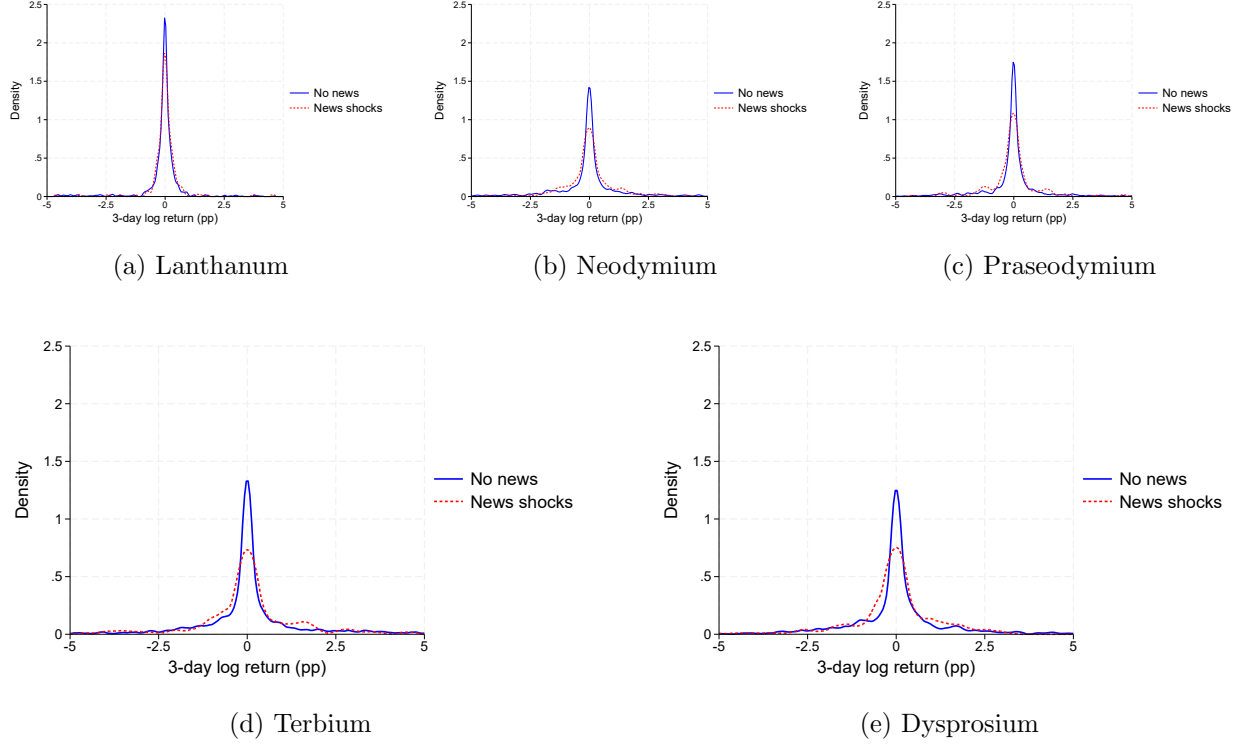
The empirical results document how firm-level informational shocks are incorporated into rare earth prices. The analysis proceeds in three steps: (i) compare the distribution of three-day log returns on event versus control days, (ii) test for variance differences, (iii) estimate local projection impulse responses. Finally, I extend the LP to broad equities to test the effect of rare earth news on financial markets.

### 5.1 Density Analysis

I begin by examining the distribution of three-day log returns around firm-specific announcements. In the main figures I pool news windows across all firms for each rare earth element. I plot kernel density estimates of  $R_{t,[-1,+1]}^M$  for pooled news windows and on strictly no news control days (as defined in the Empirical Framework section) (Silverman, 1986). This design helps isolate the contribution of firm-level information to price movements. Individual firm-metal combinations densities are reported in Appendix A.4 figures A7 - A14.

The main result, illustrated in Figure 5 for all firm news shocks, reveals an increase in return dispersion during news shock windows relative to controls. In news windows, the return density is flatter with heavier tails, indicating a higher probability of large price changes. In contrast, the control distribution peaks sharply at zero, indicating much lower volatility in the absence of firm-specific news. This pattern is robust in the pooled density analysis and holds for nearly all individual firm rare earth combinations. This is consistent with the hypothesis that informational frictions in rare earth markets lead to infrequent adjustments when new information is released. The results describe systematic price adjustments in rare earth spot prices around firm-level disclosures, consistent with information restriction accounts of price discovery (e.g., Sims, 2003; Veldkamp, 2006).

Figure 5: Kernel densities of three-day log returns: news shock window vs. no news windows



*Note:* Densities pool event windows across all firms and compare them to strict no news windows. Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman’s normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Axes are fixed across panels for comparability ( $x \in [-5, 5]$  pp;  $y$ -axis held constant). Visual differences reflect level/dispersion changes associated with news days compared to no news days.

## 5.2 Variance Analysis

To formally test whether dispersion in three-day log returns differs between event and control windows, I apply the Brown–Forsythe modification of the Levene test (Brown and Forsythe, 1974a) to all firm combined news shocks. This approach is robust to departures from normality and is used in recent high-frequency event studies to detect changes in volatility around information releases (Känzig, 2021). In this setting, the test assesses whether firm-level disclosure systematically increase return volatility within the event window in rare earth spot markets. Table 2 reports, for all firm combined news shock windows, the number of news and control windows, their respective standard deviations, and the Brown–Forsythe test  $p$ -value for equality of variances. Results for all individual firm-element combinations are provided in the Appendix A.4 Table A11 - A18.



Table 2: Variance Analysis pooled all firms: Brown–Forsythe test

| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------|
| Dysprosium   | 188               | 1.961              | 1339              | 2.820              | 0.695              | 0.310           |
| Lanthanum    | 188               | 1.582              | 1339              | 2.454              | 0.645              | 0.277           |
| Neodymium    | 188               | 1.740              | 1339              | 2.481              | 0.701              | 0.112           |
| Praseodymium | 188               | 1.720              | 1339              | 2.379              | 0.723              | 0.277           |
| Terbium      | 188               | 2.598              | 1339              | 2.888              | 0.900              | 0.625           |

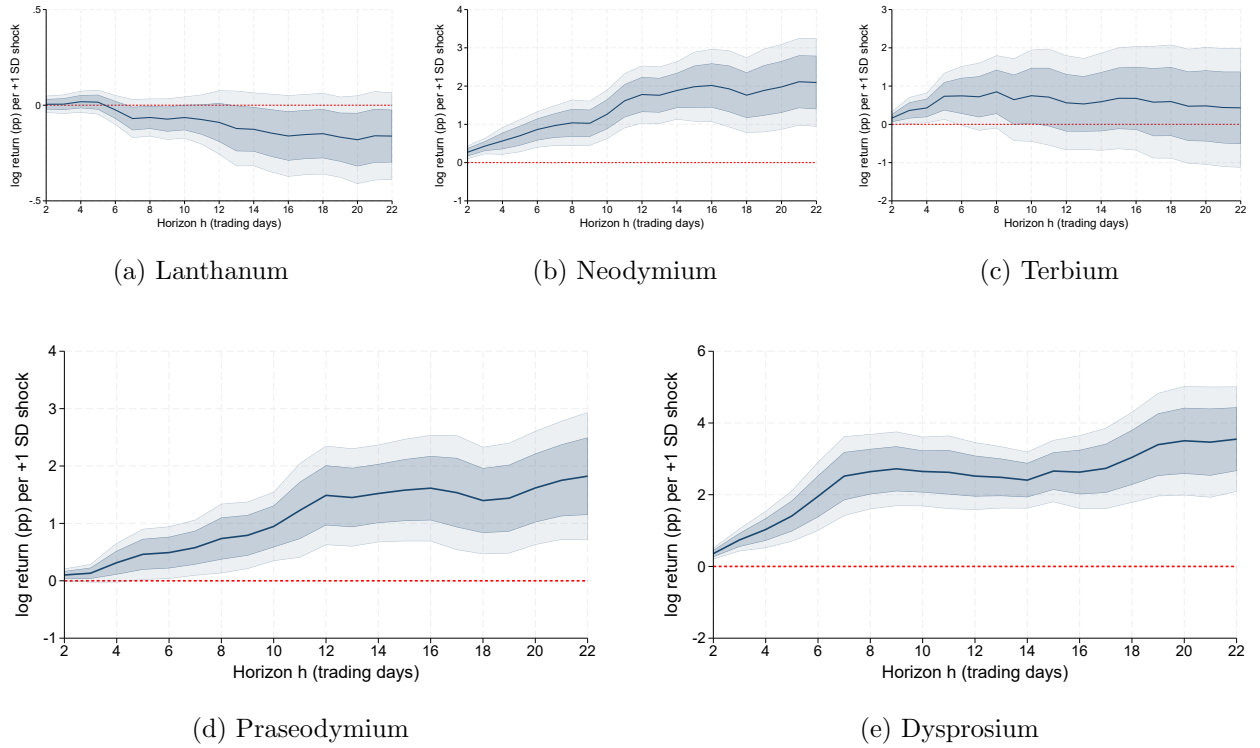
*Note:*  $N_{\text{news}}$  counts one observation per news shock window; controls are strict no-news 3-day windows. SDs are computed from 3-day log returns (percentage points, pp).  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ . Brown–Forsythe (median-centered Levene) tests equality of variances using absolute deviations;  $p$ -values of the Brown–Forsythe test (BF) come from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ . Significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

For the pooled sample Brown–Forsythe  $p$ -values are large (0.112–0.625), indicating no evidence of higher dispersion on event windows relative to strict no news controls. Looking across firms and metals, there does not seem to be a systematic increase in variance. Only a small subset of combinations comes close to conventional significance thresholds. The gap between the strong visual patterns in the density plots and the weaker statistical significance of the variance tests likely reflects both the limited number of event windows (even when pooled) and the heavy tails of return distributions. Kernel densities are sensitive to extremes and highlight infrequent large movements, whereas the Brown–Forsythe test averages over the whole distribution, so volatility driven by a few extreme observations is diluted in the variance test (Gabaix, 2009; Kelly and Jiang, 2014). Overall, volatility responses to firm announcements are modest and not systematic across firms or elements. Accordingly, rare earth spot prices show no pronounced volatility spikes on disclosure days. The evidence instead points to a pattern of weaker and less pervasive effects than anticipated. However, another possible reason for weak three-day variance differences is horizon mismatch. If rare earth prices adjust more slowly than expected, much of the dispersion may arrive after the news shock window. In that case, a Brown–Forsythe test computed on the  $[t-1, t, t+1]$  news shock window can understate the possible effect.

### 5.3 Local Projection

To describe the effect of firm-specific news shocks on rare earth spot prices, I present the local projection impulse-response specification in Eq. (1). Coefficients are interpreted as cumulative price changes (pp) per one standard deviation news shock. Horizons are  $h = 2, \dots, 22$  trading days, and 68% and 90% confidence intervals (CI)<sup>15</sup> are computed using Newey–West standard errors with bandwidth  $b_h = \min\{h, 12\}$  (with  $b_0 = 1$ ). With the  $t+2$  convention, the first plotted point is the first day after the  $[t-1, t, t+1]$  event window.<sup>16</sup> Figure 6 reports the local projection impulse responses for all firm news shocks.

Figure 6: Pooled local projection impulse responses of rare earth spot prices



*Notes:* Each panel plots the cumulative log return response (pp) to a pooled firm-news shock. The pooled regressor aggregates standardized shocks across firms for a given rare earth and is set to zero off news shock days. Local projections include  $p = 12$  lags of own daily returns; confidence bands (dark = 68%, light = 90%) use Newey–West HAC with bandwidth  $b_h = \min\{h, 12\}$ . The  $x$ -axis shows trading days since the shock. The  $y$ -axis shows cumulative log returns relative to  $t+1$ .

<sup>15</sup>The 68% band is  $\pm 1$  standard error aiding visual power in small, noisy samples; the 90% band is a stricter benchmark. This follows [Känzig \(2021, 2023\)](#), who uses the same confidence levels.

<sup>16</sup>For comparison, Appendix figure A15 shows an Local projection that starts on the news shock day ( $t$ ) and cumulates from there; it exhibits minor mechanical overlap but leaves qualitative patterns unchanged.

Figure 6 shows that a one standard deviation news shock is associated with positive and persistent cumulative responses across three of the five rare earth metals. For neodymium, praseodymium, and dysprosium, point estimates increase during the first trading week ( $h \approx 2-8$ ) and then start to flatten, but remain well above zero until  $h = 22$ . By  $h = 22$ , pooled cumulative magnitudes are roughly 2 pp for Praseodymium and neodymium, 3–4 pp for dysprosium, and no significant reaction for Terbium and Lanthanum. None of the significant rare earths exhibit reversal by the end of the 20-day window. Overall, the evidence is consistent with level shifts rather than purely transitory noise.<sup>17</sup>

Generally distributional diagnostics indicate heavier tails in event windows, yet Brown–Forsythe tests generally fail to reject equal variances, indicating that volatility itself does not systematically rise. Taken together, firm news shocks tend to move price levels by modest but persistent amounts, with magnitudes that vary by metal and firm consistent with heterogeneous information processing and gradual incorporation of news.

## 5.4 Effect on Financial Markets

Rare earth elements are critical inputs in automobiles, electronics, clean energy, and defense systems. If firm news shocks change expected availability or costs, effects may spread beyond REE spot markets to broad equities. To assess whether rare earth firm news spread beyond the sector, I extend this high-frequency analysis design to broad stock market indices. I compare S&P 500 returns on news shock days to strictly news free windows. I focus on the S&P 500 as a benchmark for the U.S. market index.<sup>18</sup> Analogous results for the NASDAQ-100, Dow Jones and MSCI World Index are reported in Appendix A.5.

Although the news shock measure is not a structural (causal) shock, the responses indicate that the constructed news shock series carries economically meaningful information for REE spot prices. This motivates testing potential effects on the financial markets, using the same standardized news shock series. I ask whether broad equity markets exhibit systematic comovement after rare earth news events. The design and interpretation remain descriptive, and coefficients trace conditional price reaction per one standard deviation of the news shock. I first report kernel densities and Brown–Forsythe tests as diagnostics and then estimate local projections as the primary dynamic specification. This specification is almost identical

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<sup>17</sup>This pattern is representative of the broader sample, figures for the individual firm–metal combinations are reported in the Appendix A.4 figure A16 - A23.

<sup>18</sup>Daily close prices (USD) for indices and equities are sourced from LSEG Workspace/Datastream.

to the rare earth price LP in Eq. (1) and is defined as

$$Y_{t,h} = \alpha_h + \beta_h s_t^{\text{pool}} + \sum_{\ell=1}^{12} \gamma_{h,\ell} r_{t-\ell} + \varepsilon_{t,h},$$

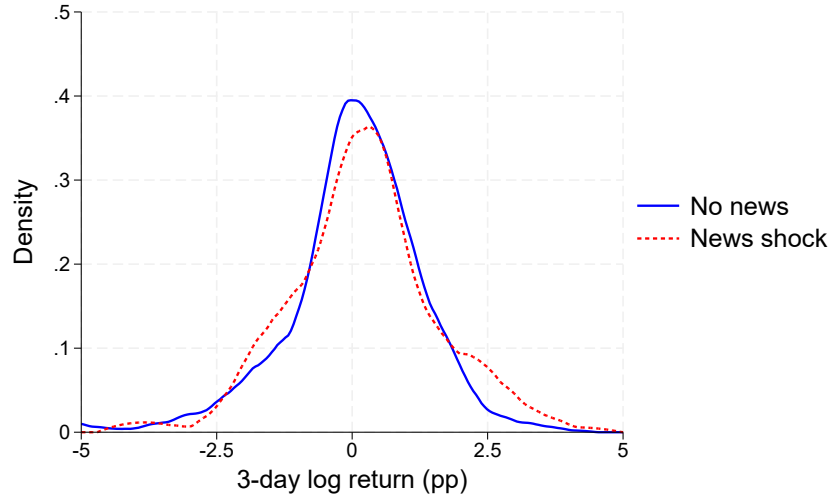
where  $Y_{t,h} \equiv \sum_{j=2}^{h+2} r_{t+j}$  is the cumulative log return (in pp) from  $t+2$  to  $t+h+2$ ,  $r_t$  denotes the daily log return of the relevant equity (index or firm). I use the same standardized news shock series defined earlier (from rare earth price changes), except that here I aggregate shocks across all metals into a single series:

$$s_t^{\text{pool}} \equiv \sum_M \sum_F s_t^{M,F},$$

The term  $\sum_{\ell=1}^{12} \gamma_{h,\ell} r_{t-\ell}$  includes twelve lags of the equity's own daily log return. The coefficients  $\gamma_{h,\ell}$  are horizon-specific and absorb short-run autocorrelation and seasonality in returns. The  $t+2$  outcome start,  $p = 12$  lags, and Newey–West HAC with bandwidth  $\min\{h, 12\}$  are kept unchanged.

Figure 7 shows the densities for news shock windows versus control windows for the S&P 500, and Table 3 the variance analysis for the S&P 500, broad equity benchmarks, and individual firms that have high demand for rare earth elements.

Figure 7: Kernel densities of three-day log return news shock window vs. no news window (S&P 500)



*Note:* Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). The  $x$ -axis is truncated to  $[-5, 5]$  pp for comparability.

Table 3: Variance Analysis: Brown–Forsythe test Financial Market

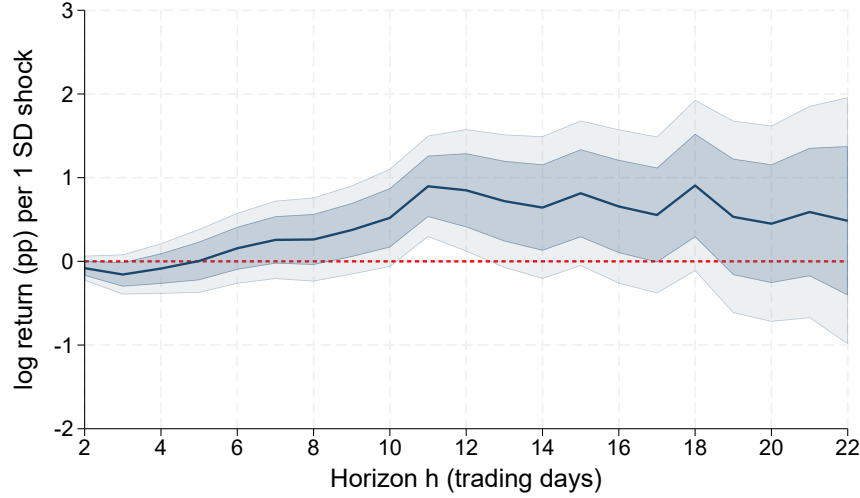
| Asset      | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | SD ratio | $p$ -value |
|------------|-------------------|--------------------|-------------------|--------------------|----------|------------|
| S&P 500    | 147               | 1.331              | 587               | 1.740              | 0.765    | 0.559      |
| Dow Jones  | 147               | 1.188              | 587               | 1.755              | 0.677    | 0.296      |
| Nasdaq 100 | 188               | 1.713              | 1221              | 2.006              | 0.854    | 0.327      |
| MSCI       | 188               | 1.322              | 1221              | 1.653              | 0.800    | 0.138      |
| Apple      | 185               | 2.630              | 1211              | 2.765              | 0.952    | 0.727      |
| AMD        | 172               | 5.479              | 1182              | 4.990              | 1.098    | 0.223      |
| NVIDIA     | 180               | 4.211              | 1196              | 4.347              | 0.969    | 0.913      |
| Tesla      | 162               | 5.744              | 935               | 5.021              | 1.144    | 0.219      |
| BYD        | 166               | 3.904              | 840               | 4.004              | 0.975    | 0.965      |

*Note:* Brown–Forsythe  $p$ -values compare variances of three-day log returns pp around pooled firm news windows vs. strictly news free control windows. “SD ratio” is  $SD_{\text{news}}/SD_{\text{ctrl}}$ . Significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Coverage of news shock windows varies because start dates differ by data availability (e.g., S&P 500 available from 2015), so the number of news shock days is not identical across series.

For the S&P 500, the news shock and no news control densities are nearly identical, with no clear location shift, and comparable tails. Consistent with this, Brown–Forsythe tests in Table 3 does not reject equal variances for any benchmark or firm. Thus, based on density and variance diagnostics, there is no evidence that broad equities react differently in the three-day event window than in no news windows.

I now present local projection impulse responses using the same pooled standardized news shock series to test for effects on financial markets; figure 8 reports the S&P 500 results.

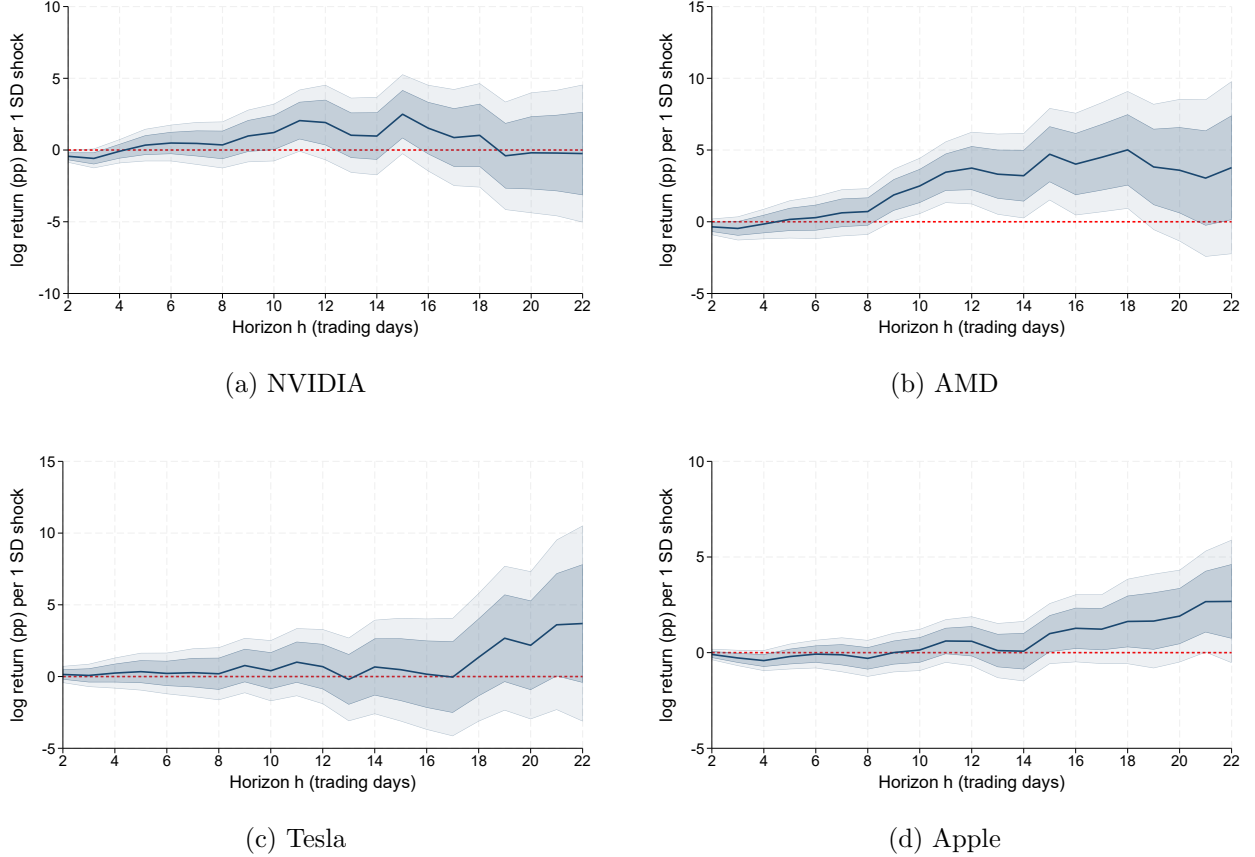
Figure 8: Local projection impulse response of the S&P 500



*Notes:* Cumulative log return (pp) over horizons  $h = 2 \dots 22$ ; outcomes cumulate from  $t+2$ . The regressor is a one standard deviation pooled rare earth news shock. Specification includes  $p = 12$  lags of own returns. Bands: 68% (dark) and 90% (light) from Newey–West HAC with  $b_h = \min\{h, 12\}$ .

Figure 8 indicates that pooled rare earth announcements are not associated with noticeable changes in broad U.S. equities. Point estimates are small at impact and remain close to zero through  $h = 22$ , and the 90% confidence bands straddle zero almost at all horizons. A mild mid-horizon uptick (around  $h \approx 10\text{--}12$ ) is not statistically significant. Overall, I find no evidence of economically meaningful or persistent spillovers to the S&P 500. One possible explanation is that the S&P 500 contains some firms with minimal or no rare earth exposure. These firms could reduce the effect size of the news shock series on the index. I therefore re-estimate the local projection specification for four firms (Apple, AMD, Nvidia, and Tesla) that are highly rare earth-intensive, identified from the main applications of rare earth elements presented earlier.

Figure 9: Local projections of log returns for selected equities/indices



*Notes:* Cumulative log return (pp) per event over horizons  $h = 2 \dots 22$ ; outcomes cumulate from  $t+2$ . Regressor is the pooled standardized rare earth news shock. Specification includes  $p = 12$  lags of own returns. Bands: 68% (dark) and 90% (light) from Newey–West HAC with  $b_h = \min\{h, 12\}$ .

For NVIDIA, Apple, and Tesla, point estimates are small at impact and remain close to zero across horizons; the 68% and 90% confidence bands include zero throughout. AMD shows a modest mid-horizon uptick (around  $h \approx 10$ ), but it is not robust across confidence bands and is not persistent, so I do not consider it a robust effect. Overall, for these four rare earth-intensive firms, the local projections provide no evidence of economically meaningful or statistically significant effects from rare earth news.<sup>19</sup> Taken together, the rare earth news shock series is not associated with measurable changes in broad U.S. equity indices or the stock returns of major rare earth-intensive firms. The results are consistent with the view that sectoral specificity and informational frictions dampen the transmission of idiosyncratic commodity news to the aggregate market (Duffie et al., 2010; Gabaix, 2011).

<sup>19</sup>Analogous estimates for the NASDAQ-100, the Dow Jones, MSCI World, and BYD likewise show no economically or statistically meaningful responses, see Appendix A.5 figure A26

By implementing this high-frequency analysis, I provide evidence that general equity market reactions to rare earth sector news appear to be negligible at best. These findings remain unchanged when alternative indices and firms are considered. Thus, the evidence indicates that the effect on the financial market from rare earth informational shocks is extremely limited.

## 6 Extension Analyses

This section examines two margins that may be related to the size and timing of rare earth price responses, complementing the results. First, the analysis tests whether political context is associated with differential responses of rare earth prices to firm news. The three institutional phases (outlined earlier) provide discrete shifts in policy and market structure that plausibly change price discovery, and risk premia, so re-estimating the baseline within each phase shows whether news shock responses differ across the institutional changes. Second, I study metals with very low substitutability, to test whether shock responses are larger or more persistent than for the economically important baseline metals.

### 6.1 Rare Earths Across Institutional Phases

To further understand how firm-level news is reflected in rare earth prices, I re-estimate the main analysis within each of the three institutional phases previously defined. This approach allows to assess whether changes in the broader policy environment are associated with measurable changes in the market’s responsiveness to informational shocks. However, due to data limitations, only the firm *Lynas Rare Earths* and some smaller companies have sufficiently complete disclosure data spanning over the three phases. Other companies either do not have historical disclosure data available for the earlier years or were not operating during those periods. Therefore, this part of the analysis is restricted to *Lynas Rare Earths*. This reduces the risk that differences in market reactions across phases are driven by sample composition and is consistent with shifts in institutional context and informational efficiency. Because the phase-based split sharply reduces the number of news shock windows, especially in Phases 1 and 2, the local projection design lacks sufficient support for reliable local projection estimation. I therefore do not report phase-specific local projections and interpret the density and Brown–Forsythe test with caution, acknowledging low statistical power and greater finite sample variability. Still, the phase-wise densities remain informative as a descriptive diagnostic of how institutional context may be associated with shifts in the location and dispersion of returns on news days relative to no news windows. I compare ker-



nel densities across the three phases for neodymium, the other rare earth elements and their phase-specific density and variance tests are reported in the Appendix Section A.6 Figure A27 - A29 and Tables A19 - A21.

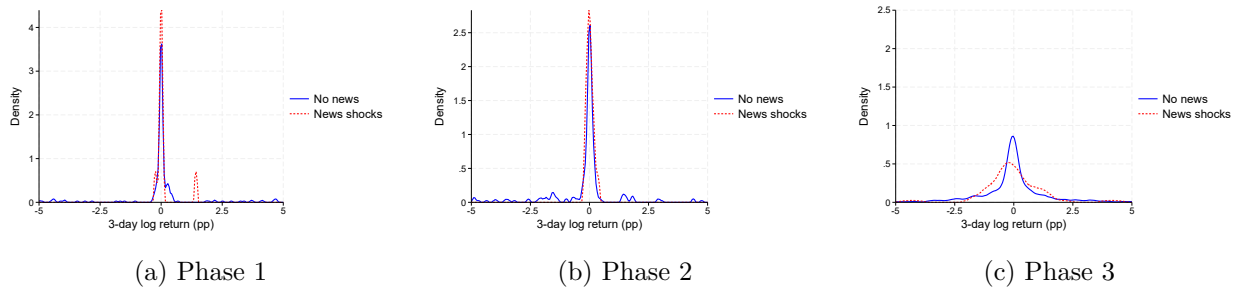
For reference, I briefly recall the three phases defined earlier:

**(i) Phase 1 (Pre-Complaint, pre-2012).** Characterized by a tightly controlled export regime, strict quotas, and segmented global pricing. This leads to a market with limited transparency and high inefficiency.

**(ii) Phase 2 (Complaint Pending, 2012–2014).** Marked by the formal WTO complaint and a transition period during which regulatory uncertainty increased and market actors began to anticipate liberalization (see [Proelss et al., 2018](#)).

**(iii) Phase 3 (Post-Ruling and Adjustment, post-2014)** Defined by the WTO’s 2014 ruling and the subsequent dismantling of quotas, resulting in improved transparency and a more liberalized market.

Figure 10: Kernel densities of three-day log returns for *Lynas* neodymium across institutional phases.



*Note:* Each panel compares days centered on *Lynas* news shocks with strict no news control windows, restricted to the respective phase’s institutional window. Kernel densities use an Epanechnikov kernel with bandwidth chosen by Silverman’s normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  ([Silverman, 1986](#)).

Across the three phases, the kernel density panels in Figure 10 suggest a gradual flattening of the distribution, with lower peaks and somewhat wider dispersion by Phase 3. Because the phase-based split leaves few news windows in Phases 1 and 2 ( $N_{\text{news}} = 11$  and  $N_{\text{news}} = 10$ , respectively), statistical power is limited and these visual contrasts should be read with caution. The variance tests in the Appendix Tables A19–A21 show little systematic difference in Phases 1 and 2. In Phase 3, Lanthanum displays a statistically significant increase in dispersion on news days relative to no news windows (SD ratio  $\approx 1.95$ ,  $p = 0.027$ , .

Table 4: Variance Analysis: Brown–Forsythe test for neodymium Across Phases

| Phase   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | SD ratio | $p$ -value |
|---------|-------------------|--------------------|-------------------|--------------------|----------|------------|
| Phase 1 | 11                | 4.428              | 287               | 3.614              | 1.225    | 0.973      |
| Phase 2 | 10                | 0.136              | 266               | 2.149              | 0.063    | 0.208      |
| Phase 3 | 43                | 1.207              | 1155              | 1.787              | 0.675    | 0.350      |

*Note:* Table reports standard deviations for three-day log returns around *Lynas* news shocks versus strictly news-free control windows, the SD ratio (news/control), and Brown–Forsythe  $p$ -values for equality of variances across phases. Significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Full variance results for other rare earths are reported in Appendix A.6 Table A19 - A21.

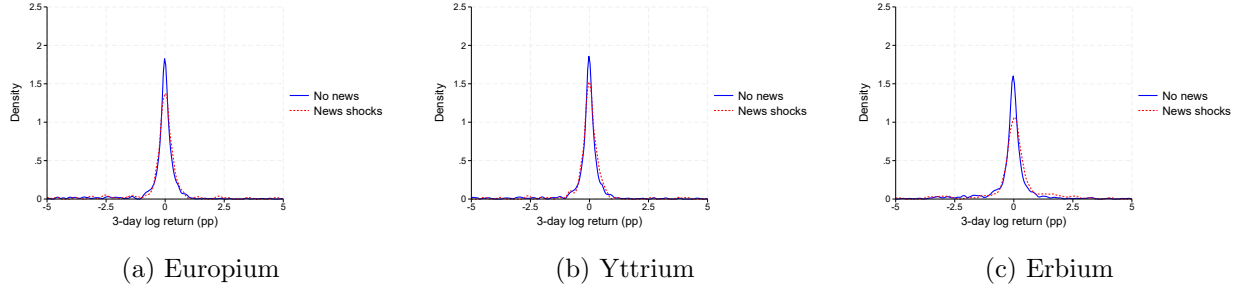
Terbium shows the opposite pattern with lower dispersion on news days than in controls (SD ratio  $\approx 0.48$ ,  $p = 0.080$ ). Neodymium, dysprosium, and praseodymium remain statistically indistinguishable from controls. I include this phase-based exercise because it helps assess whether political context shapes investor valuation and the transmission of firm news into prices. Theory suggests stronger reactions in Phase 3 under a more open market, yet continued policy control in China over rare earth production may still limit the role of firm news, with government announcements possibly more decisive for prices. Thus, even with low power, the phase-wise evidence remains useful to benchmark mechanisms: it does not indicate markedly stronger or faster reactions in Phase 3. This pattern is consistent with the possibility that, given China’s dominance across mining, separation, and refining, firm news shocks are not a primary driver of rare earth spot price changes.

## 6.2 Rare Earths with Low Substitution Index

Economic theory predicts that price reactions to supply-side informational shocks should be strongest for commodities with highly inelastic demand, since substitution is extremely limited (Fullerton and Metcalf, 2002). In the main analysis, I prioritized economically important rare earth elements, some of which also have very low substitution potential.<sup>20</sup> To further assess the robustness of the results, I extend the analysis to europium, yttrium, and erbium: rare earths classified as very hard to substitute according to the substitution index by Graedel et al. (2015). Ideally, ytterbium and thulium would be included as well, but data limitations preclude analyzing those elements. I repeat the core empirical strategy for these hard-to-substitute rare earth elements.

<sup>20</sup>Dysprosium and lanthanum exhibit low substitution indices, as documented by Graedel et al. (2015); nonetheless, given their economic importance, I include them in the main analysis.

Figure 11: Kernel densities of three-day log returns: news shock windows vs. no news windows for hard to substitute rare earth elements.



*Notes:* News windows are compared to control windows. Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman’s normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Elements shown (europium, yttrium, erbium) which are classified as very hard to substitute by Graedel et al. (2015).

Visual differences between announcement and control densities are present for all three elements: europium, yttrium, and erbium, but they are generally more muted than in the main sample of economically important rare earths. One interpretation is that, since such elements are not central to current production, firm news shocks are associated with smaller distributional shifts. Second, I conduct Brown–Forsythe variance tests to assess differences in return volatility around news shocks (see Table 6).

Table 5: Variance Analysis: Brown–Forsythe test for low-substitution rare earths

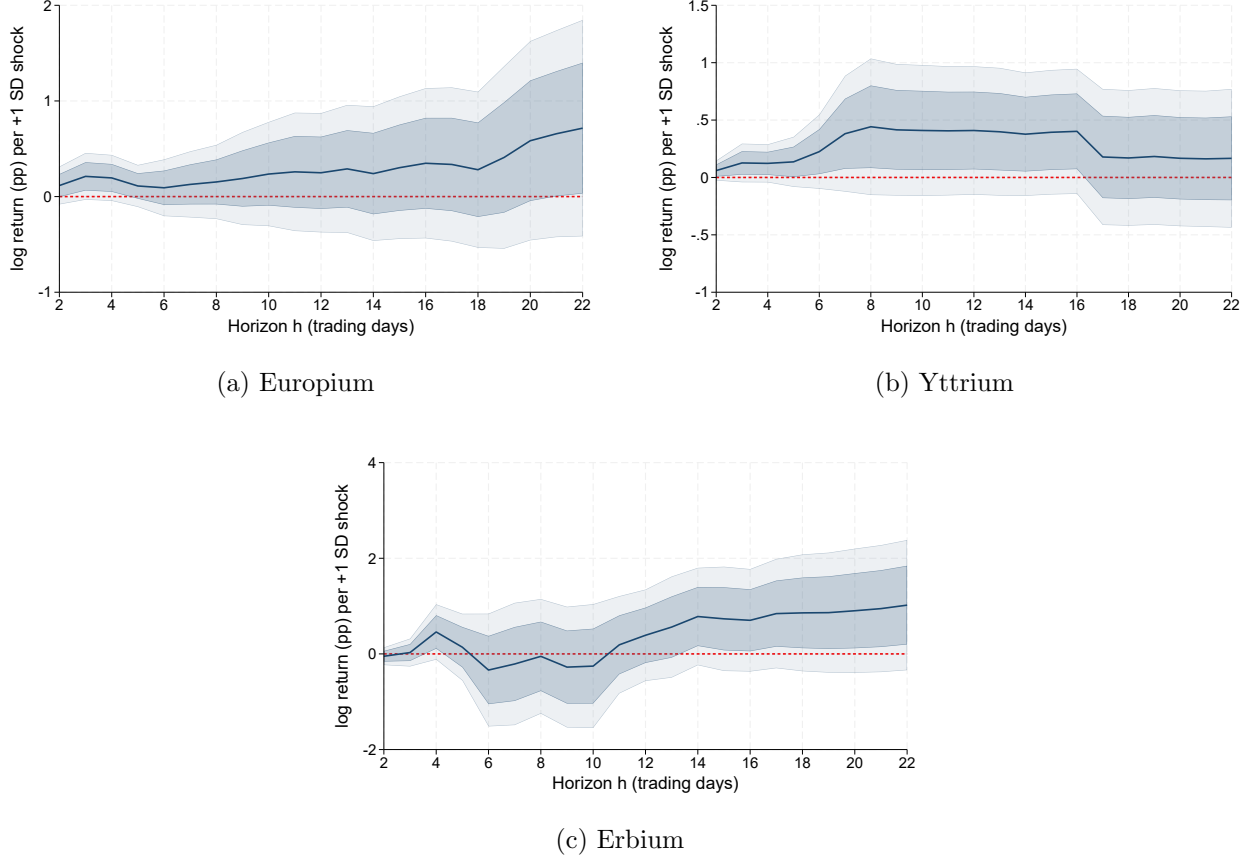
| Rare earth | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p\text{-value (BF)}$ |
|------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------------|
| Erbium     | 176               | 1.951              | 1108              | 1.456              | 1.340              | 0.147                 |
| Europium   | 176               | 1.874              | 1108              | 1.820              | 1.030              | 0.915                 |
| Yttrium    | 176               | 2.931              | 1108              | 2.398              | 1.222              | 0.698                 |

*Note:* Pooled across all firms.  $N_{\text{news}}$  counts one observation per news shock window. Controls are strict no-news 3-day windows. SDs are computed from 3-day log returns (percentage points, pp).  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ . Brown–Forsythe (median-centered Levene)  $p$ -values come from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

The Brown–Forsythe test indicates no statistically significant increase in volatility around news shocks for europium, yttrium, or erbium. Taken together, the evidence does not indicate a systematic increase in dispersion on event windows for low-substitution elements. Third, I estimate local projections using the same specification as in the main results. A one standard deviation shock (standardized within metal, pooled across firms), with  $p = 12$  lags of own

returns, and Newey–West standard errors with bandwidth  $b_h = \min\{h, 12\}$ . Horizons are  $h = 2, \dots, 22$ .

Figure 12: Pooled local projection impulse responses of hard-to-substitute rare earth prices



*Notes:* Each panel shows the cumulative log-return response (pp) to a pooled, standardized shock. Local projections include  $p = 12$  lags of own returns. Shaded regions denote 68% (dark) and 90% (light) Newey–West confidence intervals with bandwidth  $b_h = \min\{h, 12\}$ . The  $x$ -axis reports trading days since the shock. The  $y$ -axis reports cumulative log returns (percentage points) per one standard deviation shock.

Figure 12 reports local projections from  $h = 2$  onward. Across europium, yttrium, and erbium there is no statistically significant response. The 90% confidence bands include zero at all horizons and the 68% bands exclude zero only rarely. In sum, hard-to-substitute metals show no robust price reaction to firm-level news, in contrast to the economically central metals analyzed earlier. Price reactions following news shocks are notably weaker in this specification, suggesting that economically significant rare earths exhibit stronger responsiveness to firm-level news than those that are harder to substitute. One plausible reason for the muted results in the low substitution sample is, as outlined in the Background

section, the unique joint-production process of rare earths. Elements are extracted together from the same ore stream, so expansions targeted at economically central elements also increase output of others, which then tend to be stockpiled when demand is weaker. Such inventories can buffer short-run price responses to firm news for peripheral elements, even when demand for the central elements is tight. I cannot test this mechanism directly, as I lack element-level inventory or production stream data. Thus, this explanation remains a hypothesis consistent with the market’s structure, rather than a confirmed empirical finding.

## 7 Robustness Checks

To assess the validity of the constructed news shock series and the robustness of identification, I implement three checks that focus on the main design choices. First, I construct matched control windows that are more comparable to the news shock windows than the broader set of control periods used in the main analysis. Second, I exclude periods of increased geopolitical risk and apply the baseline design to verify that the results are not driven by global uncertainty spikes. Third, I extend the horizon to about five months ( $h \approx 105$  trading days) to check for delayed or highly persistent effects in spot price responses to firm news shocks.

### 7.1 Matched controls

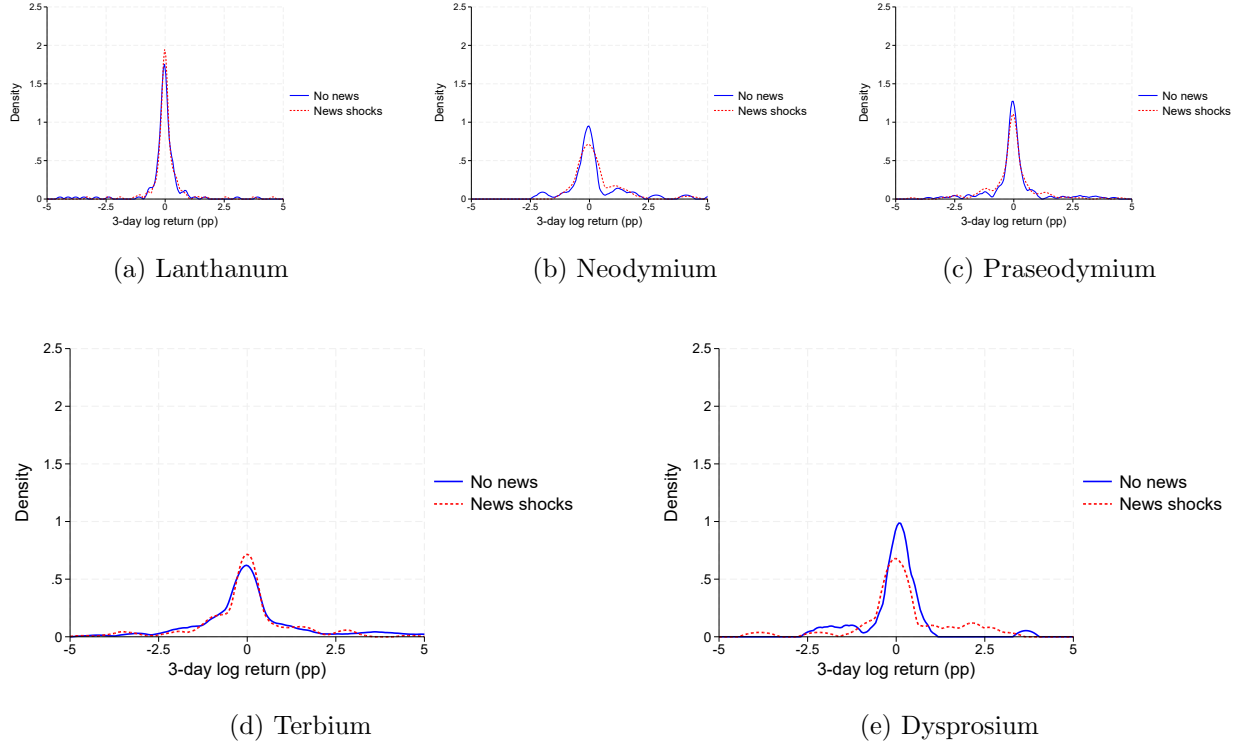
A potential concern is that news shock days may not be randomly distributed over time. If news shock days cluster by month, weekday or around recurring macro announcements, comparisons with the full set of no news days could conflate news effects with seasonal or calendar-driven patterns (Kothari and Warner, 2007). Therefore, I implement a time-matched control design. For each news shock window, I select a clean three-day control window within a prespecified band of trading days that falls in the same month and weekday, while being free of any firm-level news, following Känzig (2021). This design balances the number of news shock and control windows and aligns calendar timing so that the only systematic difference should be the presence of firm news.<sup>21</sup>

**Density analysis** I examine the distribution of three-day log returns around firm-specific news shock days using the matched control design. The event sample pools news shocks across all firms. The control sample consists of strictly no news windows matched by month, weekday, and calendar proximity.

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<sup>21</sup>Matched controls are three-day windows on the same weekday in nearby weeks (within a short horizon), chosen to be strictly no news across all firms; events without an eligible match are dropped.

Figure 13: Kernel densities of three-day log returns for news shock window vs matched no news window



*Notes:* Control uses matched windows. Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986).

Using the matched control sample, the distributional patterns are consistent with the main results but are attenuated in magnitude. For neodymium, praseodymium, and dysprosium, the return densities on news shock days remain wider with heavier tails, though the central mass of the distributions is closer to controls than in the baseline comparison. For lanthanum and terbium, the news day curves are marginally narrower around zero and show no clear excess tail mass relative to matched controls, differences are small and visually modest. Overall, matching control windows narrows the contrast between news and control days. The evidence continues to point to elevated dispersion on news shock days, although the differences are smaller once controls are calendar matched.

**Variance test** Equality of variances between news and matched-control windows is tested using the Brown–Forsythe test.

Table 6: Variance Analysis: Brown–Forsythe test with matched controls

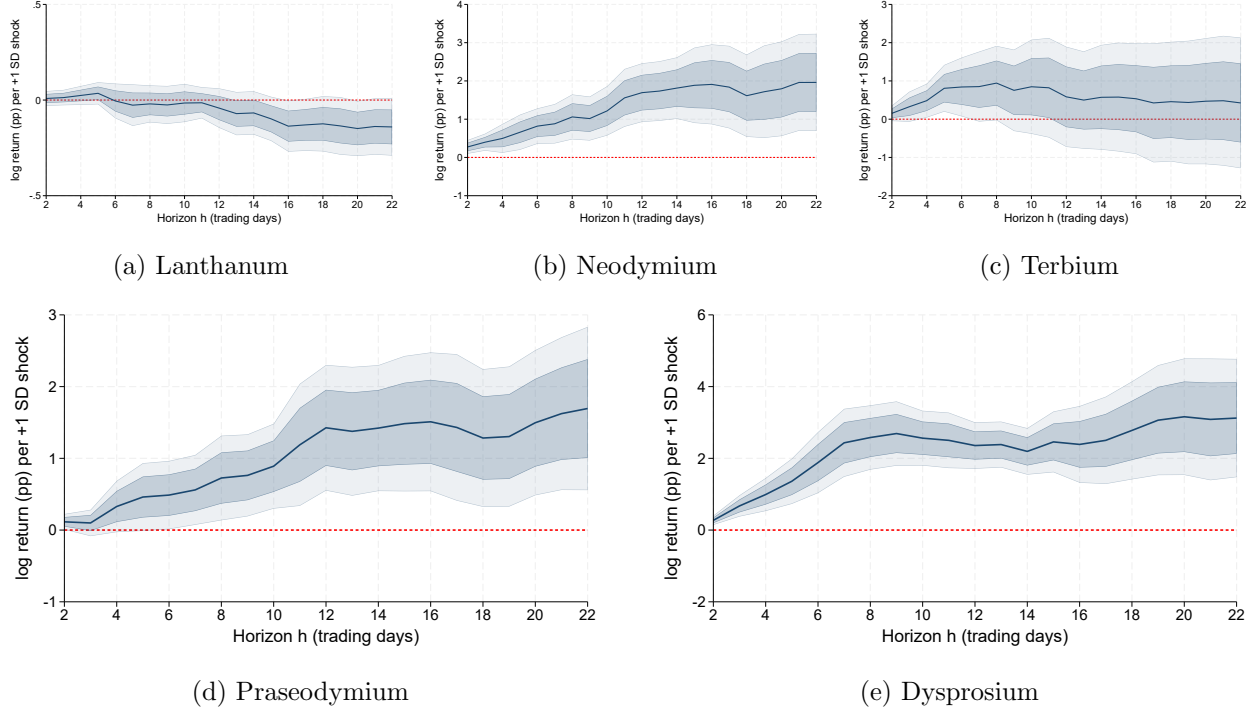
| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------|
| Dysprosium   | 169               | 2.044              | 165               | 2.023              | 1.011              | 0.806           |
| Lanthanum    | 169               | 1.639              | 165               | 1.625              | 1.008              | 0.626           |
| Neodymium    | 169               | 1.742              | 165               | 2.064              | 0.844              | 0.248           |
| Praseodymium | 169               | 1.756              | 165               | 1.753              | 1.002              | 0.490           |
| Terbium      | 168               | 2.543              | 165               | 1.729              | 1.471              | 0.359           |

*Note:* Pooled across all firms. Controls are strictly no news three-day windows matched by month, weekday, and calendar proximity. Standard deviations are computed from three-day log returns in percentage points.  $SD_{\text{Rel.}}$  equals  $SD_{\text{news}}/SD_{\text{ctrl}}$ . The Brown–Forsythe test uses absolute deviations from the group median and reports the  $p$ -value from an  $F(1, N_{\text{news}}+N_{\text{ctrl}}-2)$  reference distribution.

With matched controls, the Brown–Forsythe results closely mirror the main specification. All  $p$ -values exceed statistical significance, so the null of variances cannot be rejected. Matching on calendar time therefore does not change the qualitative conclusion: dispersion on news days is broadly similar to controls, with occasional metals showing wider tails in the density plots but without statistically significant variance differences.

**Local projections (matched controls)** I estimate impulse responses using local projections, following the approach described in the empirical framework. Since the shock regressor is nonzero only on news shock days, slope estimation relies on event time variation. Restricting the zero shock observations to matched controls therefore leaves point estimates largely unchanged, mainly affecting precision (intercept and HAC standard errors). The matched control calendar is kept fixed, and the coefficients are read as percentage point responses to a one standard deviation news shock.

Figure 14: Pooled impulse responses of rare earth spot prices with matched control windows



*Notes:* Local projections include  $p = 12$  lags of own daily returns; confidence bands (dark = 68%, light = 90%) use Newey–West HAC with bandwidth  $b_h = \min\{h, 12\}$ . The  $x$ -axis shows trading days since the news shock; the  $y$ -axis shows cumulative log returns relative to  $t+1$ .

As expected, Figure 14 shows that re-estimating with matched controls yields no material changes. Impulse responses remain positive and persistent for three of the five metals, rising the first week and then flattening without reversal by  $h = 22$ . The 68% bands exclude zero after a few days, and the 90% bands do so for the stronger cases. Thus, the qualitative conclusions from the main results remain unchanged.

## 7.2 Robustness to Geopolitical Risk Days

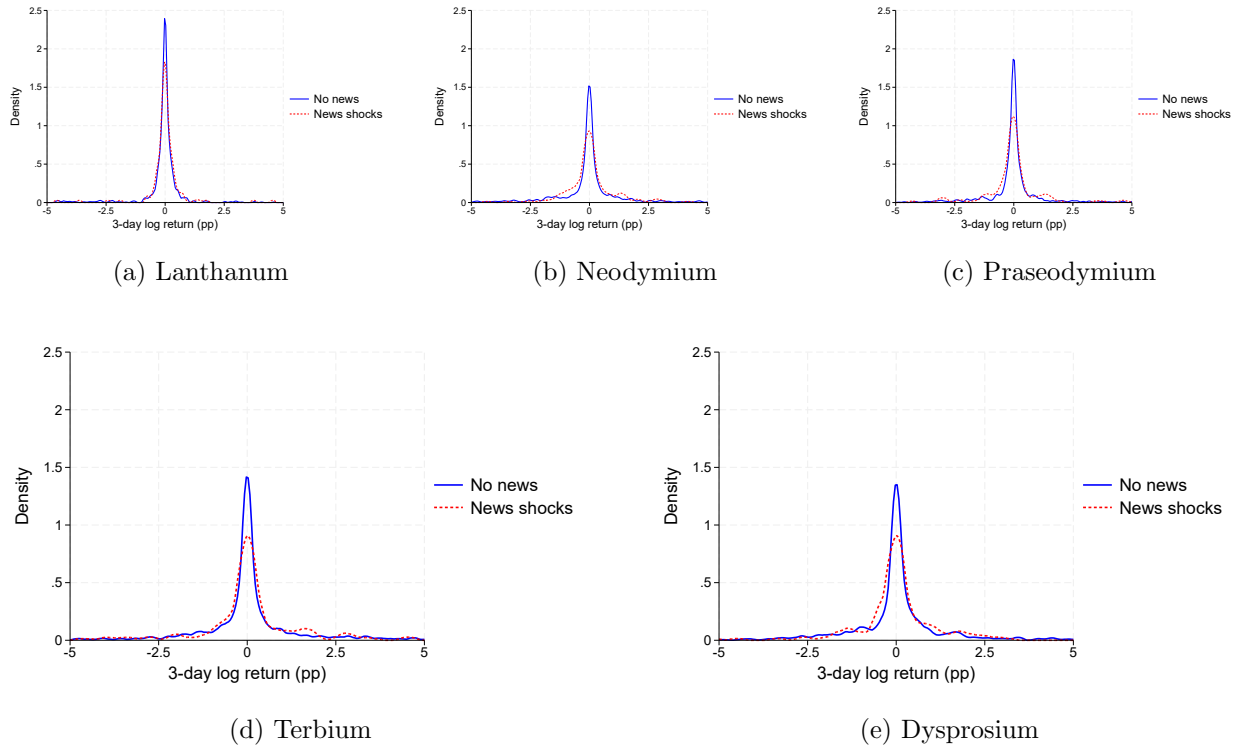
To ensure that the news shock identification is not confounded by exogenous geopolitical shocks, I implement a robustness check that excludes days of elevated geopolitical risk. Following high-frequency identification designs that limit the influence from background news, I re-estimate all results after excluding days of exceptional geopolitical risk. This parallels to Känzig (2021), who demonstrates the importance of accounting for geopolitical risk in high-frequency identification designs for commodity markets. I use the Geopolitical Risk Index (GPR), developed by Caldara and Iacoviello (2022).<sup>22</sup> From the daily GPR index,

<sup>22</sup>Dataset: Geopolitical Risk (GPR) Index, Caldara and Iacoviello. Available at <https://www.matteoiacoviello.com/gpr.htm> [accessed: 17 September, 2025].



I identify all days on which the absolute change exceeds the 95th percentile threshold over the sample period (2009 onward), capturing major spikes typically associated with wars, geopolitical crises, or trade disputes. For each identified spike day, I exclude a symmetric three-day window,<sup>23</sup> covering one trading day before and after the increased risk day, from both the news shock and the control sample.<sup>24</sup> This procedure helps to ensure that observed price dynamics are not driven by extraordinary geopolitical events rather than firm-level information shocks. After constructing this filtered sample, I replicate the main empirical strategy, including kernel density plots of three-day log returns for news shock and control days in Figure 15 and variance analysis using the Brown–Forsythe test (Table 7).

Figure 15: Kernel densities of three-day log returns on news shock windows vs. no news windows control windows, excluding periods of elevated geopolitical risk



*Note:* Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman’s normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Excluding symmetric three-day windows around dates at which the absolute change in the Caldara–Iacoviello Geopolitical Risk (GPR) Index exceeds its 95th percentile, thereby filtering out extraordinary geopolitical risk episodes from the estimation sample.

<sup>23</sup>The choice of a three-day buffer balances two concerns: geopolitical shocks often involve anticipation and short-run persistence, as emphasized by Caldara and Iacoviello (2022), but longer exclusion windows would sharply reduce the number of usable observations.

<sup>24</sup>To ensure consistency, I drop any news shock whose three-day window overlaps with a geopolitical risk window, even if only partially overlapping. The filter removes 37 of 188 news windows and 185 of 1,339 control windows.

Table 7: Variance Analysis: Brown–Forsythe test excluding extreme geopolitical risk days

| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p\text{-value (BF)}$ |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------------|
| Dysprosium   | 151               | 2.017              | 1154              | 2.717              | 0.742              | 0.346                 |
| Lanthanum    | 151               | 1.760              | 1154              | 2.525              | 0.697              | 0.568                 |
| Neodymium    | 151               | 1.851              | 1154              | 2.496              | 0.742              | 0.241                 |
| Praseodymium | 151               | 1.871              | 1154              | 2.310              | 0.810              | 0.609                 |
| Terbium      | 151               | 2.472              | 1154              | 2.830              | 0.874              | 0.805                 |

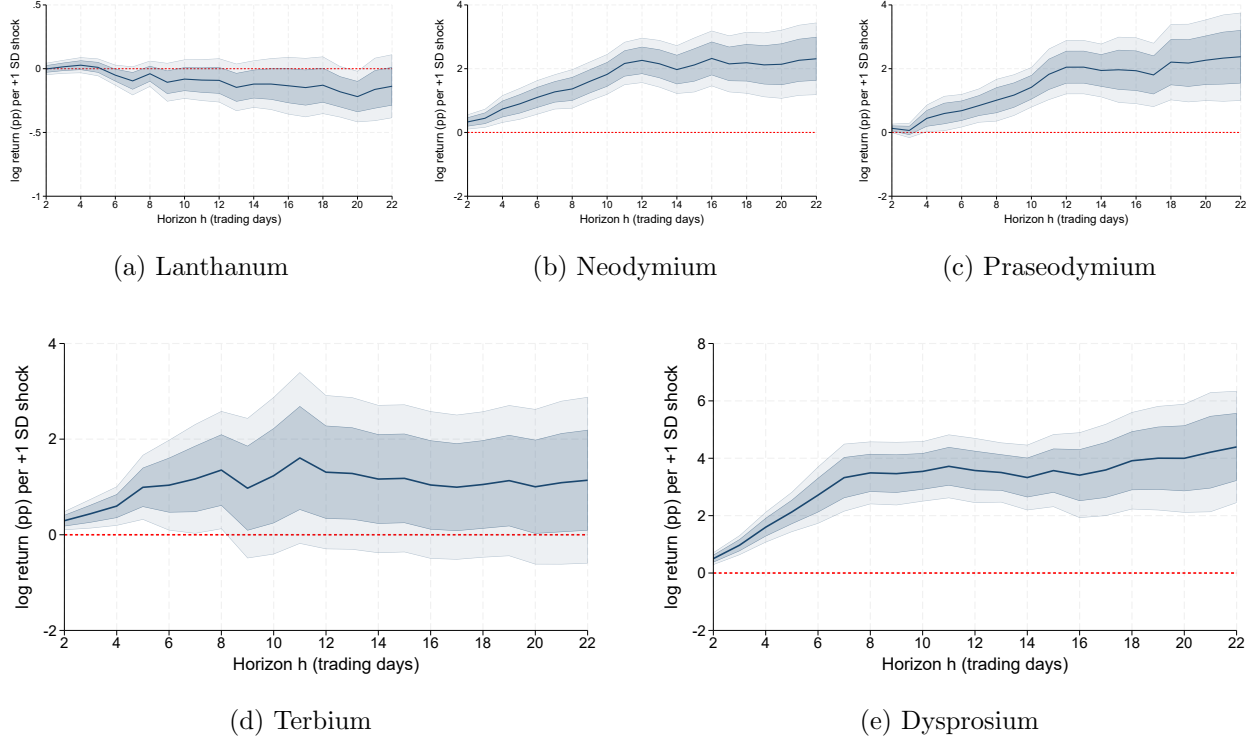
*Note:* Brown–Forsythe compares variances of three-day log returns between news shock and control windows. Days on which the absolute change in the Caldara–Iacoviello GPR Index exceeds its 95th percentile are excluded from both sets of windows.  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ .

The density plots show that after controlling for geopolitical risk, return distributions remain broadly consistent with the baseline analysis. Variance results are likewise unchanged: the Brown–Forsythe tests do not reject equality of variances between news shock and control windows. This suggests that the main results are not driven by extreme outliers associated with geopolitical events.

Figure 16 shows impulse responses re-estimated after excluding days when the Geopolitical Risk Index (GPR) records changes above its 95th percentile.

Excluding periods of elevated geopolitical risk leaves the impulse responses broadly consistent with the baseline results. Lanthanum remains flat around zero across the horizon. Terbium shows modestly stronger responses at mid-horizons, while dysprosium displays a smaller initial spike but a broadly similar later profile. Neodymium, dysprosium and praseodymium exhibit slightly larger and more monotonic cumulative gains, with both the 68% and 90% confidence bands remaining above zero throughout. Overall, the qualitative conclusions remain unchanged; if anything, excluding geopolitical-risk days marginally strengthens the evidence of gradual positive responses and increases confidence in the constructed news shock series.

Figure 16: Impulse responses of rare earth prices to firm news shocks, excluding periods of elevated geopolitical risk.



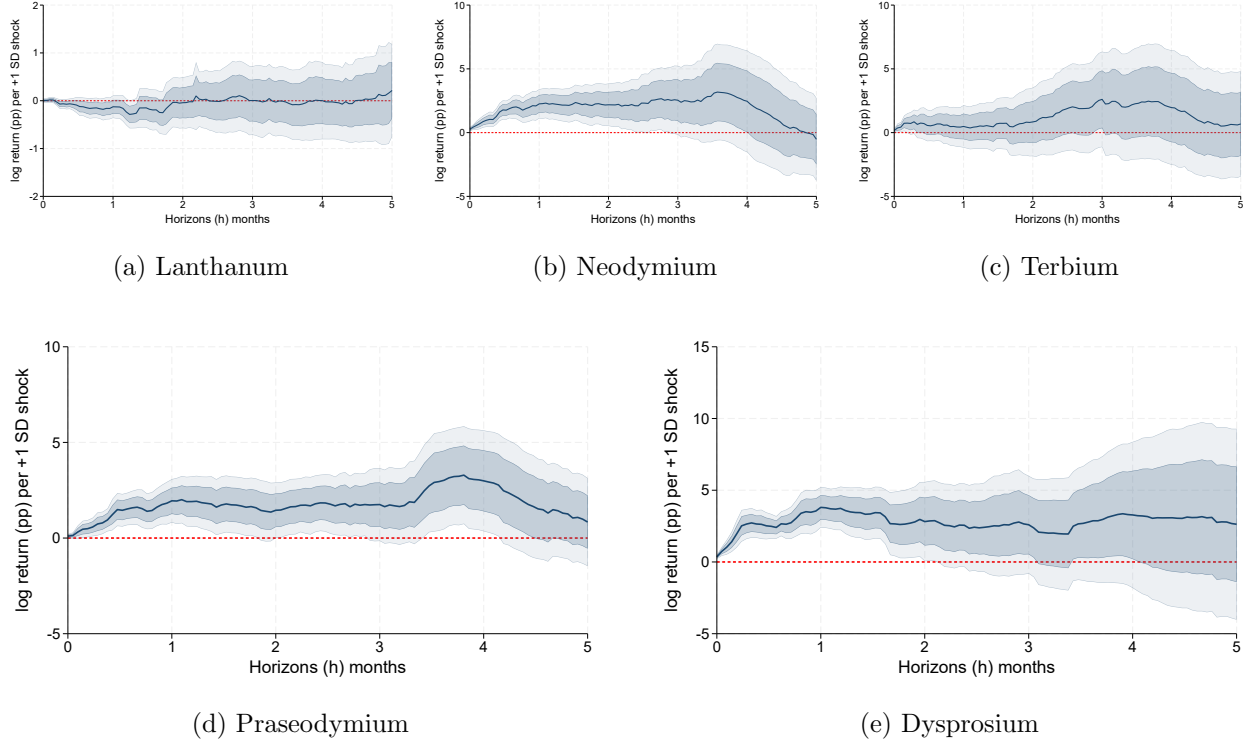
*Notes:* Panels show cumulative log price returns from  $t+2$  onward to a one standard deviation news shock. The sample excludes days in the top 5% of the Geopolitical Risk Index. Inference uses Newey–West HAC errors, bands are 68% (dark) and 90% (light).

### 7.3 Long-horizon

The one-month horizon results show persistent price level changes with little evidence of reversal. Extending the horizon is informative for two reasons. It detects delayed adjustment arising from contracting or inventory frictions, and it distinguishes persistent effects from transitory mean-reverting movements by assessing whether the cumulative response plateaus or continues to trend monthly to quarterly horizons. I therefore extend the local projection horizon to a five-month horizon approximately  $h \approx 105$  trading days. Event identification, shock construction, and standardization remain unchanged. The estimation sample is restricted so that every observation has support through  $t+h+2$ . Inference uses Newey–West HAC errors with  $b_h = \min\{h, L_T\}$ , and  $L_T = \lfloor 4(T/100)^{2/9} \rfloor$ , where  $T$  is the usable sample size at horizon  $h$ . I use this Newey–West (1994) automatic plug-in bandwidth at long horizons because it adapts to sample size and serial correlation: cumulative returns induce dependence whose span grows with  $h$ , and the selector scales with  $T$  and delivers better

small-sample size control than a fixed cap such as  $\min\{h, 12\}$  (Newey and West, 1994).<sup>25</sup>

Figure 17: Local projection impulse responses of rare earth spot prices five-month horizon



*Notes:* Horizon extends to five-months ( $h = 2, \dots, 105$ ). Specification as before: pooled standardized shock and  $p = 12$  lags. Newey–West HAC uses  $b_h = \min\{h, L_T\}$  with  $L_T = \lfloor 4(T/100)^{2/9} \rfloor$  (where  $T$  is the usable sample at horizon  $h$ ); bands denote 68% (dark) and 90% (light).

A one standard deviation news shock is associated with positive and persistent cumulative responses for three of the five metals over  $h = 2, \dots, 105$ . Point estimates rise over the first few weeks and then flatten. For neodymium, praseodymium, and dysprosium the 68% bands move above zero after the short-run, and the 90% bands typically exclude zero at intermediate horizons. For lanthanum and terbium, the bands cross zero throughout. Cross-metal heterogeneity mirrors the baseline: responses are stronger for dysprosium, praseodymium, and neodymium, and comparatively small for lanthanum and terbium. Overall, the long horizon evidence is consistent with the earlier results in showing persistent responses for several weeks. However, toward the end of the horizon, the profiles for neodymium, praseodymium, and dysprosium display clear attenuation, with confidence bands drifting back toward zero. This pattern is consistent with medium run effects that attenuate over time rather than permanent shifts in price levels.

<sup>25</sup>As a robustness check, I also re-estimate the local projections using the conventional  $b_h = \min\{h, 12\}$ . Impulse responses and confidence bands are essentially unchanged; see Appendix A.6 Figure A30.

## 8 Discussion

Several limitations of this study should be acknowledged when interpreting the findings. The primary constraint is data availability. News shocks are identified only for listed producers with sufficient disclosures, leaving unlisted firms and private contracts outside the sample. Coverage is uneven across firms and over time, as some producers entered late in the sample or disclosed irregularly. The analysis relies on daily spot prices for selected rare earth oxides. Price formation through long-term contracts, inventories, or private negotiations is not observed. In addition to uneven coverage, transparency in rare earth markets is limited. Firm announcements that can be used as a potential news shock with timestamps are unavailable for many firms (hence the restricted firm sample), element-level inventories are rarely disclosed, and there is no centralized futures benchmark. These gaps restrict identification and statistical power.

A second limitation concerns identification. The news shock series is constructed from observed spot price returns rather than from an exogenous instrument, so the responses reported here should be interpreted as associations in event time rather than causal effects. In principle, one could construct a high-frequency surprise series from very tight windows in a liquid benchmark market, as in [Känzig \(2021, 2023\)](#). In the rare earth setting, however, the absence of a centralized futures market, asynchronous firm disclosures, and thin spot quotes make this infeasible at present and would risk measurement error or weak instruments. If a liquid futures benchmark were to develop, or if other valid external instruments were identified, future research could pursue this approach to obtain stronger causal inference and to test more precisely for effects on the financial markets.

A third limitation concerns the treatment of policy announcements. The present analysis does not incorporate a set of policy announcements, and this is particularly relevant for Chinese policy measures and supply interventions, which are likely to have strong effects on prices. Future work could construct a policy announcement shock series in parallel to the firm news series to measure their separate contributions to price formation. It would also be possible to exclude policy announcement days from the firm-based series to reduce the risk that observed responses to firm news are confounded by concurrent policy effects.

Taken together, these limitations suggest that the results should be interpreted as first-step evidence on how firm news is reflected in rare earth spot markets and what effects news shocks could have on the financial markets. The findings indicate modest, metal-specific price shifts that persist for several weeks but remain small on average, and no statistically significant effects on the financial markets are detected in this setting. Future research could expand the event database to include unlisted producers and a systematic chronology of

policy communications. Automated text analysis could classify each announcement along two dimensions. First, by sign: supply expanding versus supply constraining. Second, by horizon: near-term operational changes with immediate market relevance versus forward-looking capacity plans with limited short-run impact. Attaching these tags to the event windows would allow direct tests for asymmetry (positive versus negative news) and for timing (near-term versus forward-looking) in spot prices and, importantly, in effects on the financial markets.

Improvements in data would strengthen identification. Standardized announcement timestamps, intraday quotes, and better coverage of contract terms would reduce measurement error in the news shock series and make event-time comparisons tighter. Taken together, these improvements would enable credible causal identification of firm news shock effects on rare earth prices and broad equity markets.

## 9 Conclusion

Rare earth markets are strategically important for national defense and clean energy production, yet there is little transparent evidence on how rare earth firm announcements affect rare earth spot prices and broader financial markets. This paper develops the first high-frequency news shock series for the rare earth market by aligning dated firm disclosures from listed producers with symmetric trading-day event windows and daily spot prices. The contribution is the new event-time database and its use to study how rare earth spot prices and financial markets respond to firm news.

I assess the information content of the news shock series using kernel density comparisons, Brown–Forsythe variance tests, and local projection impulse responses. Relative to strict no news windows, event windows display flatter densities with heavier tails. Formal variance tests rarely reject equality of variances. Local projection estimates are small on impact but cumulate to modest medium-horizon increases for a subset of metals; others show no robust response. These patterns are robust to calendar-matched controls, to excluding days with elevated geopolitical risk, and to extending the projection window to roughly five months.

I then use the constructed news shock series to examine effects on the financial markets. Equity benchmarks outside the rare earth sector, as well as selected rare earth intensive firms, show no statistically significant reaction to rare earth news. Taken together, firm disclosures are reflected only weakly in daily spot prices and are not associated with systematic movements in broad equities.

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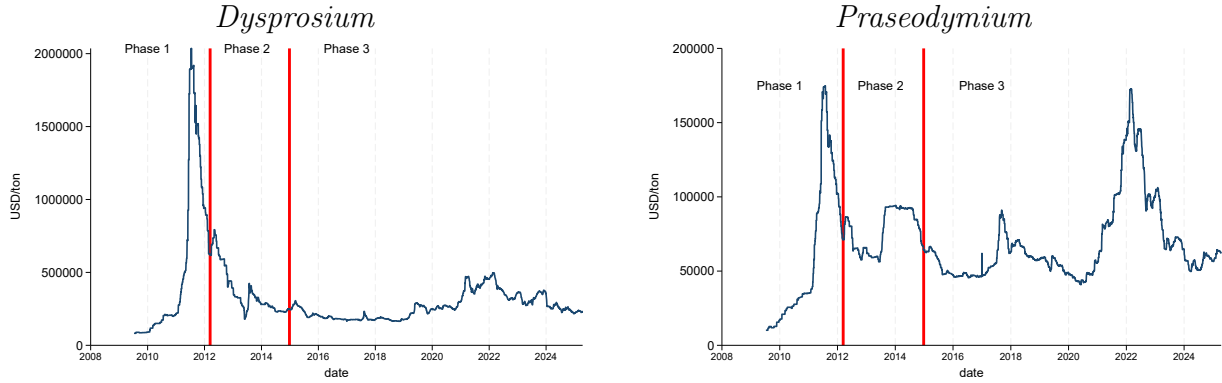
# Appendix

This appendix provides supplementary materials that complement the main analysis of the paper. Here, I present additional figures, tables, and robustness checks that are referenced but not shown in detail in the main text.

## A.1 Spot Price Trajectories of Rare Earth Elements (2009–2024)

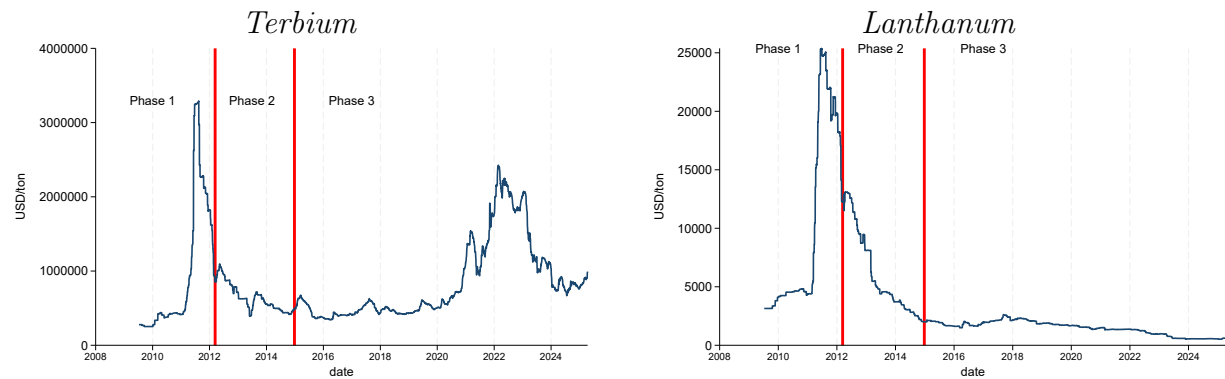
The following figures present the monthly spot prices (USD per ton) for the remaining four major rare earth elements used in this study. The figures highlight the three phases analyzed in this study: Phase 1 (start to 13 March 2012), Phase 2 (14 March 2012 to 26 December 2014), and Phase 3 (01 May 2015 onward). This structure enables a visual assessment of the price dynamics in relation to regulatory changes and significant global market developments. The comparative perspective across all elements and the fund provides evidence of coordinated price shifts and systemic responses to external shocks in the rare earth sector.

Figure A1: Rare Earth Spot Prices: Dysprosium and Praseodymium



*Note:* Spot price series for dysprosium and praseodymium, segmented by the three institutional phases discussed in Section 2.2. Both series exhibit sharp price increases during Phase 1 and early Phase 2, with pronounced corrections beginning even before the official WTO filing and policy adjustment in Phase 3. These patterns suggest that market expectations and anticipatory behavior may have contributed to the observed price dynamics, rather than regulatory actions alone. (Source: Author's own illustration) with Shanghai Metals Market (SMM) prices accessed via LSEG Datastream.

Figure A2: Rare Earth Spot Prices: Terbium and Lanthanum



*Note:* Monthly spot prices for terbium and lanthanum, segmented by the three institutional phases discussed in Section 2.2. Both series exhibit sharp price increases during Phase 1 and early Phase 2, with pronounced corrections beginning even before the official WTO filing and policy adjustment in Phase 3. These patterns suggest that market expectations and anticipatory behavior may have contributed to the observed price dynamics, rather than regulatory actions alone. (Source: Author’s own illustration) with Shanghai Metals Market (SMM) prices accessed via LSEG Datastream.

## News shock days by firm

Table A1: News shock days by firm

| Firm     | News shock days |
|----------|-----------------|
| Lynas    | 64              |
| MP       | 16              |
| Iluka    | 48              |
| Arafura  | 10              |
| Northern | 63              |
| RareElem | 37              |
| ChinaN   | 29              |
| Shenghe  | 31              |

Note: Days with firm news from quarterly or annual reports, production updates, or material regulatory filings/press releases [Arafura Rare Earths Ltd, 2025](#); [Northern Minerals Ltd, 2025](#); [Iluka Resources Ltd, 2025](#); [Rare Element Resources Ltd, 2025](#); [Shenghe Resources Holding Co., Ltd, 2025](#); [China Northern Rare Earth \(Group\) High-Tech Co., Ltd, 2025](#); [MP Materials Corp., 2025](#); [Lynas Rare Earths Ltd, 2025](#). [accessed 01 July 2025].

## A.2 Granular Granger Causality Tests: Motivation and Implementation

This appendix reports firm by metal tests of whether lagged rare earth returns help predict the timing of firm news shocks. For each firm  $F$  and metal  $M$ , I estimate a logit in which the dependent variable is the event window around firm news shock. The key regressors are the first and second lags of the daily log return (in percentage points) of metal  $M$ ,  $r_{t-1}^M$  and  $r_{t-2}^M$ . All specifications include weekday and month dummies, and standard errors are clustered by calendar month. This setup implements a Granger-style timing-predictability test (Granger, 1969). For each  $(F, M)$  pair, I test the joint null that the coefficients on  $r_{t-1}^M$  and  $r_{t-2}^M$  are zero using a Wald  $\chi^2$  test with two degrees of freedom. The exercise assesses predictability of event timing from prior price movements. Because many tests are conducted across firms and metals, occasional marginal rejections are expected by chance.

Table A2: Timing-predictability test (Granger-style) all rare earth elements

| Firm     | $N_{\text{news shock}}$ | $N_{\text{control}}$ | $p$ -value |
|----------|-------------------------|----------------------|------------|
| Lynas    | 50                      | 528                  | 0.0941*    |
| MP       | 14                      | 617                  | 0.2840     |
| Iluka    | 40                      | 526                  | 0.1561     |
| Arafura  | 9                       | 614                  | 0.1482     |
| Northern | 47                      | 521                  | 0.5391     |
| RareElem | 36                      | 675                  | 0.2181     |
| ChinaN   | 25                      | 534                  | 0.4883     |
| Shenghe  | 29                      | 599                  | 0.0849*    |

*Notes:* Firm-level logit on two lags of rare earth returns, with weekday and month dummies; standard errors clustered by calendar month. Reported  $p$  values are from the joint Wald test. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .



Table A3: Predictability of News Shocks: Lynas

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 0.19        | 0.911                 |
| Lanthanum    | 0.37        | 0.830                 |
| Neodymium    | 0.52        | 0.772                 |
| Praseodymium | 2.91        | 0.233                 |
| Terbium      | 4.40        | 0.111                 |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A4: Predictability of News Shocks: MP Materials

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 0.58        | 0.750                 |
| Lanthanum    | 1.02        | 0.601                 |
| Neodymium    | 4.12        | 0.128                 |
| Praseodymium | 1.22        | 0.544                 |
| Terbium      | 2.92        | 0.232                 |

Table A5: Predictability of News Shocks: Iluka

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 2.61        | 0.272                 |
| Lanthanum    | 5.11        | 0.078*                |
| Neodymium    | 0.84        | 0.657                 |
| Praseodymium | 1.75        | 0.417                 |
| Terbium      | 3.46        | 0.177                 |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A6: Predictability of News Shocks: Arafura

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 1.72        | 0.424                 |
| Lanthanum    | 1.88        | 0.390                 |
| Neodymium    | 0.52        | 0.770                 |
| Praseodymium | 2.26        | 0.323                 |
| Terbium      | 2.72        | 0.257                 |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A7: Predictability of News Shocks: Northern Minerals

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 0.52        | 0.769                 |
| Lanthanum    | 5.33        | 0.070*                |
| Neodymium    | 1.34        | 0.512                 |
| Praseodymium | 3.79        | 0.150                 |
| Terbium      | 1.00        | 0.606                 |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A8: Predictability of News Shocks: Rare Element Resources

| <b>Metal</b> | $\chi^2(2)$ | <b><i>p</i>-value</b> |
|--------------|-------------|-----------------------|
| Dysprosium   | 0.36        | 0.836                 |
| Lanthanum    | 1.83        | 0.396                 |
| Neodymium    | 0.64        | 0.727                 |
| Praseodymium | 1.09        | 0.579                 |
| Terbium      | 0.51        | 0.775                 |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A9: Predictability of News Shocks: China Northern Rare Earth

| <b>Metal</b> | $\chi^2(2)$ | <i>p</i> -value |
|--------------|-------------|-----------------|
| Dysprosium   | 3.23        | 0.199           |
| Lanthanum    | 1.96        | 0.375           |
| Neodymium    | 2.89        | 0.236           |
| Praseodymium | 4.71        | 0.095*          |
| Terbium      | 5.57        | 0.062*          |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table A10: Predictability of News Shocks: Shenghe

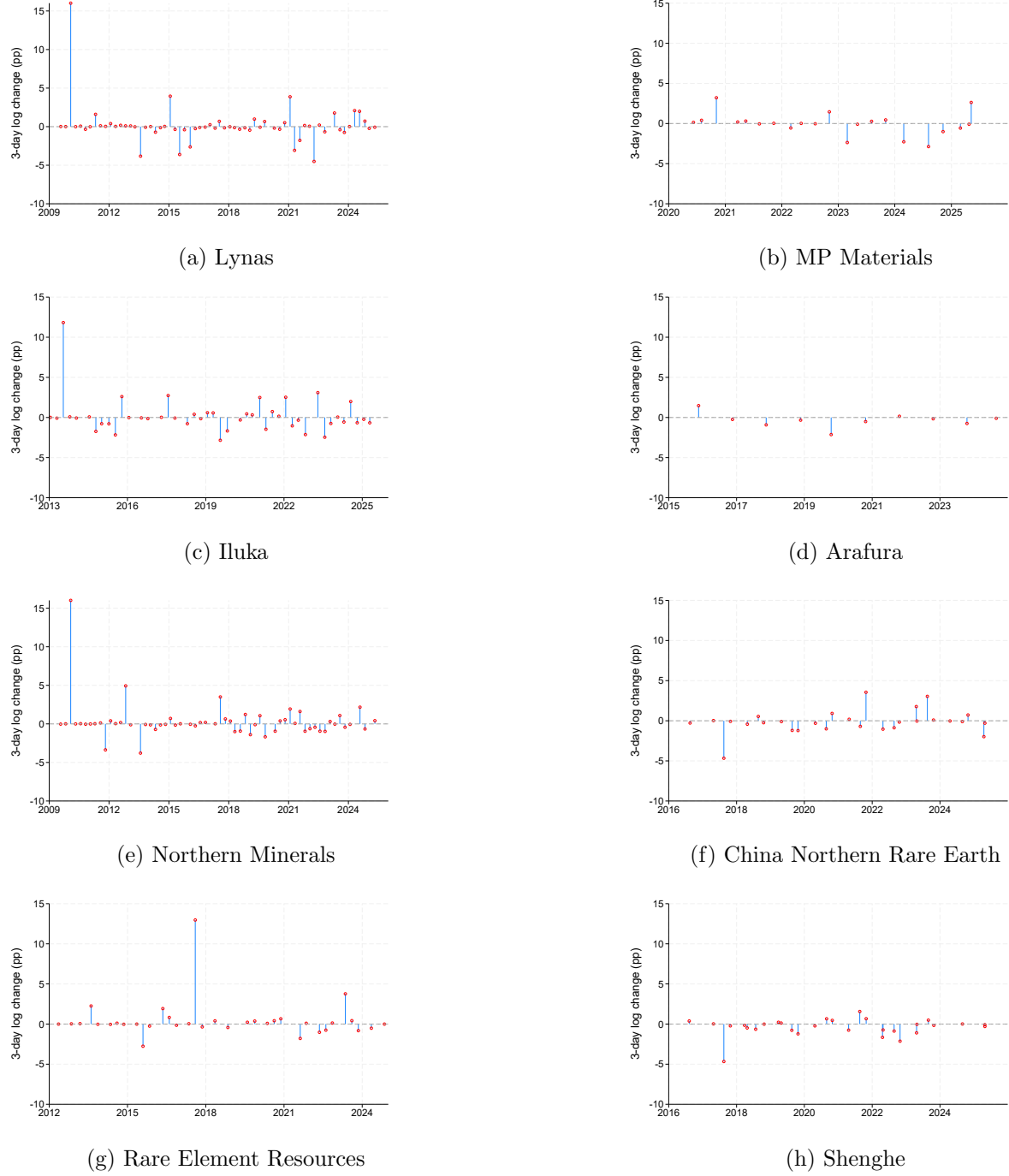
| <b>Metal</b> | $\chi^2(2)$ | <i>p</i> -value |
|--------------|-------------|-----------------|
| Dysprosium   | 3.12        | 0.210           |
| Lanthanum    | 5.06        | 0.080*          |
| Neodymium    | 0.00        | 1.000           |
| Praseodymium | 1.19        | 0.551           |
| Terbium      | 1.79        | 0.409           |

*Note:* For each firm metal pair, a logit predicts whether day  $t$  is the announcement day (center of the three-day window) from the metal's lagged daily log returns (pp),  $r_{t-1}$  and  $r_{t-2}$ , with weekday and month dummies; standard errors are clustered by month. Reported are the joint Wald  $\chi^2(2)$  and  $p$ -value for  $H_0 : \beta_1 = \beta_2 = 0$ . Rejection indicates that returns in the two days before the window help predict announcement timing. Significance: \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## A.3 News Shock Series Across Rare Earth Elements and Firms

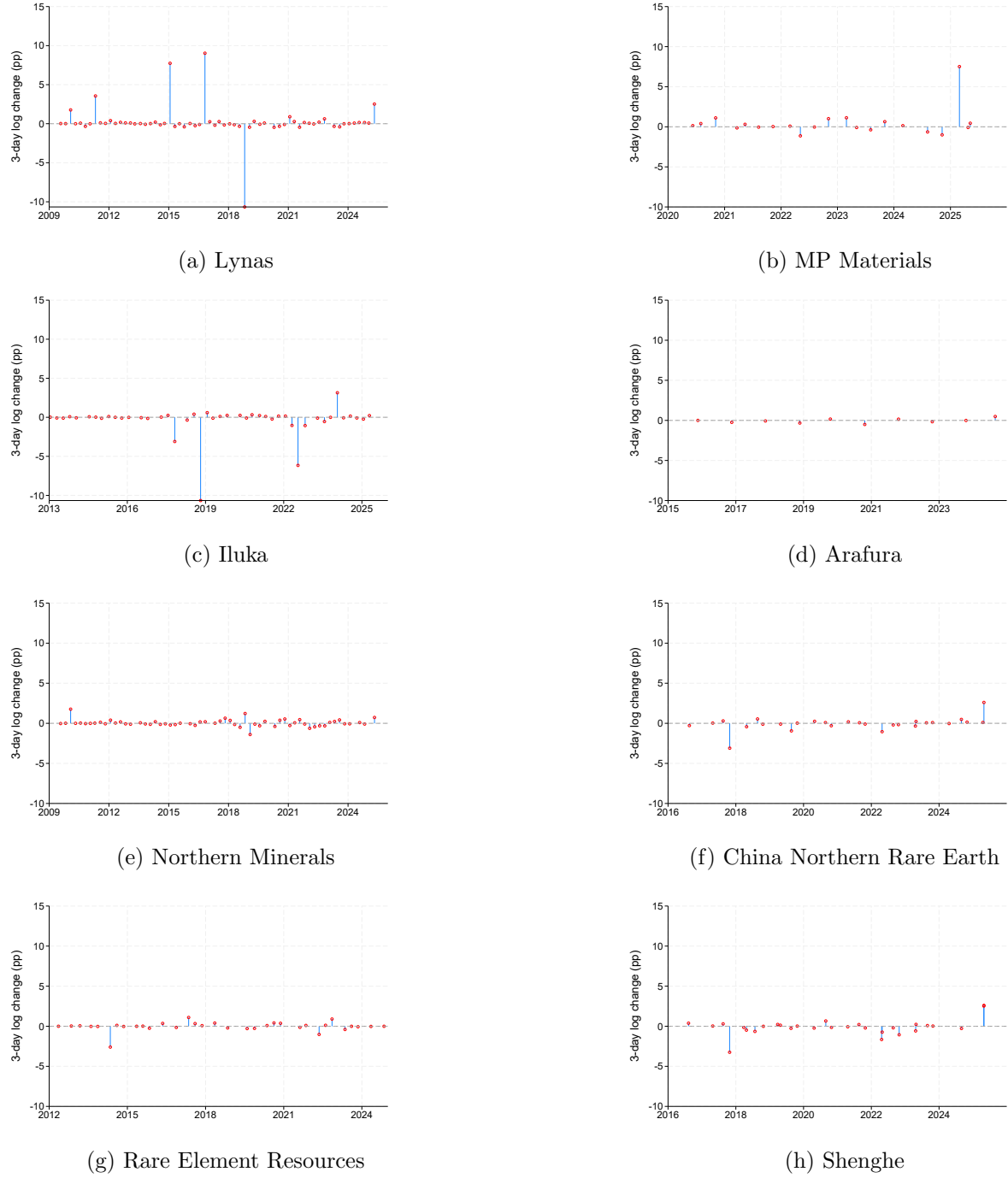
**News-Shock Series.** The following figures present the news shock series for each metal firm pair. For each panel, spikes equal the three-day log return in percentage points on the firm's event news day,  $S_t^{M,F} \equiv 100(\log P_{t+1}^M - \log P_{t-1}^M)$  if a news event for firm  $F$  occurs at  $t$ , and 0 otherwise.

Figure A3: Dysprosium news-shock series across firms.



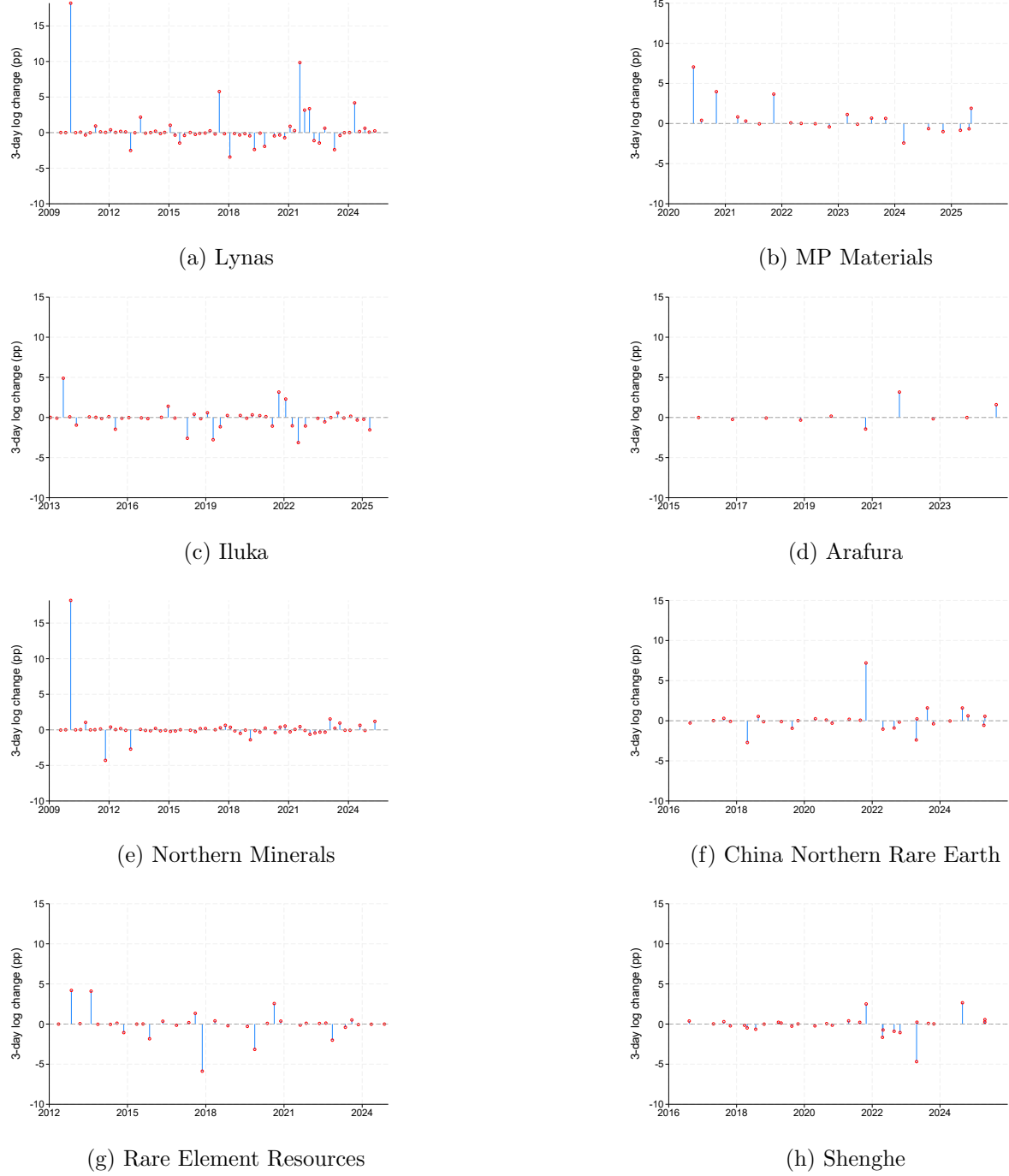
*Note:* Each panel plots the three-day log change (pp) on firm news days; values are zero on non-event days.

Figure A4: Lanthanum news-shock series across firms.



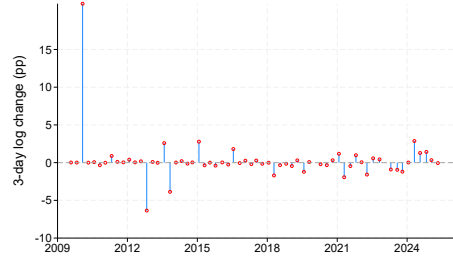
*Note:* Three-day log change (pp) at news day shocks; zeros on non-event days.

Figure A5: Praseodymium news–shock series across firms.

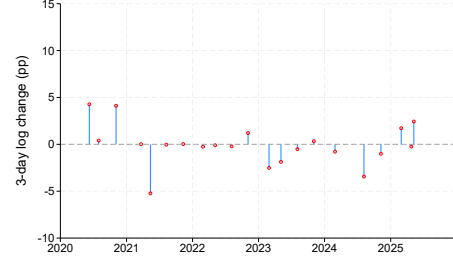


*Note:* Three-day log change (pp) at firm news days; zero elsewhere.

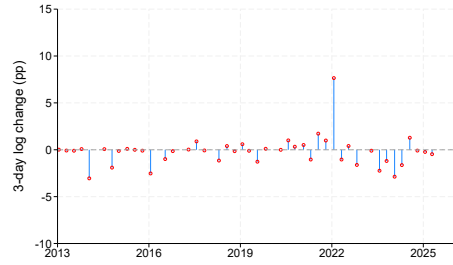
Figure A6: Terbium news-shock series across firms.



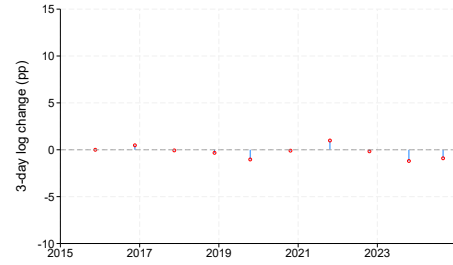
(a) Lynas



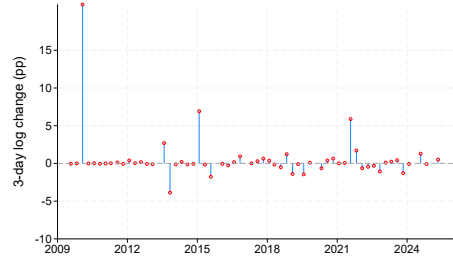
(b) MP Materials



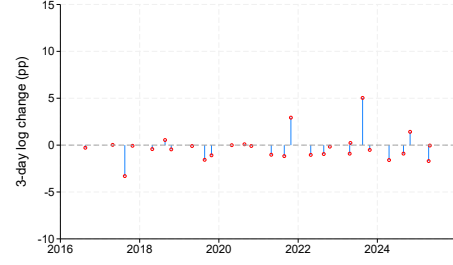
(c) Iluka



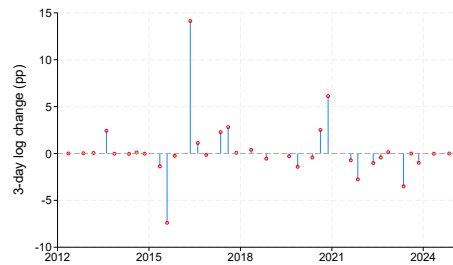
(d) Arafura



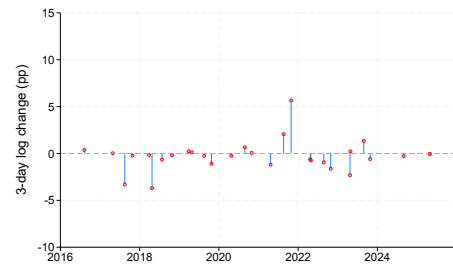
(e) Northern Minerals



(f) China Northern Rare Earth



(g) Rare Element Resources



(h) Shenghe

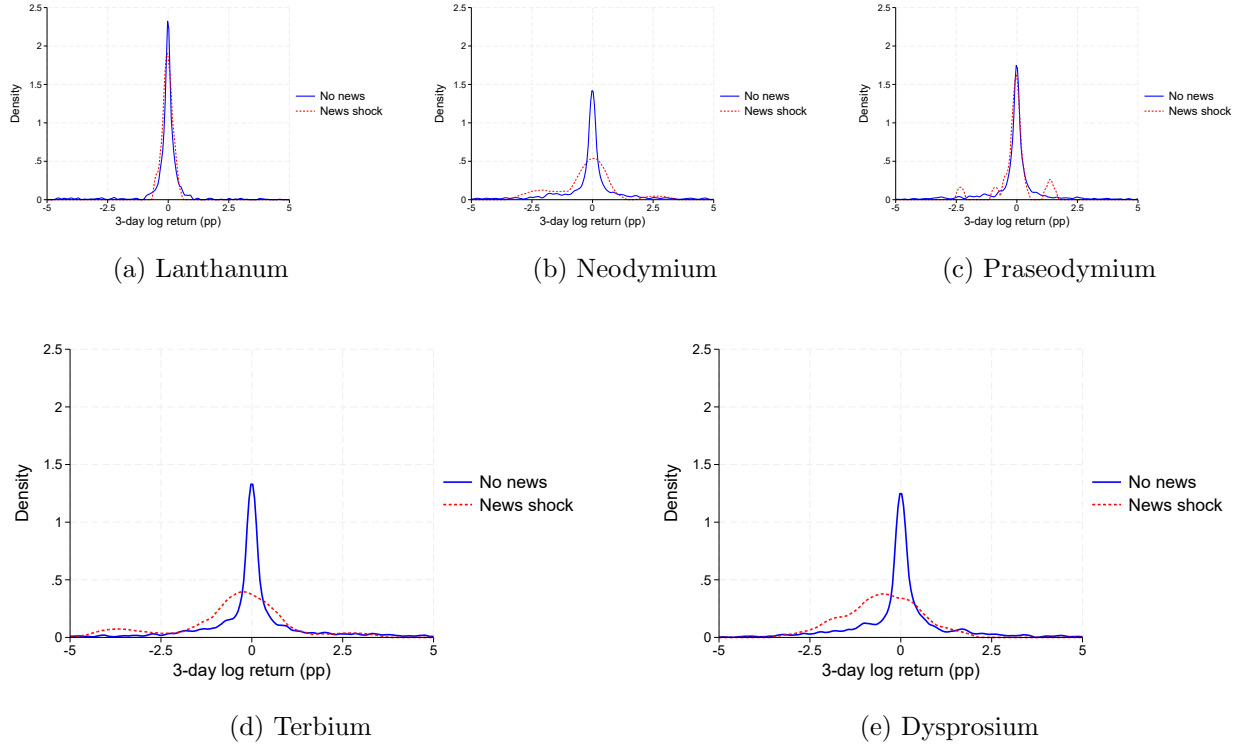
*Note:* Three-day log change (pp) on news day event dates; zero elsewhere.



## A.4 Additional Results—Density, Variance, Local Projections (All Firms)

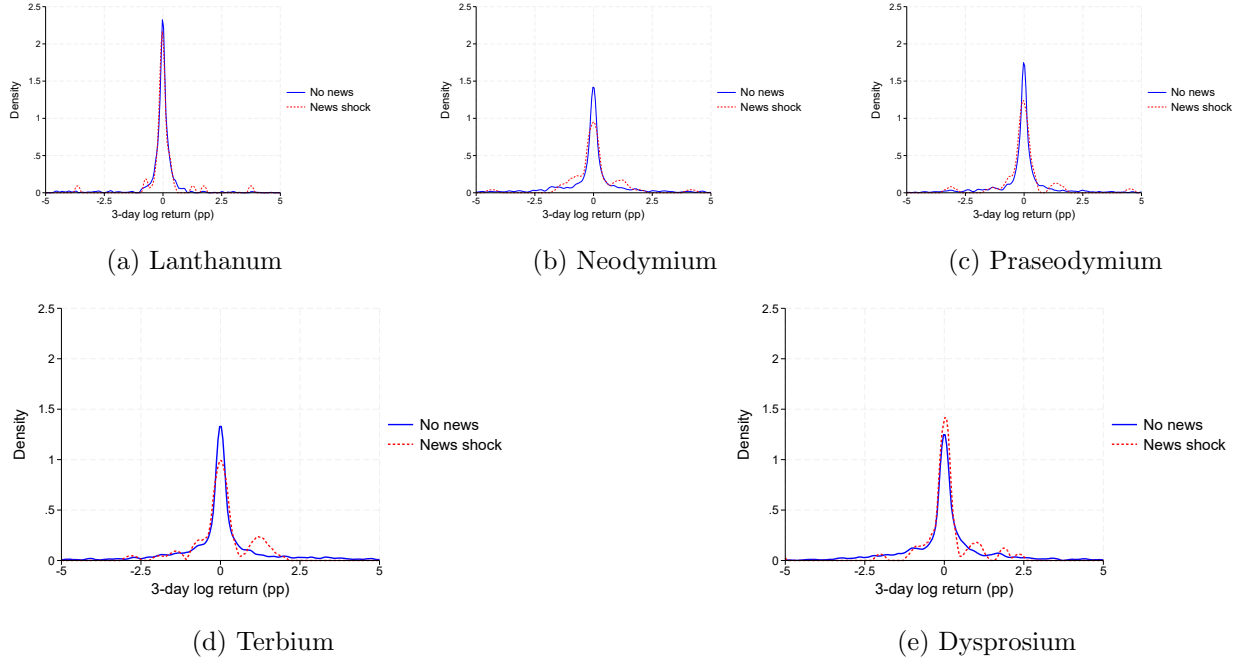
**Density Analysis** These figures plot kernel densities of three-day log returns (pp) on firm-level news shock days vs. strict no news controls. The axes are comparable across the panels.

Figure A7: Kernel densities of three-day log returns (pp) for news shocks vs. strict no-news controls—China Northern Rare Earth



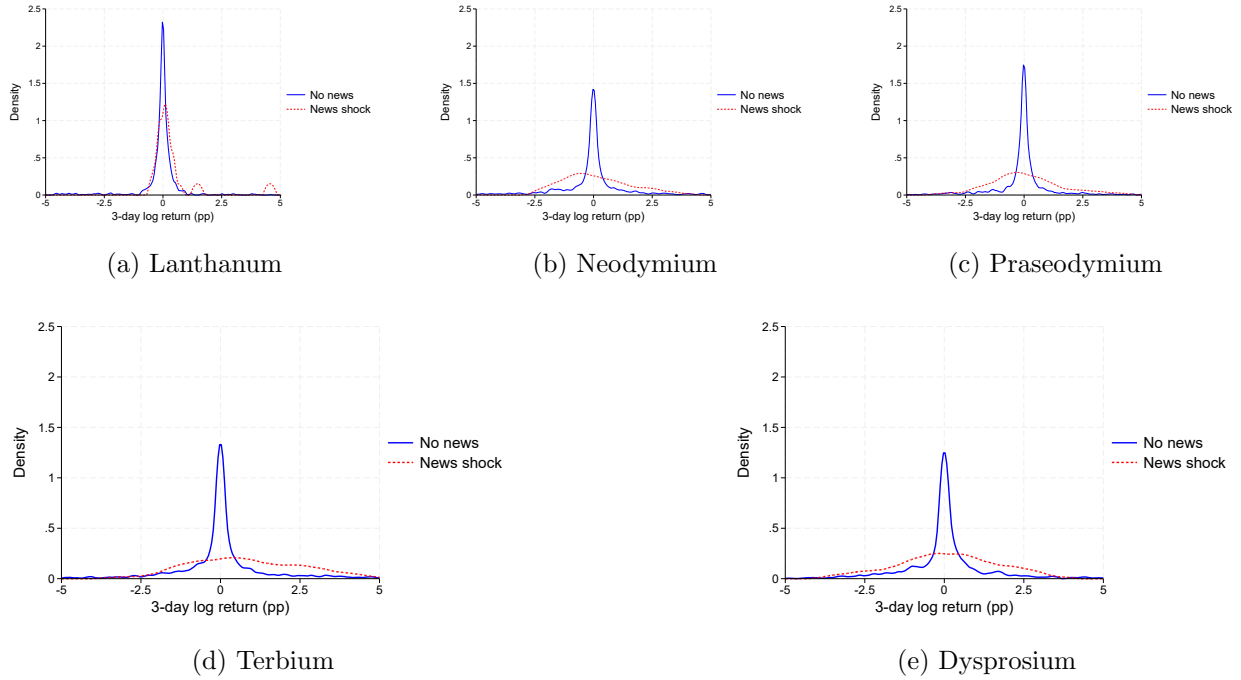
*Notes:* Densities pool event windows across all firms and compare them to strict no news windows. Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Axes are fixed across panels for comparability ( $x \in [-5, 5]$  pp;  $y$ -axis held constant). Visual differences reflect level/dispersion changes associated with news days compared to no news days.

Figure A8: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Lynas.



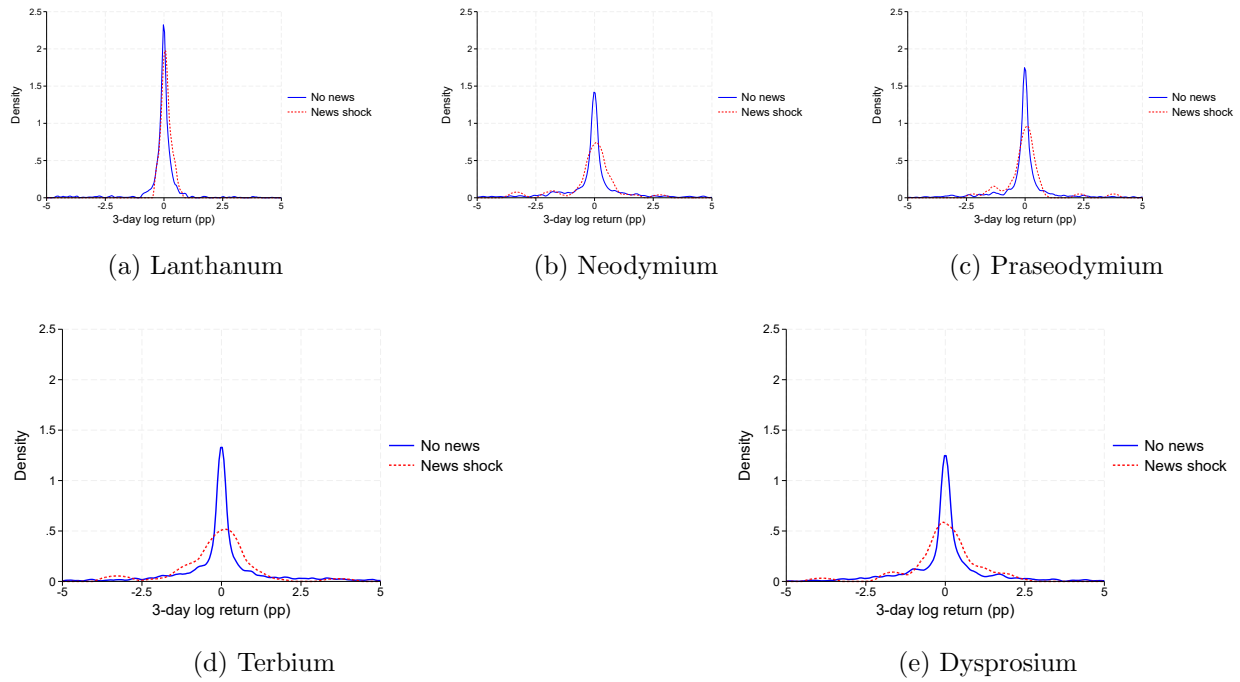
Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A9: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—MP Materials.



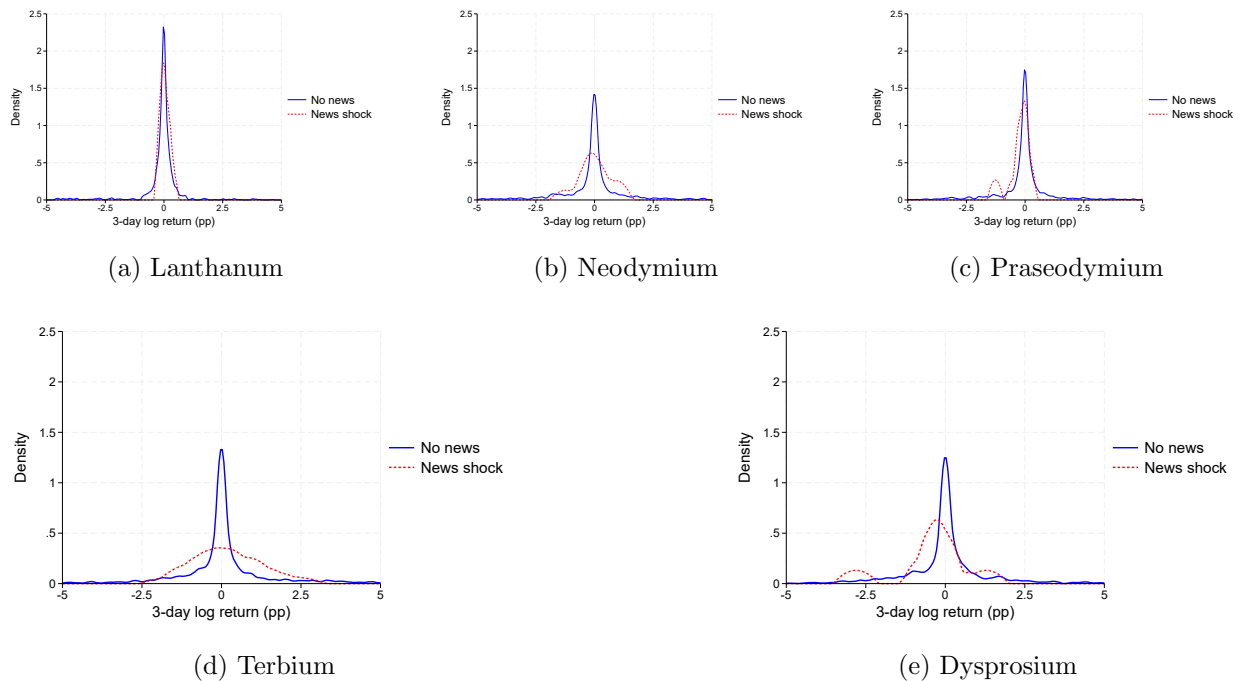
Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A10: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Iluca.



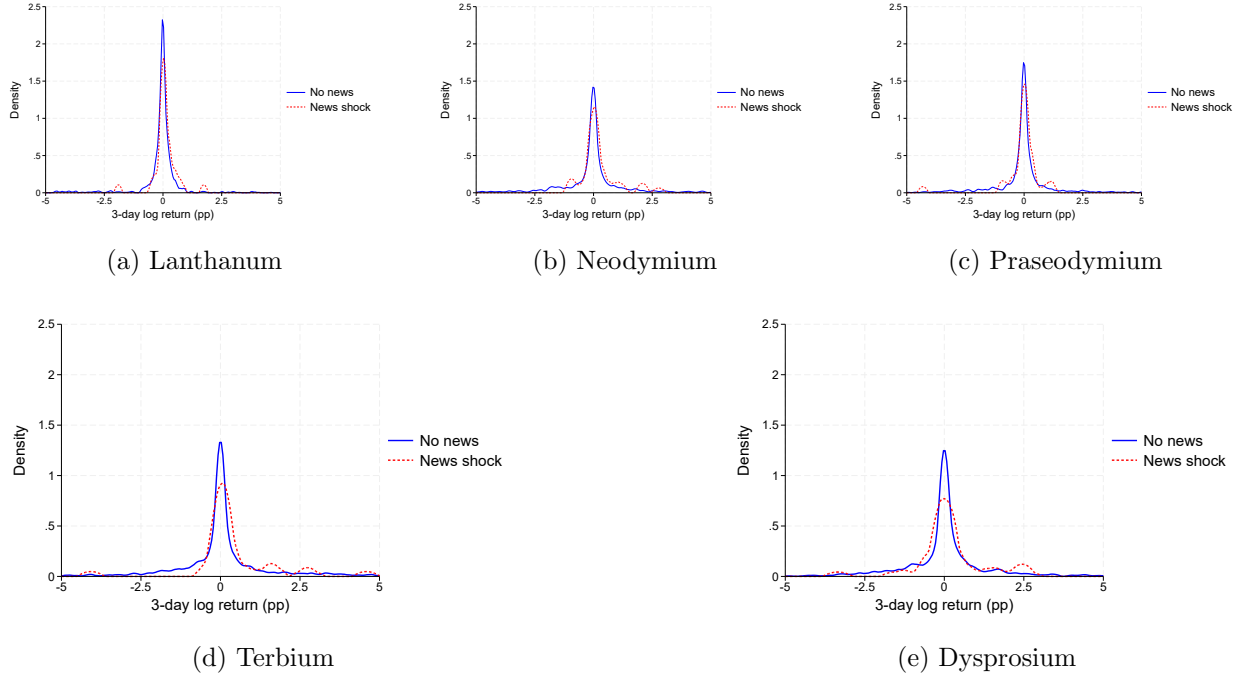
Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A11: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Arafura.



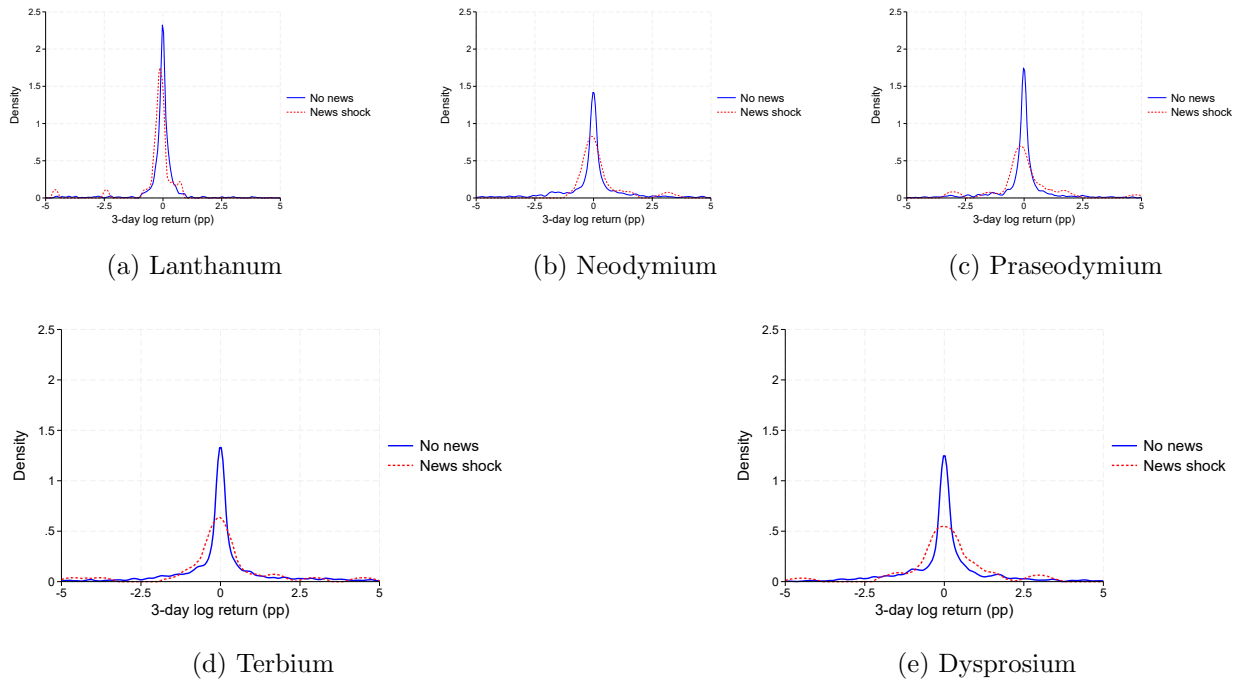
Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A12: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Northern Minerals.



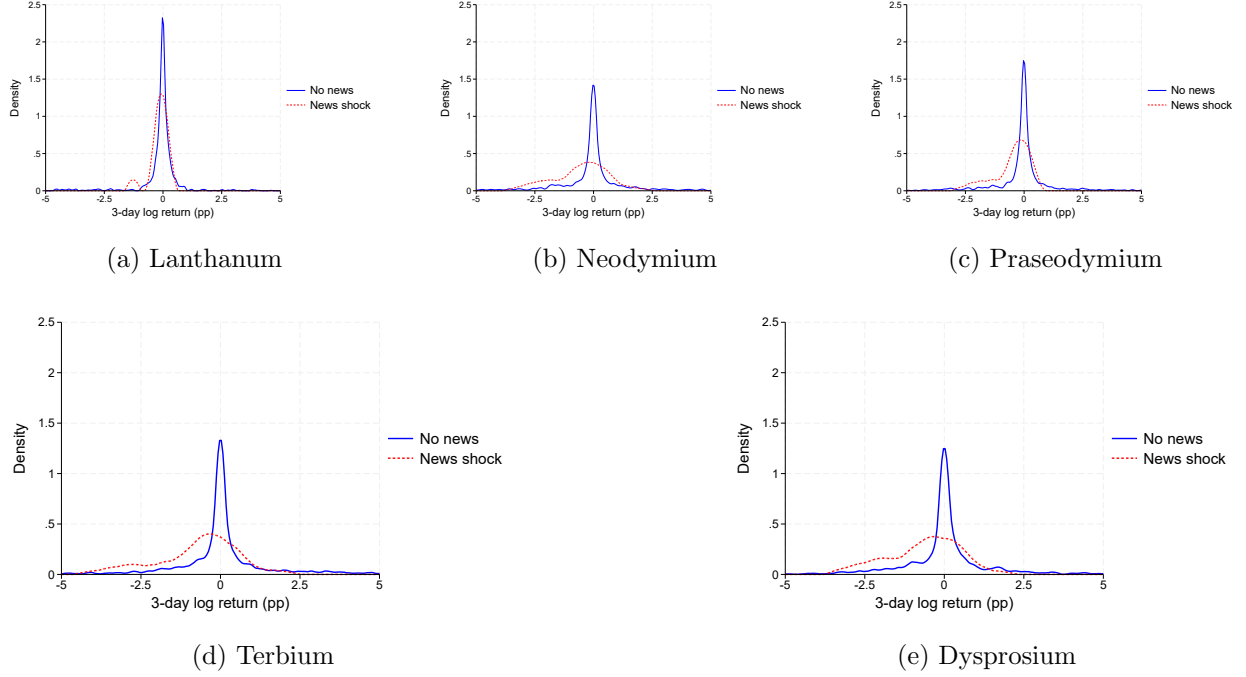
Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A13: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Rare Element Resources.



Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

Figure A14: Kernel densities of three-day log returns (pp), news-shock days vs. strict no-news controls—Shenghe.



Notes: Epanechnikov kernel; Silverman bandwidth; axes harmonized across panels.

**Variance Analysis.** The following tables report Brown–Forsythe tests comparing the dispersion of three-day log returns (pp) in news windows to strictly news-free control windows. For each firm–metal pair I show  $N_{\text{news}}$ ,  $N_{\text{ctrl}}$ , the standard deviations  $SD_{\text{news}}$  and  $SD_{\text{ctrl}}$  of  $100 \times r_{t,[-1,+1]}^M$ , the ratio  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ , and the Brown–Forsythe  $p$ -value:  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ . Significance: \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A11: Variance Analysis: Brown–Forsythe Test for China Northern Rare Earth

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 19                | 1.606                     | 1339              | 2.820                     | 0.570                     | 0.760           |
| Lanthanum    | 19                | 0.211                     | 1339              | 2.454                     | 0.086                     | 0.285           |
| Neodymium    | 19                | 1.182                     | 1339              | 2.481                     | 0.476                     | 0.505           |
| Praseodymium | 19                | 0.764                     | 1339              | 2.379                     | 0.321                     | 0.321           |
| Terbium      | 19                | 1.923                     | 1339              | 2.888                     | 0.666                     | 0.885           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A12: Variance Analysis: Brown–Forsythe Test for Lynas

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 49                | 2.638                     | 1339              | 2.820                     | 0.936                     | 0.642           |
| Lanthanum    | 49                | 2.517                     | 1339              | 2.454                     | 1.026                     | 0.575           |
| Neodymium    | 49                | 2.374                     | 1339              | 2.481                     | 0.957                     | 0.568           |
| Praseodymium | 49                | 2.616                     | 1339              | 2.379                     | 1.100                     | 0.905           |
| Terbium      | 49                | 3.353                     | 1339              | 2.888                     | 1.161                     | 0.784           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A13: Variance Analysis: Brown–Forsythe Test for MP Materials

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 16                | 1.497                     | 1339              | 2.820                     | 0.531                     | 0.941           |
| Lanthanum    | 16                | 1.182                     | 1339              | 2.454                     | 0.482                     | 0.759           |
| Neodymium    | 16                | 1.350                     | 1339              | 2.481                     | 0.544                     | 0.900           |
| Praseodymium | 16                | 1.483                     | 1339              | 2.379                     | 0.624                     | 0.803           |
| Terbium      | 16                | 2.971                     | 1339              | 2.888                     | 1.029                     | 0.321           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A14: Variance Analysis: Brown–Forsythe Test for Iluka

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 38                | 1.783                     | 1339              | 2.820                     | 0.632                     | 0.539           |
| Lanthanum    | 38                | 1.826                     | 1339              | 2.454                     | 0.744                     | 0.474           |
| Neodymium    | 38                | 1.447                     | 1339              | 2.481                     | 0.583                     | 0.415           |
| Praseodymium | 38                | 0.966                     | 1339              | 2.379                     | 0.406                     | 0.295           |
| Terbium      | 38                | 1.819                     | 1339              | 2.888                     | 0.630                     | 0.748           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A15: Variance Analysis: Brown–Forsythe Test for Arafura

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 8                 | 1.155                     | 1339              | 2.820                     | 0.410                     | 0.614           |
| Lanthanum    | 8                 | 0.193                     | 1339              | 2.454                     | 0.079                     | 0.487           |
| Neodymium    | 8                 | 0.691                     | 1339              | 2.481                     | 0.279                     | 0.417           |
| Praseodymium | 8                 | 0.463                     | 1339              | 2.379                     | 0.195                     | 0.407           |
| Terbium      | 8                 | 1.021                     | 1339              | 2.888                     | 0.353                     | 0.667           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A16: Variance Analysis: Brown–Forsythe Test for Northern Minerals

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 36                | 3.020                     | 1339              | 2.820                     | 1.071                     | 0.829           |
| Lanthanum    | 36                | 1.987                     | 1339              | 2.454                     | 0.810                     | 0.966           |
| Neodymium    | 36                | 2.542                     | 1339              | 2.481                     | 1.024                     | 0.563           |
| Praseodymium | 36                | 2.922                     | 1339              | 2.379                     | 1.228                     | 0.851           |
| Terbium      | 36                | 4.006                     | 1339              | 2.888                     | 1.387                     | 0.268           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A17: Variance Analysis: Brown–Forsythe Test for Rare Element Resources

| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 34                | 1.905                     | 1339              | 2.820                     | 0.676                     | 0.746           |
| Lanthanum    | 34                | 1.418                     | 1339              | 2.454                     | 0.578                     | 0.711           |
| Neodymium    | 34                | 1.964                     | 1339              | 2.481                     | 0.791                     | 0.662           |
| Praseodymium | 34                | 1.827                     | 1339              | 2.379                     | 0.768                     | 0.861           |
| Terbium      | 34                | 3.072                     | 1339              | 2.888                     | 1.064                     | 0.386           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A18: Variance Analysis: Brown–Forsythe Test for Shenghe

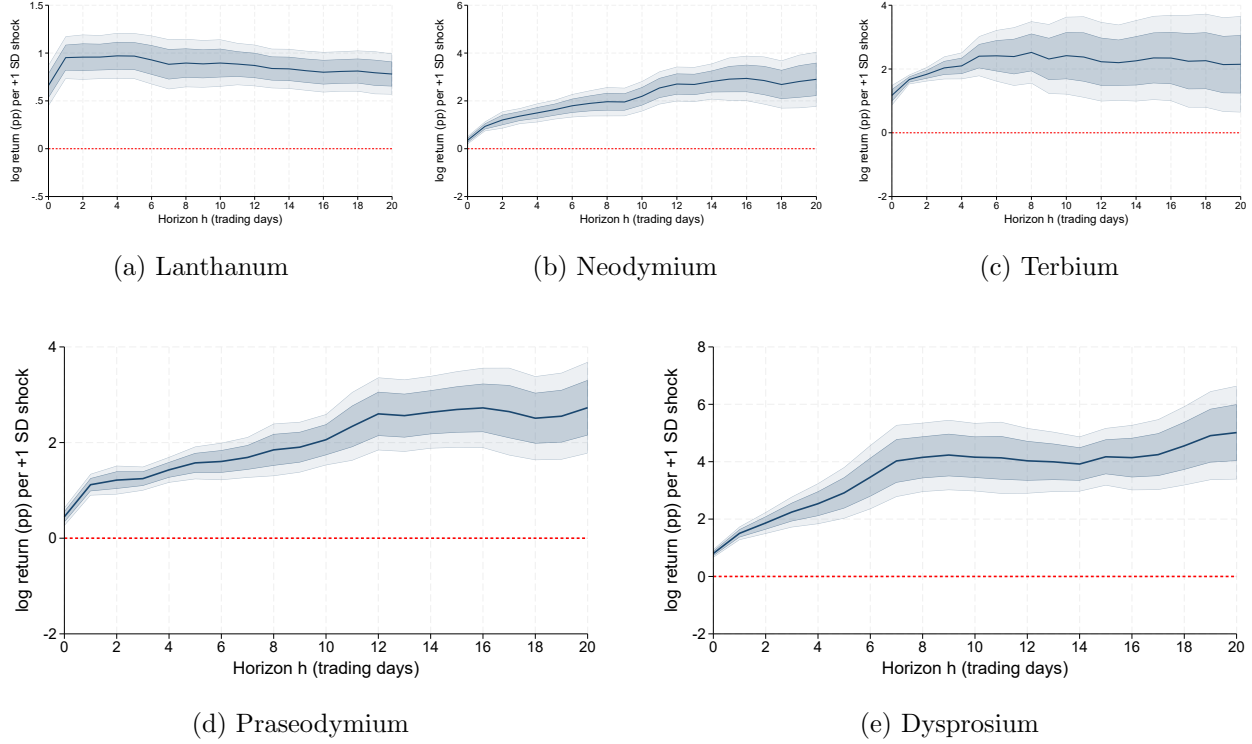
| Rare earth   | $N_{\text{news}}$ | $\text{SD}_{\text{news}}$ | $N_{\text{ctrl}}$ | $\text{SD}_{\text{ctrl}}$ | $\text{SD}_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|---------------------------|-------------------|---------------------------|---------------------------|-----------------|
| Dysprosium   | 15                | 1.072                     | 1339              | 2.820                     | 0.380                     | 0.587           |
| Lanthanum    | 15                | 0.377                     | 1339              | 2.454                     | 0.154                     | 0.423           |
| Neodymium    | 15                | 1.105                     | 1339              | 2.481                     | 0.445                     | 0.591           |
| Praseodymium | 15                | 0.747                     | 1339              | 2.379                     | 0.314                     | 0.449           |
| Terbium      | 15                | 1.252                     | 1339              | 2.888                     | 0.434                     | 0.690           |

*Note:*  $\text{SD}_{\text{Rel.}} = \text{SD}_{\text{news}}/\text{SD}_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .



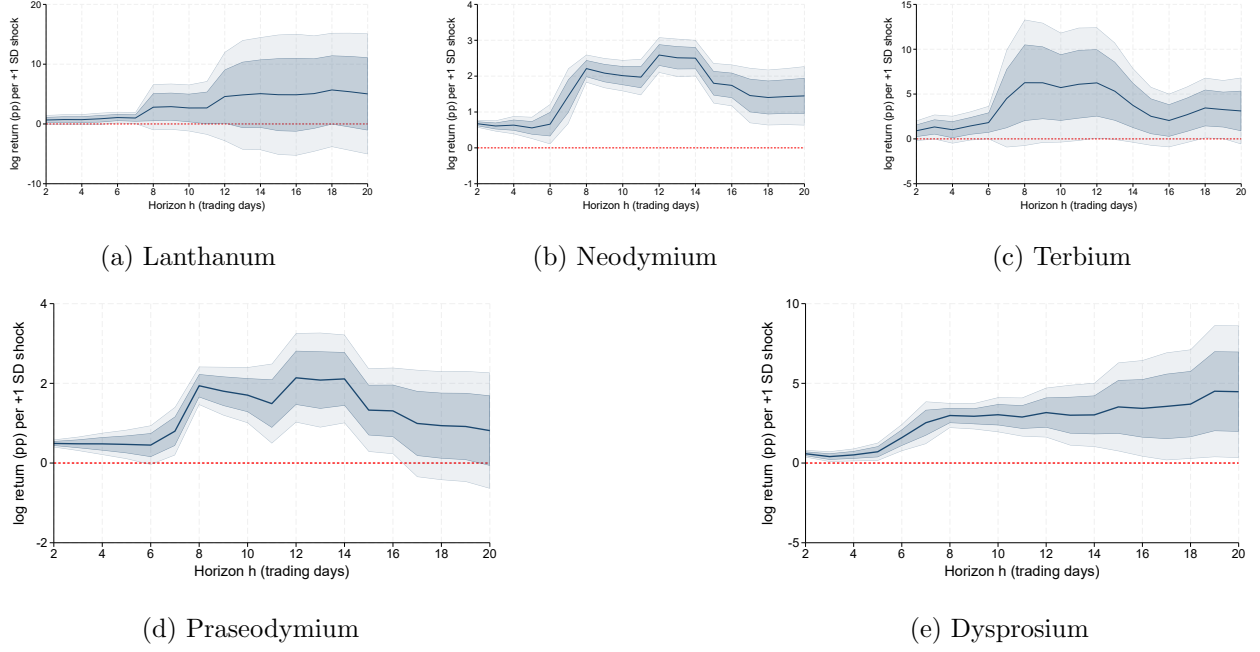
**Local Projections.** Cumulative responses (percentage points) to a one standard deviation news shock day, by metal and firm. Shaded areas: 68% (dark) and 90% (light) CIs; Newey–West  $b_h = \min\{h, 12\}$ .

Figure A15: Pooled impulse responses of rare earth spot prices cumulating from the news shock day  $t$  ( $h = 0 \dots 20$ )



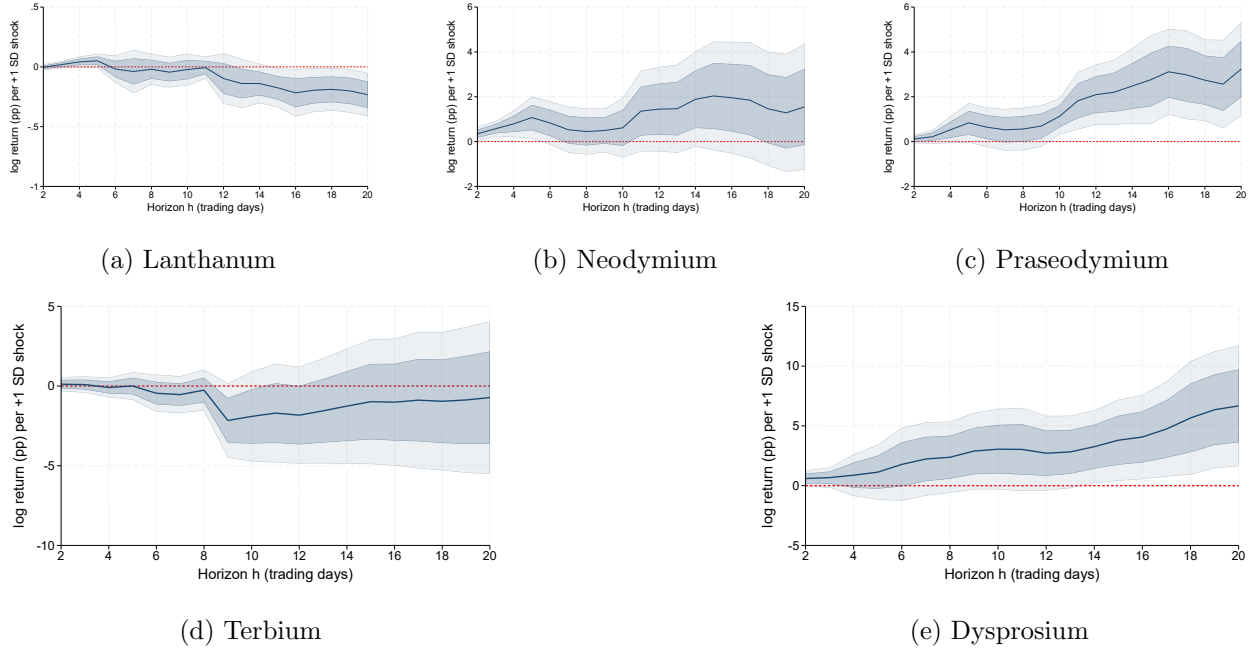
*Notes:* Each panel shows cumulative log returns (pp) starting at the event day  $t$  (no  $t+2$  shift). The pooled regressor sums standardized news shocks across firms and is zero off event windows; coefficients read as pp per one standard deviation. Local projections include  $p = 12$  lags; bands (dark = 68%, light = 90%) use Newey–West HAC with  $b_h = \min\{h, 12\}$ . Because cumulation begins at  $t$ , outcomes mechanically overlap with the shock, so interpret precision with this caveat in mind.

Figure A16: Local projections of log returns (pp) to a one standard deviation news shock—China Northern Rare Earth



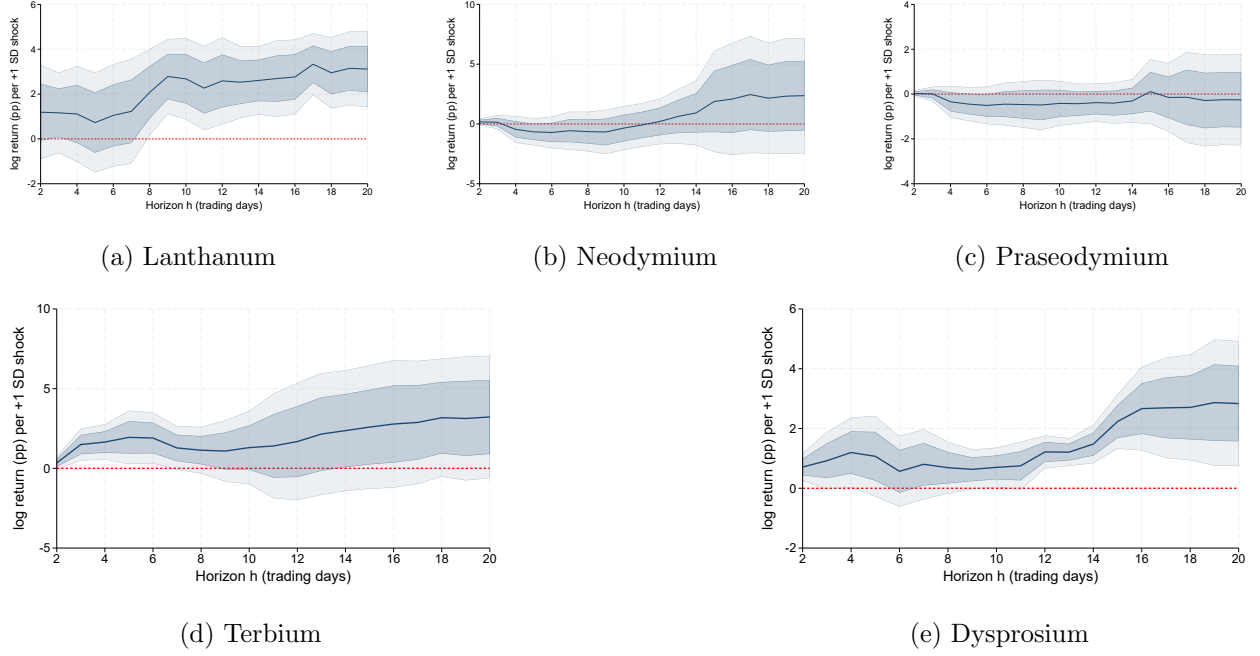
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A17: Local projections of log returns (pp) to a one standard deviation news shock—Lynas



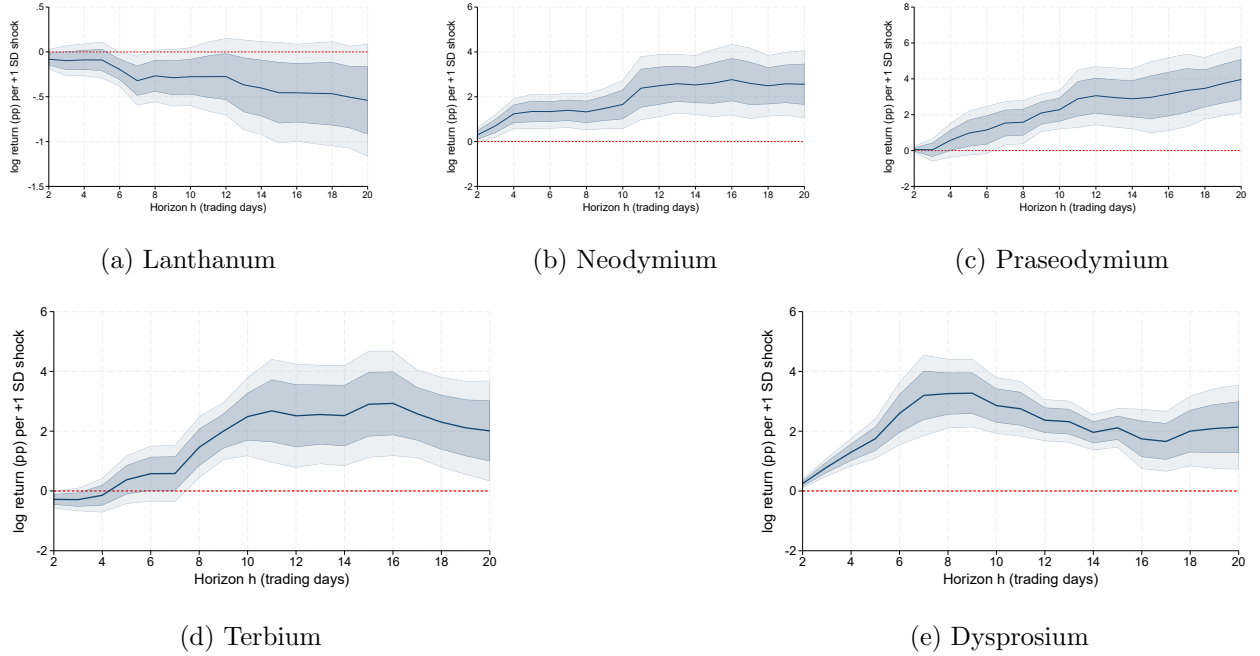
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A18: Local projections of log returns (pp) to a one standard deviation news shock—MP Materials



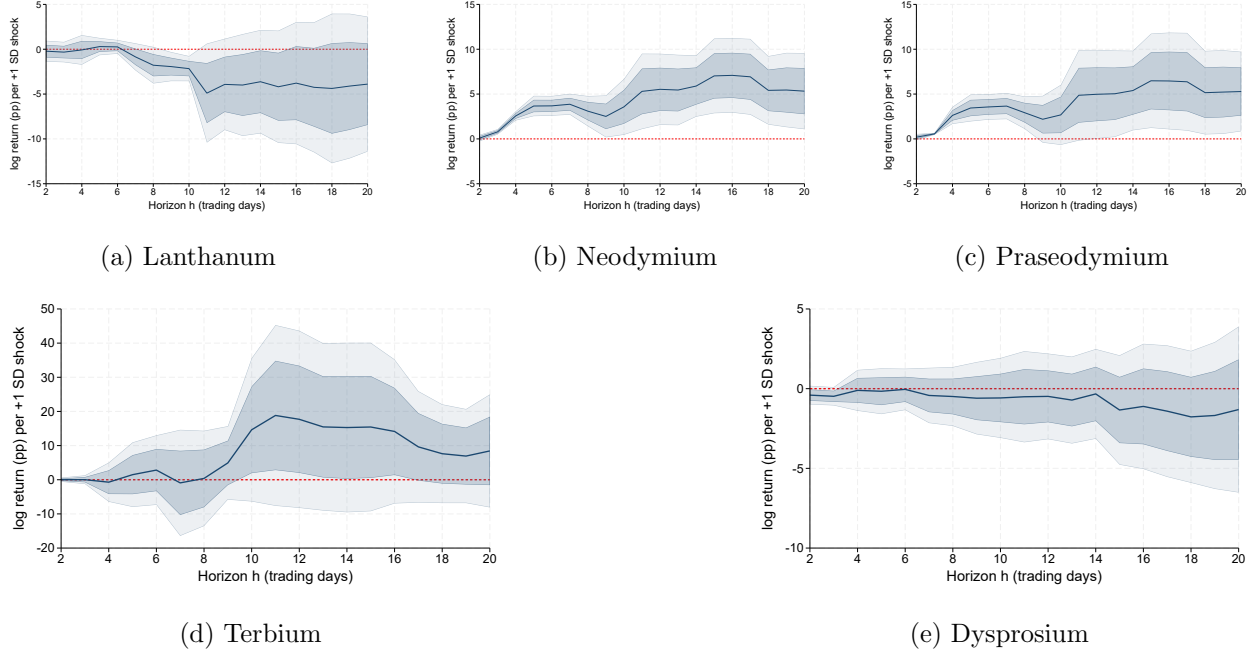
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A19: Local projections of log returns (pp) to a one standard deviation news shock—Iluka



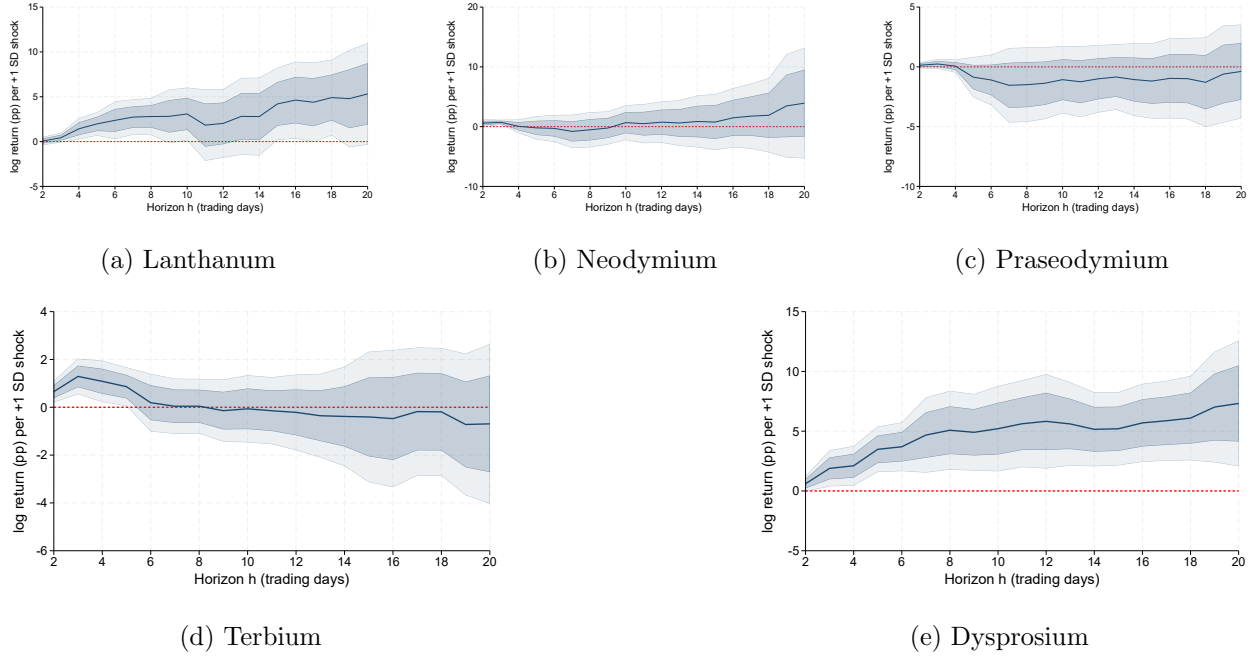
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A20: Local projections of log returns (pp) to a one standard deviation news shock—Arafura



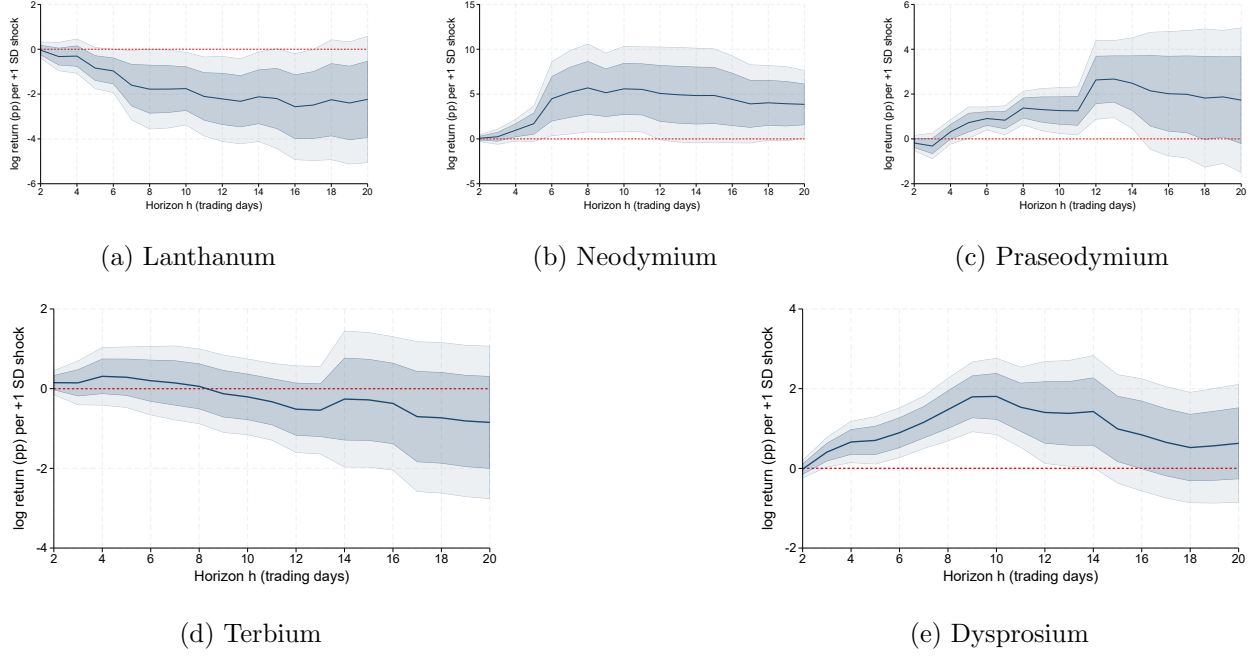
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A21: Local projections of log returns (pp) to a one standard deviation news shock—Northern Minerals



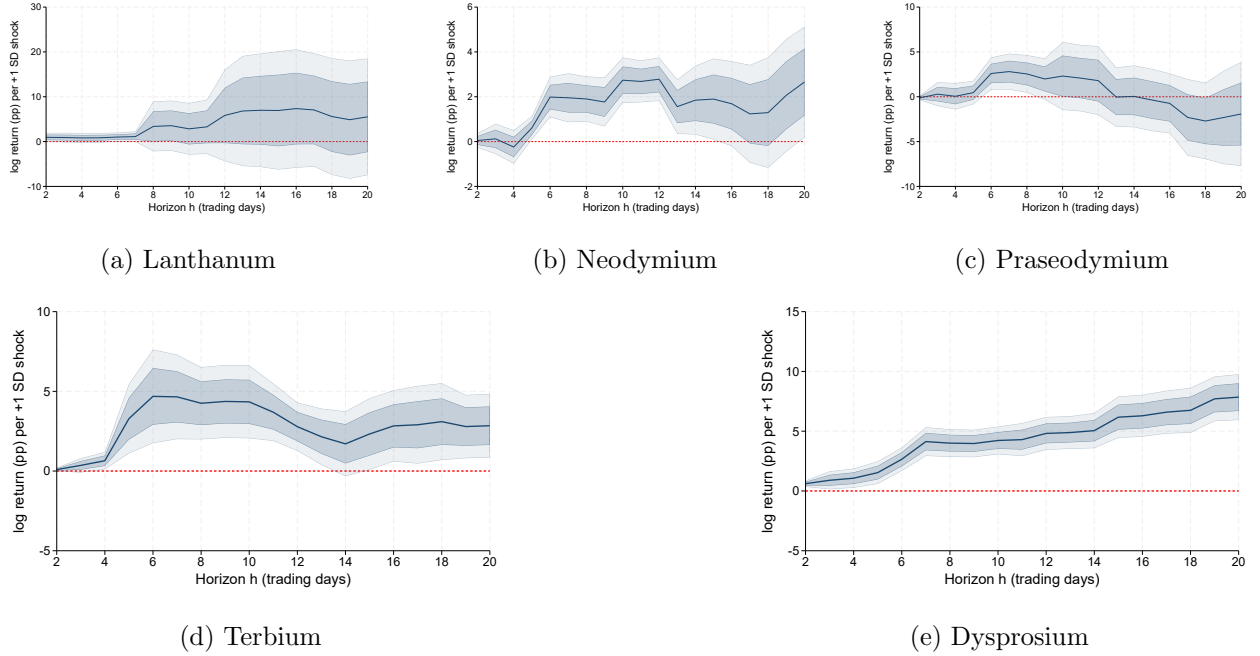
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

Figure A22: Local projections of log returns (pp) to a one standard deviation news shock—Rare Element Resources



Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

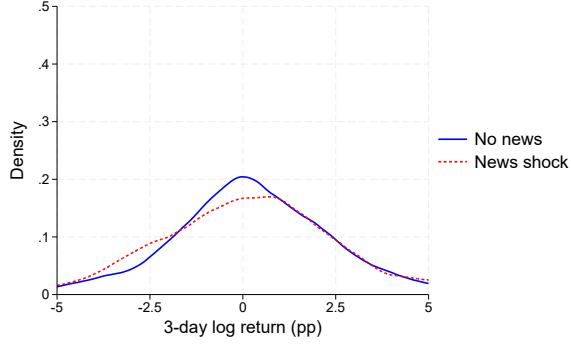
Figure A23: Local projections of log returns (pp) to a one standard deviation news shock—Shenghe



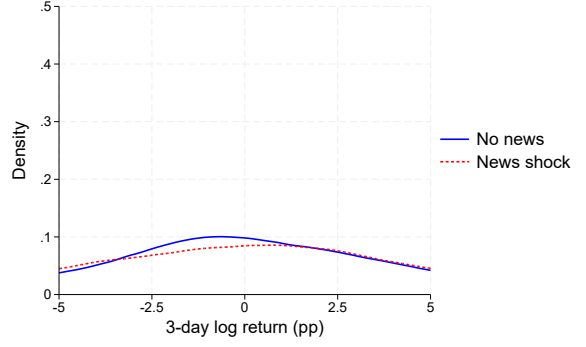
Notes: LP with  $p = 12$  lags; Newey–West with  $b_h = \min\{h, 12\}$ . CI: 68% (dark), 90% (light).

## A.5 Financial Market Additional results

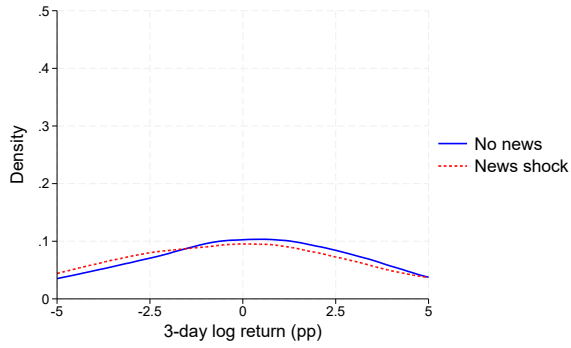
Figure A24: Kernel densities of three-day log returns (pp): news day vs. strict no news windows (general equities)



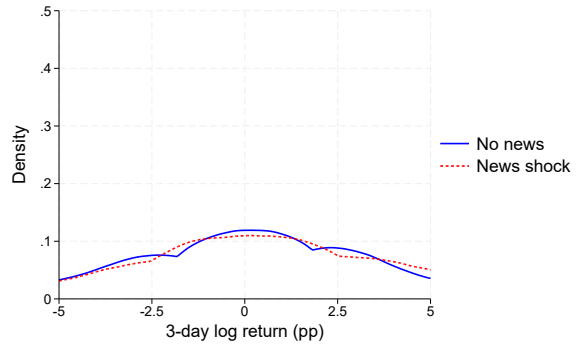
(a) Apple



(b) AMD



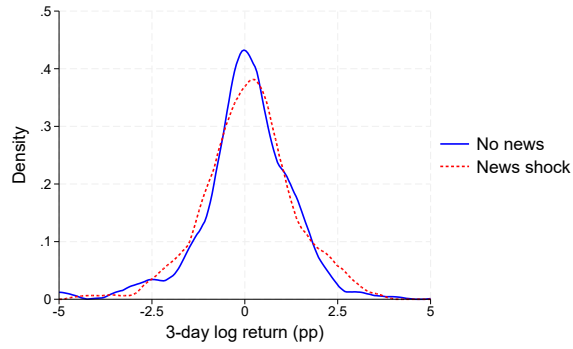
(c) Tesla



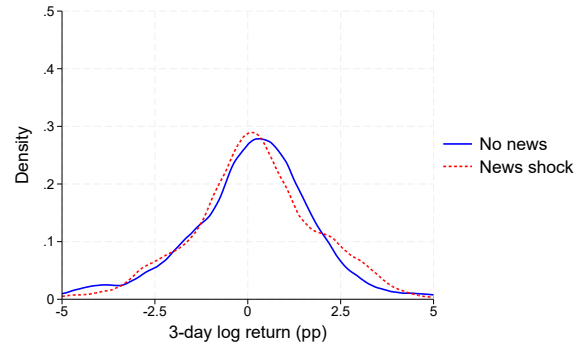
(d) NVIDIA

*Notes:* Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman's normal reference rule  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Axes are fixed across panels for comparability ( $x \in [-5, 5]$  pp;  $y$ -axis held constant).

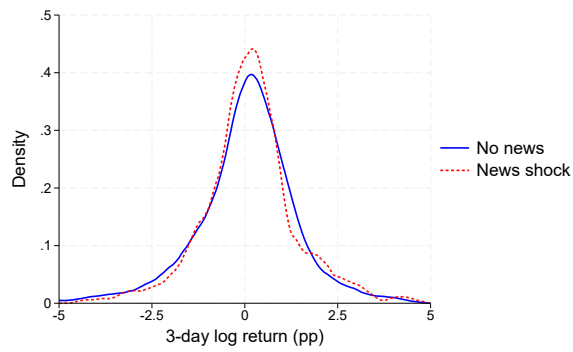
Figure A25: Kernel densities of three-day log returns (pp): news day vs. strict no news windows (general equities)



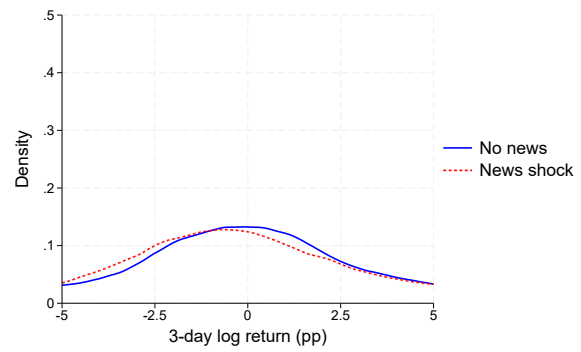
(a) Dow Jones



(b) NASDAQ-100



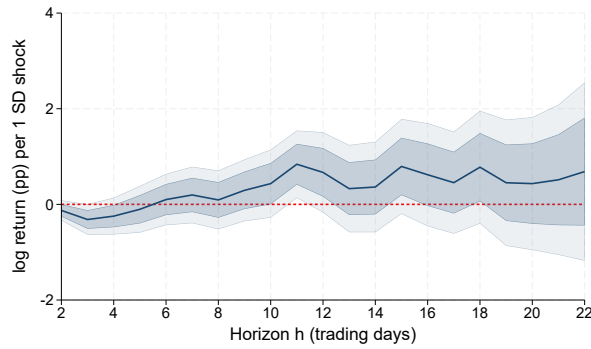
(c) MSCI World Index



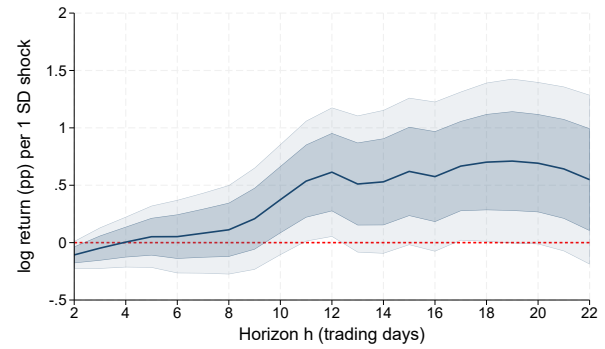
(d) BYD

*Notes:* Kernel density estimates use an Epanechnikov kernel with a single common bandwidth per panel, selected by Silverman's normal reference rule  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986). Axes are fixed across panels for comparability ( $x \in [-5, 5]$  pp;  $y$ -axis held constant).

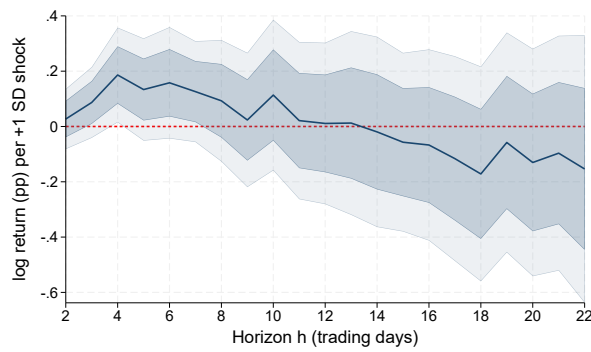
Figure A26: Local projections of log returns (pp) to a one standard deviation news shock for selected equities/indices



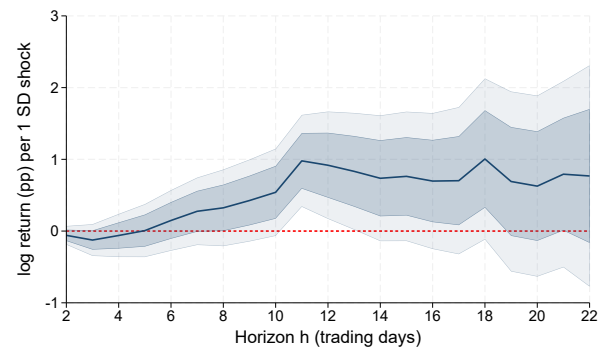
(a) NASDAQ-100



(b) MSCI World Index



(c) BYD



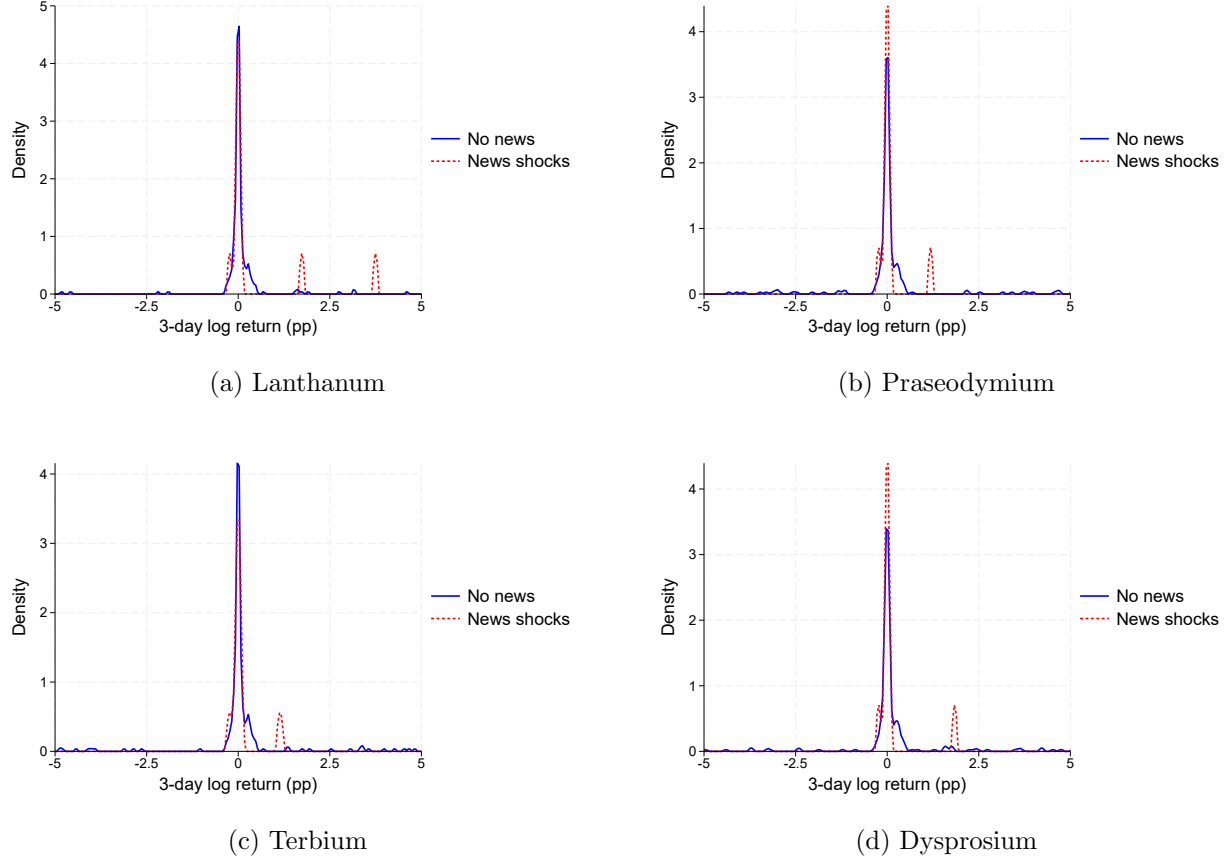
(d) Dow Jones

*Notes:* Cumulative log return (pp) per event over horizons  $h = 2 \dots 22$ ; outcomes cumulate from  $t+2$ . Regressor is an any-firm news-shock indicator. Specification includes  $p = 12$  lags of own returns. Bands: 68% (dark) and 90% (light) from Newey–West HAC with  $b_h = \min\{h, 12\}$ .



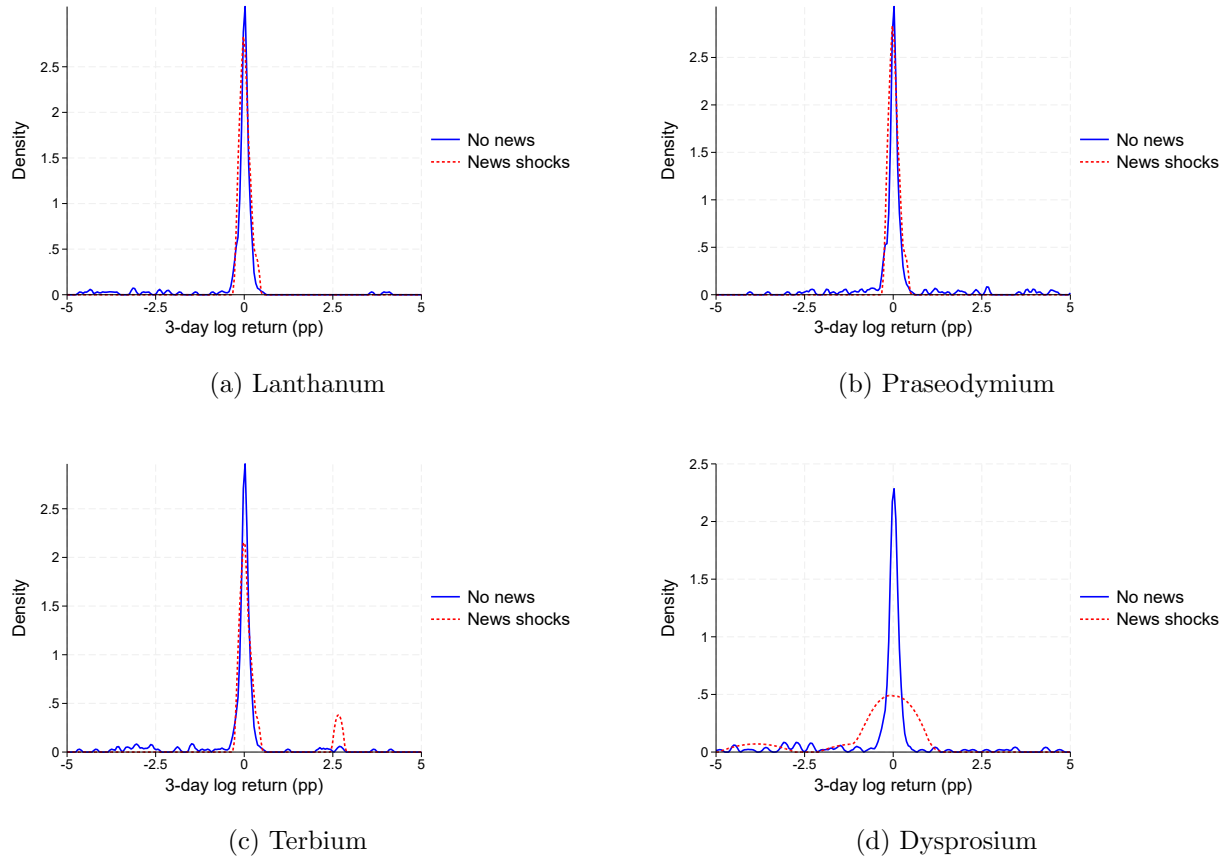
## A.6 Extensions & Robustness checks

Figure A27: Kernel densities of three-day log returns for *Lynas* across Phase 1



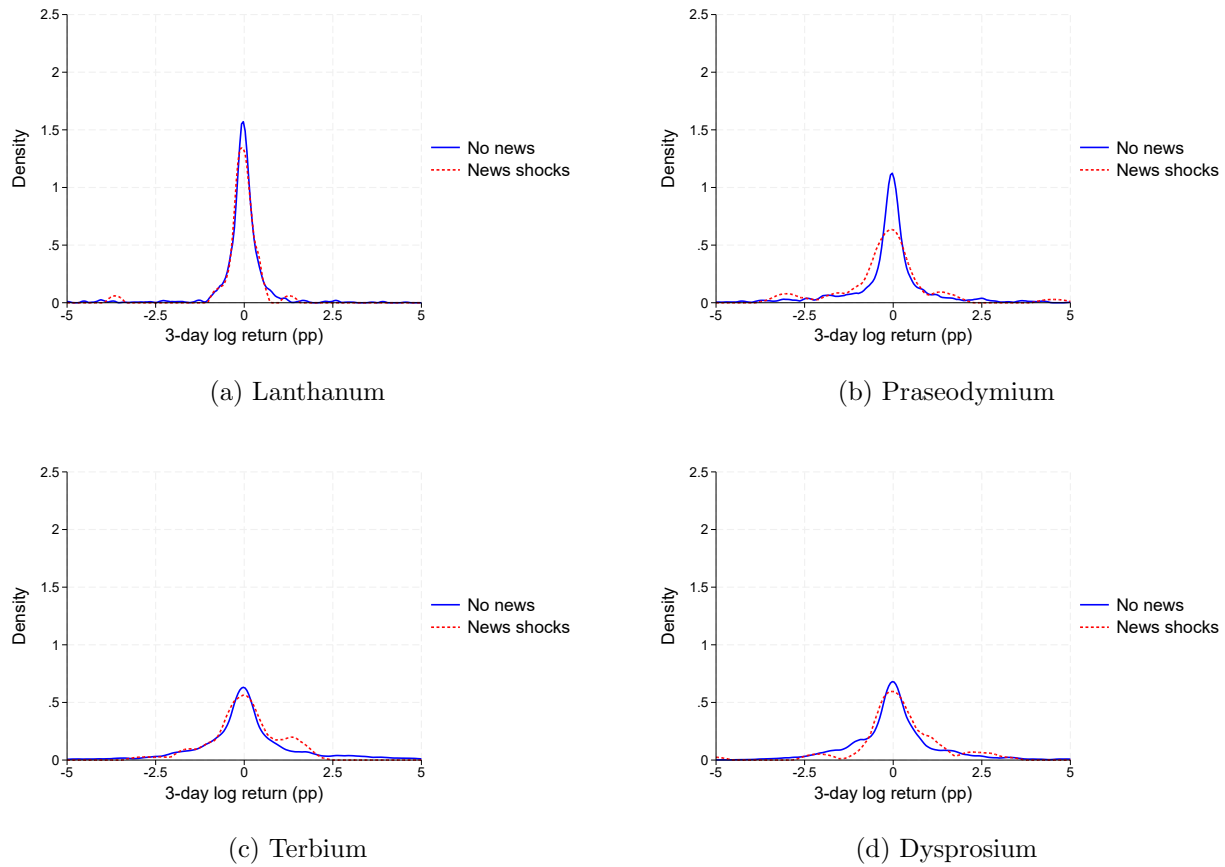
*Note:* Each panel compares days centered on *Lynas* news shocks with strict no news control windows, restricted to Phase 1. Kernel densities use an Epanechnikov kernel with bandwidth from Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986).

Figure A28: Kernel densities of three-day log returns for *Lynas* across Phase 2



*Note:* Each panel compares days centered on *Lynas* news shocks with strict no news control windows, restricted to Phase 2. Kernel densities use an Epanechnikov kernel with bandwidth from Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986).

Figure A29: Kernel densities of three-day log returns for *Lynas* across Phase 3



*Note:* Each panel compares days centered on *Lynas* news shocks with strict no news control windows, restricted to Phase 3. Kernel densities use an Epanechnikov kernel with bandwidth from Silverman's normal reference rule,  $h = 1.06 \hat{\sigma} n^{-1/5}$  (Silverman, 1986).

Table A19: Variance Analysis: Brown–Forsythe Test for Lynas (Phase 1, Pre-2012)

| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------|
| Dysprosium   | 11                | 4.462              | 287               | 4.543              | 0.982              | 0.955           |
| Lanthanum    | 11                | 1.210              | 287               | 4.070              | 0.297              | 0.596           |
| Neodymium    | 11                | 4.428              | 287               | 3.614              | 1.225              | 0.973           |
| Praseodymium | 11                | 5.022              | 287               | 3.538              | 1.419              | 0.747           |
| Terbium      | 11                | 6.432              | 287               | 4.758              | 1.352              | 0.441           |

*Note:* Lynas Phase 1 (pre-2012)  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A20: Variance Analysis: Brown–Forsythe Test for Lynas (Phase 2, 2012–2014)

| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------|
| Dysprosium   | 10                | 2.658              | 266               | 3.258              | 0.816              | 0.915           |
| Lanthanum    | 10                | 0.136              | 266               | 2.422              | 0.056              | 0.314           |
| Neodymium    | 10                | 0.136              | 266               | 2.149              | 0.063              | 0.208           |
| Praseodymium | 10                | 0.136              | 266               | 1.543              | 0.088              | 0.226           |
| Terbium      | 10                | 3.049              | 266               | 2.640              | 1.155              | 0.753           |

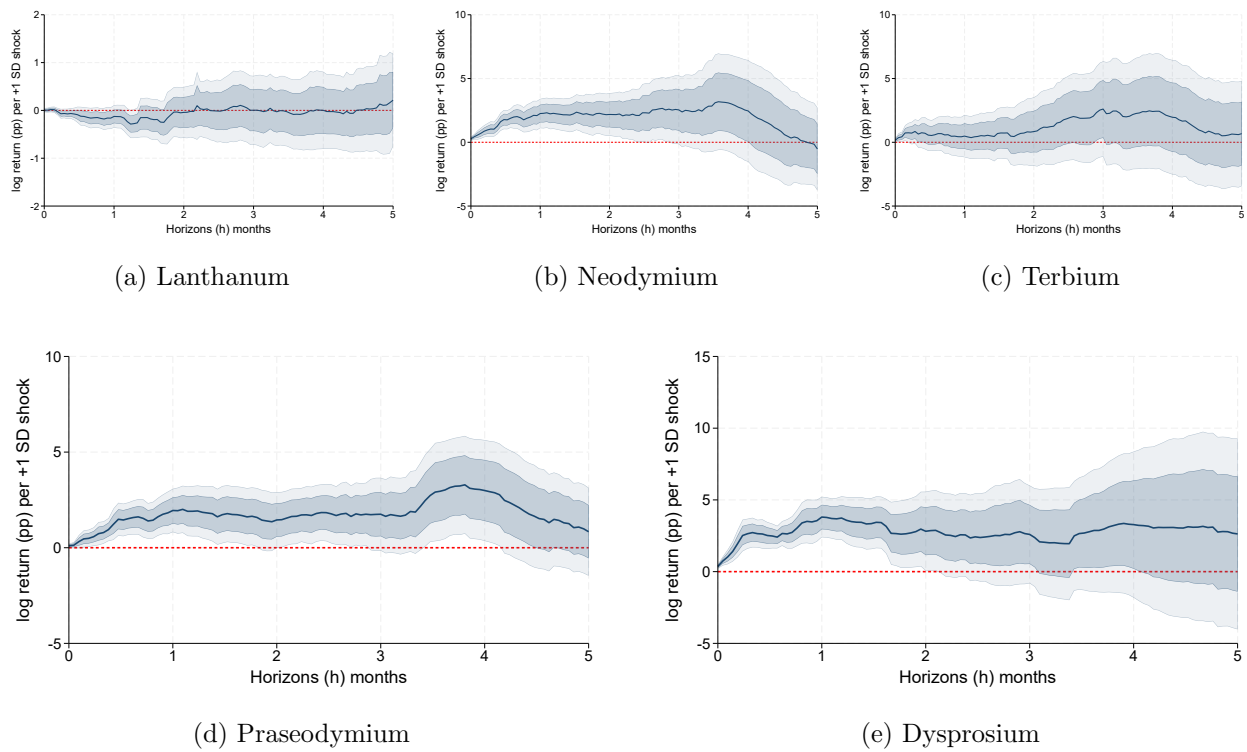
*Note:* Lynas Phase 2 (2012–2014)  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Table A21: Variance Analysis: Brown–Forsythe Test for Lynas (Phase 3, post-2014)

| Rare earth   | $N_{\text{news}}$ | $SD_{\text{news}}$ | $N_{\text{ctrl}}$ | $SD_{\text{ctrl}}$ | $SD_{\text{Rel.}}$ | $p$ -value (BF) |
|--------------|-------------------|--------------------|-------------------|--------------------|--------------------|-----------------|
| Dysprosium   | 43                | 1.290              | 1155              | 1.515              | 0.851              | 0.477           |
| Lanthanum    | 43                | 2.621              | 1155              | 1.342              | 1.953              | 0.027**         |
| Neodymium    | 43                | 1.207              | 1155              | 1.787              | 0.675              | 0.350           |
| Praseodymium | 43                | 1.263              | 1155              | 1.524              | 0.828              | 0.871           |
| Terbium      | 43                | 0.952              | 1155              | 1.992              | 0.478              | 0.080*          |

*Note:* Lynas Phase 3 (post-2014)  $SD_{\text{Rel.}} = SD_{\text{news}}/SD_{\text{ctrl}}$ . Brown–Forsythe test on absolute deviations of 3-day log returns;  $p$ -values from  $F(1, N_{\text{news}} + N_{\text{ctrl}} - 2)$ .

Figure A30: Local projections of log returns (pp) to a one standard deviation news shock five month horizon with  $(h = 0, \dots, 105)$ . Bandwidth  $b_h = \min\{h, 12\}$ .



*Notes:* Local Projection  $p = 12$  lags; Newey–West HAC with  $b_h = \min\{h, 12\}$ . Bands: 68% (dark) and 90% (light). Outcomes are cumulative log returns (pp).

## **Declaration of Academic Integrity**

I, Philip Kroos, hereby declare that the master's thesis entitled "Rare Earth Elements Supply and Financial Markets: Evidence from High-Frequency Analysis" is entirely my own work, and that I have employed no sources or aids other than the ones listed. I have clearly marked and acknowledged all ideas and illustrations that have been taken directly or indirectly from the works of others. I also confirm that this thesis has not been submitted in this form, or a similar form, to any other academic institution.